# Repairing Safety After Benign Fine-Tuning: A Compute-Matched Comparison of Data Repairs

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## Abstract

Fine-tuning a safety-tuned language model on benign instruction data can erode its refusals of harmful requests. When only the training data may change, the owner must choose a repair, such as removing rows or adding safety data, yet repairs are rarely compared with training seeds, side effects and compute counted. We measure which data repair restores the most safety for the compute spent without raising over-refusal or losing task skill; lost safety is measured as refusal erosion as Qwen3Guard labels it and as harmful compliance as the HarmBench classifier defines it. After a benign LoRA fine-tune that partly erodes OLMo-2-1B-Instruct, six pre-registered repairs (rows removed by training-data attribution, a representation filter or a free length rule; random, selected or self-generated refusal examples added), each over six seeds, are compared with random removal and with no repair, labelled better, equivalent within 5 percentage points, worse or inconclusive, and set against five references that change no data. Pre-registered gates test whether the erosion is localized in data categories; a positive control tests whether each training-signal diagnostic finds a concentrated, harmful-prompt cause [PLACEHOLDER: if funded]; and predictions of when diagnosis pays are written down in advance and scored on a second family, Llama-3.2-1B-Instruct, at a matched erosion dose. We find [FINDING 1: the labels of the six repairs]; [FINDING 2: how the best data repair compares with the references]; and [FINDING 3: whether the erosion is localized, and the control's result].

## 1 Introduction

Fine-tuning a safety-tuned ("Instruct") language model on data that contains nothing harmful can still erode its refusals of harmful requests. Qi et al. (2024) observed this after one epoch on ordinary instruction data sets (Alpaca, Dolly) for Llama-2-7b-Chat and GPT-3.5 Turbo; He et al. (2024) showed that benign rows selected for their similarity to harmful examples erode safety far more than random rows; and Guo et al. (2026) report that 100 random benign rows of full fine-tuning raise attack success sharply on recent 8B-class Instruct models. The size of the effect depends on the recipe (Kim et al., 2025). When a fine-tune does erode safety, its owner often controls only the data and a few training settings: commercial fine-tuning services screen training data, gate the tuned model on a safety evaluation, and report harm categories rather than rows, leaving the customer to fix the data and resubmit (Microsoft, 2026).

Repair is then a data decision with side effects. One can *remove* rows: published filters score benign rows by their contribution to erosion (He et al., 2024; Guan et al., 2025; Li et al., 2025; Wang et al., 2026b; Zhang et al., 2026c), and training-data attribution ranks rows by gradient similarity to harmful examples (Jiao et al., 2025; Pan et al., 2026). One can *add* safety data, but which data are added decides the outcome: a random 1% mix of safety rows made Llama3-8B-Instruct less safe than no safety data, while DataRx's selected mix made it safer (Zhang et al., 2026b). Safety data can also cause over-refusal (Bianchi et al., 2024; Luo et al., 2026b). And some repairs need no data change at all: a safety prompt used only at test time (Lyu et al., 2024), a gentler recipe (Kim et al., 2025), a post-hoc projection of the fine-tuned adapter (Hsu et

al., 2024), or a fine-tuning objective that constrains updates on the first answer tokens (Qi et al., 2025). Every data repair costs compute, and the only direct evidence about it is a retrain.

Which repair works best is an empirical question that published results do not settle. Each repair was evaluated in its own setting, with its own models, judges and baselines, mostly with one or a few training seeds, and seldom with its scoring compute charged against the alternative of retraining [CITE-CHECK]. The closest comparison we found, DualSelect (Chen et al., 2026), compares data selectors and a safety-instruction mix over three seeds, with a false-refusal rate and its selection overhead reported, but on training data that include red-teaming pairs, and without model-level references or equivalence-bounded verdicts. Three results suggest that expensive diagnosis may not pay. Once selection compute is charged, powerful data-selection methods are almost never compute-optimal (Yin & Rush, 2025) [CITE-CHECK]. Gradient-based attribution tracks the answer format of training examples more than their task (Kim et al., 2026) [CITE-CHECK], so a free format or length rule may match it. And the erosion may be diffuse: even subsets selected for low conflict with safety still erode it (Guo et al., 2026), and Springer et al. (2026) argue from theory that better curation cannot remove the fragility.

We therefore measure repairs directly (§3, §4). OLMo-2-1B-Instruct is fine-tuned on 4,800 Dolly rows with a fixed LoRA recipe (recipe B: a constant learning rate, one epoch). Six repairs fixed in advance, each run over six seeds, are compared with random removal of the same size and with no repair: three removals of 600 rows (ranked by training-data attribution, a LARF-style representation filter or answer length) and three additions of 48 refusal examples (random, selected in the style of DataRx, or written by the untuned model). Each contrast is labelled *better*, *equivalent* (its 90% interval within $\pm 5$ percentage points, pp), *worse* (significantly adverse on the unsafe rate, over-refusal or task loss) or *inconclusive*, after multiplicity correction; over-refusal and held-out task loss are feasibility limits, and cost is measured in full-run equivalents (FRE) under two conventions. A data repair counts as dominant only if it beats five references that change no data, none of them ours: a test-time safety prompt, the adapter scaled down, its Safe LoRA projection (Hsu et al., 2024), recipe G (a standard warmup and cosine schedule) and recipe B with the constrained objective of Qi et al. (2025). The second family, Llama-3.2-1B-Instruct, uses recipe B at its own learning rate, so that both families erode partly (a matched erosion dose; §4.1).

Pre-registered parts surround the comparison. Gate A requires erosion beyond seed noise. Gate B asks whether the erosion is *localized*: does removing one Dolly category restore more safety than removing as many random rows? Gate C asks which diagnostic picks that category, and whether a format rule explains attribution's pick. A positive control plants a concentrated, harmful-prompt cause in nested menus of 8, 32 and 128 cells [TO BE FINALISED] and asks whether each training-signal diagnostic finds it, and at which menu size diagnosis beats trial runs [PLACEHOLDER: if funded]. A secondary, model-based analysis predicts when paying for a diagnostic beats spending the same compute on trial runs selected by successive halving (Jamieson & Talwalkar, 2016; Li et al., 2018). Every rule is committed before the data it judges; outcomes are replicated over seeds, confirmed on held-out prompts, and reported as *too close to call* when the evidence cannot separate options.

Our contributions are:

- **A measured comparison of data repairs with equivalence bounds** (§3.2, §4.3–§4.4): six repairs over six seeds each, labelled better, equivalent, worse or inconclusive, with over-refusal, task loss and measured cost, against five references that change no data, on OLMo-2-1B and, in stage 2, on Llama-3.2-1B. [FINDING 1: the six labels, and which repair, if any, beats the references]. We release the menus, seeds, outcome tables and costs [ARTIFACT].

- **A positive control for diagnosis** (§4.5): [FINDING 2: whether each training-signal diagnostic finds a concentrated, harmful-prompt cause, and the runs-only cost of finding it, at menu sizes 8, 32 and 128] [PLACEHOLDER: if funded]. Text-only scorers' ranks are construction checks, published before the pilot.
- **Where the erosion lives** (§4.6, §4.10): [FINDING 3: the Gate B verdict with its detection limit; Gate C's reading with the format control; how much of each repair's effect runs through refusal openings].
- **Pre-registered predictions of when diagnosis pays** (secondary; §3.4): [FINDING 4: predicted against measured value of diagnosis on the category and planted menus, and on the second family].

[FIGURE 1: the repair comparison. For each repair, the safety restored against random removal or no repair, with its 90% interval and the ±5 pp equivalence band, plotted against over-refusal (left) and measured cost in FRE (right); the adapter-scaling reference curve is drawn in the left panel.]

## 2 Related Work

**Safety erosion under benign fine-tuning.** Qi et al. (2024) showed that fine-tuning aligned models on benign instruction data degrades safety, though less than harmful data does, with no training seeds reported. He et al. (2024) selected benign rows by representation and gradient similarity to harmful anchors; in their best case, 100 such rows gave over 70% unsafe answers against under 20% for random rows. This shows that benign rows *can* erode safety, not that ordinary fine-tunes have a localized cause. Kim et al. (2025) change the size of the erosion several-fold through optimization choices alone. Guo et al. (2026) find that 100 random benign rows of full fine-tuning erode recent 8B-class Instruct models, and that a handful of safety rows restores safety after a small attack. Under LoRA their results depend on the family: in their Table 2, Qwen2.5-7B-Instruct reads 7.27% before fine-tuning, 4.50% after 100 rows and 9.09% after 5,000 rows (two seeds), while Llama-3.1-8B-Instruct rises to 44.80% at 5,000 rows. Springer et al. (2026) explain erosion geometrically, as benign updates curving into a sharp, safety-relevant subspace.

**Removal repairs: filters and attribution.** Published filters score benign rows for their contribution to erosion. DataShield projects rows onto a compliance direction (Zhang et al., 2026c); LARF uses representations of safety-sensitive layers (Li et al., 2025); SQSD projects each row's parameter update onto danger and safety directions (Wang et al., 2026b); bi-anchoring compares a row's gradients or representations with harmful and safe anchors (He et al., 2024); and Self-Inf-N, proposed as an attack, finds outlier benign rows whose fine-tuning breaks safety (Guan et al., 2025) [CITE-CHECK]. Row filters report their scoring cost without charging it against runs (Zhang et al., 2026c; Hu et al., 2026) [CITE-CHECK]; other defences select tokens (Li et al., 2026) or use continual learning (Alssum et al., 2025) [CITE-CHECK]. A concurrent submission finds risk rankings taken at initialization nearly uncorrelated with those after two epochs, and re-scores rows during training (Anonymous, 2026). For attribution, Pan et al. (2026) find that removing rows flagged by gradient-based attribution gives a safer retrained model than removing WildGuard's picks, from one run per arm and with no compute reported. DATE-LM (Jiao et al., 2025) finds that with safety-aligned look-alikes in the data, attribution's detection drops while LLM classifiers are largely unaffected; without them, the strongest classifiers detect better, yet removing attribution's picks gives the safer retrained model. Luo et al. (2026b) report weak, inconsistent effects from deleting or reweighting influential rows, large effects from rewriting them, and an over-refusal trade-off for safety rewriting.

**Addition and model-level repairs.** Adding about 3% safety examples to instruction tuning improves safety but can cause exaggerated refusals (Bianchi et al., 2024) [CITE-CHECK]. Which rows are added matters: DataRx selects safety rows by the gap between a model's own response and a safe reference (Zhang et al.,

2026b) [CITE-CHECK]; on Llama3-8B-Instruct, averaged over seven tasks, its random 1% mix raised attack success from 20.24% (no safety data) to 59.23%, while its selected mix lowered it to 13.70%. Its baselines, all additions, include self-distillation, a precedent for our self-generated arm. Guo et al. (2026) restore safety with a handful of safety rows, and Llama 3's safety mix was tuned by repeated experiments that traded violations against false refusals (Grattafiori et al., 2024). Model-level repairs change no data: a safety prompt used only at test time reduces erosion (PTST; Lyu et al., 2024), SafeTune compares repairs of safety drift in single deterministic runs (Seth et al., 2026) [CITE-CHECK], a gentler recipe shrinks erosion (Kim et al., 2025), Safe LoRA projects the LoRA weights of selected layers onto a safety-aligned subspace without training or data (Hsu et al., 2024), and an objective that constrains updates on the first answer tokens makes alignment more persistent under fine-tuning (Qi et al., 2025). We use such repairs as references that a data repair must beat. Other repairs act after fine-tuning and are not among our references (§6): Safe Delta adjusts the parameter change of a harmful or benign fine-tune (Lu et al., 2025), Antidote prunes harmful weights after harmful fine-tuning (Huang et al., 2025), and Zhang et al. (2026a) restore safety by further training on a single safety example. Huang et al. (2024) survey harmful fine-tuning attacks and defences.

**Attribution, its cost and what it tracks.** Training-data attribution ranges from cheap gradient similarity on LoRA gradients (LESS; Xia et al., 2024) [CITE-CHECK] to estimates of finite removal effects by unrolled differentiation (SOURCE; Bae et al., 2024), which are reported to cost several training runs [CITE-CHECK]. Xiao & Aranguri (2026) compare the cost of diagnosis methods for undesirable post-training behaviours; probe attribution is over ten times cheaper than gradient attribution and LLM-judge baselines in their study. Yin & Rush (2025) charge selection compute against training compute and find that powerful selection methods are almost never compute-optimal; gradient-based selection pays only when the trained model is about ten times larger than the selection model [CITE-CHECK]. Kim et al. (2026) find that gradient similarity tracks answer format more than task [CITE-CHECK]; we therefore include a free answer-length rule as a repair and as a control for attribution's picks.

**Prediction and decisions under a budget.** Persona vectors (Chen et al., 2025) project training data onto activation directions to flag data that will shift a trait [CITE-CHECK]; TuneAhead (Luo et al., 2026a) and Data2Behavior (Wang et al., 2026a) predict a fine-tune's outcome from its data before training; MNEME predicts side effects of fine-tuning and unlearning by sparse model diffing (Kassem et al., 2025) [CITE-CHECK]; any such predictor can enter our comparison as a scorer at its measured cost. Successive halving and Hyperband allocate a fixed budget by repeatedly discarding the worst candidates at increasing fidelity (Jamieson & Talwalkar, 2016; Li et al., 2018); multi-fidelity methods model how cheap observations relate to full ones (e.g. Ouyang et al., 2025). DataDecide (Magnusson et al., 2025) scores small pretraining runs by decision accuracy against compute, and Ivison et al. (2025) charge selection compute against training on more data [CITE-CHECK].

**How we differ.** We do not claim a first benchmark of safety filters or defences, a first comparison of removal against addition, or a first pricing of data selection: SafeTune and SafeTuneBed (Hossain et al., 2025) compare defences; DualSelect compares selectors and a safety-data mix over seeds, with a false-refusal rate and selection cost; DataRx compares ways to choose added safety data; and DataShield's cost table and Yin & Rush (2025) price selection [CITE-CHECK]. Our contribution is the combination; to our knowledge, and not found by our search, no prior work combines, for purely benign data, equivalence-bounded contrasts over six training seeds for removal and addition repairs; over-refusal and task loss as limits fixed in advance; every arm's full cost, scoring included, under two conventions; model-level references that a data repair must beat; a positive control with a planted cause; and predictions about the value of diagnosis written down before the data.

## 3 Problem Setup

### 3.1 Setting and endpoints

Let $\theta_0$ be a safety-tuned Instruct model, fine-tuned with a fixed recipe $\mathcal{R}$ (adapter, loss, optimizer, schedule, length rule) on benign data $\mathcal{D} = D_1 \cup \cdots \cup D_K$, split into $K$ *units* by metadata fixed before any outcome. A *repair* $a$ maps $(\mathcal{D}, \mathcal{R})$ to a training set and recipe $(a(\mathcal{D}), \mathcal{R}_a)$; a data repair keeps $\mathcal{R}_a = \mathcal{R}$. The model trained with seed $s$ (adapter initialization and data order) is $\theta(a, s)$. For a model $\theta$, $f(\theta)$ is the *unsafe rate* (the share of sampled answers to a harm bank that a frozen judge labels unsafe), $\ell(\theta)$ is the held-out *task loss*, and $r(\theta)$ is the *over-refusal rate* on benign prompts that resemble harmful ones. Endpoints average over seeds: $F(a) = \mathbb{E}_s[f(\theta(a, s))]$, and likewise $\bar{\ell}(a)$ and $R(a)$. With $a_0$ the fine-tune without repair, the *erosion*, refusal erosion as Qwen3Guard labels it (§4.2), is $E = F(a_0) - f(\theta_0)$; repair is meaningful only when $E$ exceeds seed noise (Gate A, §4.6). Two actors can repair: a fine-tuning customer, who controls the data, learning rate and epochs, and the service provider, who also holds the weights and can run every scorer and reference. We report the best repair within each actor's menu (Table 1).

### 3.2 Repair contrasts, limits and labels (primary)

Let $a_{\mathrm{R}}$ remove the same number of random rows as the removal repairs. Removals are matched in rows; a token-matched analysis beside the row-matched one, and token-matched random removals for A1, L1 and H1 as secondary arms in the funded tier, are pre-registered (Appendix C.3). A removal repair $m$ is compared with random removal, $\Lambda_m = F(a_{\mathrm{R}}) - F(m)$: the safety it restores beyond removing the same amount of data at random. An addition repair is compared with no repair, $\Lambda_m = F(a_0) - F(m)$, and a selected or self-generated addition with the random addition of the same size. The restoration $G_m = F(a_0) - F(m)$ is reported for every arm. The primary model, HB-C, pairs arms by seed within two design clusters (no repair and the additions; random removal and the removals) and crosses prompts with seeds; HB, S-a and S-aW are pre-registered sensitivity analyses (Appendix C.11) [PLANNED; the stage-1 pre-registration is not yet frozen].

Side effects are the *over-refusal excess* $O(m) = R(m) - r(\theta_0)$, against the untuned model ($R(m) - R(a_0)$ is also reported), and the *task-loss increase* $L(m) = \bar{\ell}(m) - \bar{\ell}(a_0)$. A repair is *feasible* at limits $(\tau_O, \tau_L)$ if the two-sided 90% intervals for $O(m)$ and $L(m)$ end at or below them, *infeasible* on a measure whose interval lies above its limit, else *undetermined* (Appendix C.11) [PLANNED; the stage-1 pre-registration is not yet frozen]. Results are reported at $\tau_O \in \{2, 5, 10\}$ pp; $\tau_L$ is two standard errors of a seed-to-seed difference in held-out loss, fixed before the comparison runs. The *cost* of a repair is $C(m) = c(m) + 1$ FRE: $c(m)$ is the measured cost of building the repaired data (scoring rows, selecting or generating safety rows), and one FRE is the measured cost of one random-removal run (§4.7).

With a smallest effect of interest $\delta = 5$ pp (scaled with the erosion $E$ in another family; Appendix C.11) and one closed testing family at family-wise level 0.05 (§4.9), each contrast receives one label: *better* if its test rejects $\Lambda_m = 0$ and the estimate is at least $\delta$; *equivalent* if the 90% interval for $\Lambda_m$ lies inside $(-\delta, \delta)$ (two one-sided tests); *worse* if it is significantly adverse on the unsafe rate ($\Lambda_m < 0$) or, by rule R1.3, on over-refusal or task loss: infeasible against $\theta_0$ at the primary over-refusal bar $\tau_O^* = 5$ pp (task loss: against $a_0$ at $\tau_L$) and significantly above the comparator; and *inconclusive* otherwise. *Better* and *equivalent* exclude each other (Appendix C.11). *Worse* is checked first, and the full row is always printed (Appendix C.11) [PLANNED; the stage-1 pre-registration is not yet frozen]. A margin of 4 pp is reported as a sensitivity. Five references change no data (§4.4). One, *adapter scaling*, $\theta_\lambda = \theta_0 + \lambda\,(\theta(a_0, s) - \theta_0)$ for $\lambda \in \{1, 0.75, 0.5, 0\}$, traces $(F, O, L)$ from the fine-tune ($\lambda = 1$) to the untuned model ($\lambda = 0$). A data repair *dominates* only if it lies below this curve: a lower $F$ than the curve at equal $O$ and $L$.

### 3.3 Localization

Let $a_{-k}$ remove unit $D_k$ and $a_{\text{rand},k}$ remove $|D_k|$ random rows drawn across units. Erosion is *localized* at level $\delta_B$ if some $\Lambda_k = F(a_{\text{rand},k}) - F(a_{-k})$ reaches $\delta_B$ beyond seed noise (the random arm controls for the smaller training dose), and *diffuse* if every $\Lambda_k$ is credibly below $\delta_B$. Diffuse erosion leaves the repair comparison unchanged: unit removals then act like random removal up to $\delta_B$, and a unit-level diagnostic has little to find.

### 3.4 The value of diagnosis (secondary, model-based)

A *selector* with budget $B$ (in FRE) buys trial runs on a menu of $M$ repairs and returns a pick; its regret is the expected unsafe rate of the pick minus that of the menu's best feasible repair. The value $\Delta_d$ of a diagnostic $d$ is the regret saved by paying its cost before the same frozen trial-run rule, and its break-even cost $c_d^*$ is the largest cost with $\Delta_d \geq 0$. At the menus we can afford, these are mostly outputs of a model, frozen before Gate B's category runs; we report measured against predicted values and score OLMo-calibrated predictions on Llama (Appendix C.7).

## 4 Experimental Design

### 4.1 Models, data and recipes

**Families and scale.** At 1B, OLMo-2-0425-1B-Instruct (Team OLMo et al., 2024) is the stage-1 model and Llama-3.2-1B-Instruct (Grattafiori et al., 2024) the second family (stage 2; §4.8). At 7–8B, the full comparison runs on OLMo-2-1124-7B-Instruct and Llama-3.1-8B-Instruct, at learning rates fixed by the same rule (below) [PLANNED; funding-gated, fallback one family]. Olmo-3-7B-Instruct (Team Olmo et al., 2025) is a third 7–8B model under the same rule [PLANNED; funding-gated; first 7–8B model cut]. A headline check at the largest single-GPU size comes last [PLACEHOLDER: if resources allow]. Qwen2.5 (Qwen et al., 2024) is outside this study (§6). Every model trains on one GPU, and every comparison stays on one GPU type.

**Data.** Gate B and stage 1 train on 4,800 rows of Dolly-15k (Conover et al., 2023), 600 from each of 8 task categories (the units; $K = 8$), sealed per tokenizer with a manifest; 200 further rows, 25 per category, measure task loss. The add repairs draw their safety rows from PKU-SafeRLHF (§4.3). Alpaca is the planned second benign dataset, after Gate B [NOT YET FIXED: its units and learning rate].

**Recipes.** *Recipe B*, the main recipe: LoRA (Hu et al., 2022) of rank 64 and scale 128 on all linear layers; AdamW at a constant learning rate of $10^{-4}$, no warmup; 128 rows per step; one epoch (38 steps); a per-row loss (each row's mean over its response tokens). Gate A, Gate B and stage 1 use it. *Recipe G*, a reference arm that changes only the recipe (§4.4), adds a 4-step warmup and cosine decay, which halve the mean learning rate, and a token-mean loss. It is not a general fix: its effect on erosion differs across families and sizes [CITE: concurrent submission, anonymized]. Training is deterministic on matched hardware.

**A matched erosion dose.** Each family uses recipe B with only its learning rate changed: the highest rung of a ladder ($10^{-4}$, $5 \times 10^{-5}$, $2 \times 10^{-5}$) whose probe reads partial (20–55 pp) under the frozen one-seed Gate A bars (§4.6; a probe too close on $E$ gets one more seed; any other too-close bar goes to the user). The rate is chosen from probes only, never from a repair result. Families respond differently to the same recipe [CITE: concurrent submission, anonymized]; we fixed this rule after reading a one-recipe development ladder, on which no rung left both 1B families partly eroded, and Llama's first probe at recipe B, and we state cross-family claims at a matched erosion dose. The rungs are $10^{-4}$ for OLMo-2 (1B and 7B) and $5 \times 10^{-5}$ for

Llama (3.2-1B and 3.1-8B). Both Llama rungs were accepted on a too-close bar (one seed's answers-end bar at 1B; the window's upper edge at 8B); §5 reports both.

### 4.2 Measures, judges and the claim rule

- **Unsafe rate $U$.** The share of 600 sampled answers (200 development harm prompts × 3 samples at temperature 0.7, up to 2,048 new tokens) that Qwen3Guard-Gen-4B (Zhao et al., 2025) labels Unsafe. The prompts come from HarmBench (Mazeika et al., 2024), StrongREJECT (Souly et al., 2024), SimpleSafetyTests (Vidgen et al., 2023) and OR-Bench's toxic set (Cui et al., 2025). Standard errors are clustered by prompt. $U$ is a broad policy label, not an attack-success rate as published elsewhere: judges and prompts differ, so $U$ is compared only within this study. A rise in $U$ is refusal erosion as Qwen3Guard labels it. Gates and selectors read development banks only; final claims are confirmed on held-out prompts (§4.8).
- **Compliance and the claim rule.** HarmBench's validation classifier (512-token window; Appendix C.10) asks whether an answer carries out the requested behaviour. A contrast in $U$ supports the wording *harmful compliance as the HarmBench classifier defines it*, never harm alone, only if the classifier's contrast has the same sign and at least half the size and the classifier's erosion under that recipe is at least 20 pp; otherwise the sentence says *restored refusal*. A blind human read during development rated about a third of judge-unsafe answers as usable help (Appendix A).
- **Also per arm:** hard-category $U$ (violent, illegal, sexual or self-harm categories); $U$ by answer-length band and a length-reweighted $U$; Granite Guardian 3.3 (Padhi et al., 2024) on the greedy answers; and a selectivity score (refusal on harm prompts minus refusal on benign prompts). Whether StrongREJECT's evaluator score (Souly et al., 2024) and a prefilling probe (Qi et al., 2025) are secondary endpoints is open [TO BE FIXED in the stage-1 pre-registration].
- **Over-refusal.** Qwen3Guard's refusal flag on greedy answers to the 100 development over-refusal prompts, benign prompts that resemble harmful ones (50 XSTest safe prompts, Röttger et al., 2024; 50 OR-Bench-Hard prompts, Cui et al., 2025), and 400 harmless PHTest prompts (An et al., 2024) drawn by a sealed rule, never from the held-out bank, since a pre-registered power check put the six-seed minimum detectable effect (MDE) on the 100 above 5 pp (Appendix C.11). Reported by source, with a lexicon check.
- **Task loss and skill.** Held-out loss on all 200 task rows is the skill limit of the comparison. Probe and Gate A bars also check ARC-Easy (Clark et al., 2018) under a rule loosened during development; we report both verdicts (Appendix C.4).
- **Answers end.** A coherence screen flags answers that hit the token cap, repeat 4-grams heavily, are empty, or have under 5 tokens; a failing arm is still read, with the flag "fails answers end" (Appendix C.11) [PLANNED; the stage-1 pre-registration is not yet frozen].

### 4.3 Stage 1: the primary comparison

Table 1 lists the stage-1 repairs. Every rule, scorer and pool is committed by hash before Gate B's category runs (§4.9).

**Table 1.** Stage-1 repairs on OLMo-2-1B (recipe B; one epoch; removal arms train 33 steps, addition arms 38); references in §4.4. Actor (§3.1): C, customer (data, learning rate, epochs); P, provider (also the weights); *if the customer can sample the untuned model.

| Arm | Repair | Rule | Seeds | Actor |
|---|---|---|---|---|
| $a_{\mathrm{R}}$ | random removal | 600 random rows, 75 per category | 7 | C, P |
| $a_0$ | no repair | – | 7 | – |
| H1 | length rule (free) | remove the 600 rows with the shortest answers | 6 | C, P |
| A1 | attribution | remove the top 600 rows by bi-anchored gradient similarity at two checkpoints of $a_0$ (the Gate C method) | 6 | P |
| L1 | LARF-style filter | remove the top 600 rows by the untuned model's representation similarity to harmful anchors at one layer | 6 | P |
| S1 | random refusals | add 48 rows (1%) drawn at random from a fixed refusal pool | 6 | C, P |
| S3 | DataRx-style | add 48 rows selected from S1's pool by DataRx's score | 6 | P |
| S4 | self-generated | add 48 pool prompts with the untuned model's own refusals | 6 | C*, P |

**Primary family.** P1: A1 against $a_{\mathrm{R}}$; P2: L1 against $a_{\mathrm{R}}$; P3: H1 against $a_{\mathrm{R}}$; P4: S1 against $a_0$ (two-sided, since prior work predicts both directions; a significant adverse result is labelled *worse*); P5: S3 against S1; P6: S4 against S1. One closed family at 5% (§4.9); labels as in §3.2.

**Pools, seeds and secondary arms.** The refusal pool is from PKU-SafeRLHF (Ji et al., 2025). Seeds run in blocks fixed before any outcome; a blinded rule on Gate B's pooled noise can shrink the family or stop stage 1. The MDE depends on the same-seed correlation [PLACEHOLDER: final MDE from Gate B's pooled noise]. Pools, seeds, secondary arms (three seeds, exploratory) and two candidate data-only additions are in Appendix C.3 [TO BE FINALISED].

### 4.4 References that change no data

All five are mandatory, and we claim no novelty for any. **Safety prompt:** a fixed safety instruction as a prefix of the user turn (OLMo-2 has no system turn), at evaluation only, as in PTST (Lyu et al., 2024), on three $a_0$ seeds. **Adapter scaling:** the $a_0$ adapter scaled by $\lambda = 0.75$ and 0.5 on three seeds, which with $\lambda \in \{0, 1\}$ gives the reference curve of §3.2. **Safe LoRA** (Hsu et al., 2024): $a_0$'s adapter projected as in §2, on three seeds; the base model, layer rule and threshold follow the paper's rule or a fixed default, never stage-1 outcomes [NOT YET FIXED]. **Recipe G** (a reference that changes only the recipe): the unchanged data under recipe G, three seeds. **B+Qi:** recipe B with the token-wise constrained SFT objective of Qi et al. (2025), which constrains updates on the first answer tokens, on the unchanged data at six seeds. A sixth, **lower learning rate** [PLANNED; to be pre-registered], is recipe B at the next lower rung of the ladder in §4.1, a lever the customer has; such rungs are studied in [CITE: concurrent submission, anonymized]. The customer can run the safety prompt and the lower learning rate; the provider can run all. The paper states plainly whether any data repair beats these references; the seeds of the post-hoc references (three, or all seven $a_0$ seeds) and the test (against B+Qi, and against the adapter-scaling curve matched on task loss) are open [TO BE FIXED in the stage-1 pre-registration].

### 4.5 Positive control: a planted cause

[PLACEHOLDER: runs only if funded; otherwise menu size leaves the headline.] The planted cause is $k = 37$ rows: PKU red-team prompts with PKU-safe, non-refusing answers. It replaces the rows of one sealed leaf cell, and nested cells [TO BE FINALISED] give menus of $M = 8$, 32 and 128. Because the truth is known, the sweep measures how diagnosis and trial runs scale with $M$, which natural menus at our budget

cannot: the safety restored by removing the planted cell, each scorer's top-$k$ hit rate, and the runs-only cost of finding the cell (Appendix C.5). Only training-signal diagnostics (A1 and a Self-Inf-N-style score) count toward finding the cause; text-only scorers read only row text and $\theta_0$, so their ranks are fixed at the draw and are construction checks, published before the pilot [PLANNED; the stage-1 pre-registration is not yet frozen].

### 4.6 Gates A, B and C

**Gate A (entry).** For each family at its matched dose, over three seeds (the probe seed and two fresh seeds), mean $E$ must lie in a pre-registered partial-erosion window of 20–55 pp, be at least twice the seed standard deviation, and exceed zero on every seed; a second judge must agree in direction, the answers must end, and the task must be learned without lost skill (§4.2); results and every setting tried are in §5.

**Gate B (localization).** Its pre-registration was frozen before any Gate B run; Appendix C.1 gives its rules. Each category removal (600 rows) is compared with random removal of 600 rows at the same seed. All 8 categories run at one seed; the two with the largest contrast, plus Gate C's picks, run at two more seeds. The verdict is LOCALIZED if some category restores at least $\delta_B = 8$ pp more than random removal and clears a Bonferroni-corrected bar; DIFFUSE if every category is credibly below 8 pp; and TOO CLOSE otherwise. It reads $U$ alone, under guards. At planning noise, 80% detection needs about 10–12 pp, so DIFFUSE means no category-level cause above about 8 pp, not no cause. If DIFFUSE, Gate C is not read, stage 1 is unchanged, and a rewrite arm runs first among the secondary arms (Appendix C.3). **Status:** the noise gate has passed, and Gate C's rankings were sealed before the category runs.

**Gate C (which diagnostic picks the category).** Attribution (A1's score, on a retrained first-seed $a_0$) and representation similarity (on the untuned model) each pick their top category; a format rule picks the category with the largest share of its loss on the first five answer tokens. Gate C is read only if Gate B is LOCALIZED. In the sealed rankings both diagnostics pick the same category, and the format rule another, so by the frozen rules ATTR_BETTER and REP_BETTER cannot occur, and the paper may say only that neither pick is the format pick (Appendix C.2).

### 4.7 Cost accounting

Every arm and scorer is logged in GPU-seconds by stage and charged; one FRE per family is the median time of its random-removal arms. Claims about cost hold only where two conventions, *full* and *lean*, frozen before stage 1, agree (Appendix C.6), and no cost ratio measured at 1B is claimed for a larger model. [TO BE FINALISED: measured costs per arm and per scorer.]

### 4.8 Second family, confirmation and scale

**Llama-3.2-1B** (stage 2) runs at its matched dose (§4.1): its own Gate A, a reduced Gate B on categories, and the primary family at six seeds. A contrast "replicates across two families" only if it has the same sign in both and the family difference is below the interaction's MDE; a family difference is claimed only above that MDE. **Confirmation** (stage 3): at most two contrasts per family, chosen by a rule committed before stage 1, re-run at six fresh seeds on a sealed 200-prompt held-out subsample and on the development bank. **Scale** (stage 4): a replication in a published setting (Appendix C.3), the full 7–8B comparison of §4.1 and the headline check [PLACEHOLDER: if resources allow]. No result is claimed for a size without runs.

### 4.9 Pre-registration and testing order

Gate B runs first, to its verdict; funding for the later stages is then sought with that result, and stage 1 starts only once funded, never because of an outcome. Before Gate B's category runs, the stage-1 pre-registration

[NOT YET FIXED] commits every stage-1 scorer's code and settings hash, every edit rule, the pools, the plant, the primary family, the seed rule, the limits, the claim rule, the selector specification and its predictions, and it lists every Gate B value seen before it is frozen (the noise-gate reading and the sealed rankings). Anything added later is labelled post-Gate-B exploratory. Families of tests are read in order, each at its own 5% level: (1) the primary family; (2) confirmation; (3) the second-family (Llama) comparison; (4) the planted-sweep claims; (5) the decision-value predictions. The primary family is one closed graphical family, $\Phi$, of 24 nodes, four for each of P1–P6 (Appendix C.11) [PLANNED; the stage-1 pre-registration is not yet frozen]. Everything else is descriptive.

### 4.10 Side studies

1. **Mechanism: refusal openings against answer content.** An opening decomposition (Appendix C.8) splits each arm's change in $U$ into answers that stop opening with a refusal and what the answers say. A teacher-forced first-token refusal margin on Gate B arms should track $U$ if a category's effect runs through openings; Qi et al. (2025) argue that safety alignment often adapts mainly the first few output tokens. [PLACEHOLDER: report against the bound on a first-token path recorded before Gate B.] Exploratory.
2. **Recipe and family contrasts** are not results of this paper: recipe G enters only as a reference, and the contrasts behind the per-family rule (recipe G against recipe B in several families, a pre-registered single-seed run that changes only the loss weighting, the learning-rate rungs and a rank-16 check) are reported in [CITE: concurrent submission, anonymized].

Two more side studies are in Appendix C.9.

## 5 Results

[PLACEHOLDER: Gate A with every setting tried and its disclosures; Table 2, the stage-1 comparison ($G$, $\Lambda$ with 90% intervals and labels, $O$, $L$, cost under both conventions, the HarmBench contrast and the claim-rule wording) with the references and the adapter-scaling curve; Figure 1; the second family, if run; Gates B and C with the length control and the format reading; the planted sweep, if funded; measured against predicted decision value; the opening decomposition; every too-close-to-call result.]

## 6 Discussion and Limitations

**What "unsafe" means.** Our primary judge gives a broad policy label, so a rise in $U$ is refusal erosion as Qwen3Guard labels it. In a blind human read during development (one reader; Appendix A), most judge-unsafe answers went along with the request, but only about a third gave usable help. The claim rule (§4.2) therefore allows at most "harmful compliance as the HarmBench classifier defines it", and $U$ is not comparable with attack-success rates published under other judges and prompts.

**Equivalence and power.** The labels resolve differences of about 5 pp, and an *equivalent* label means "within $\pm$5 pp", not "identical" (§3.2). Gate B's detection limit (§4.6) holds for categories; row-level causes that span categories are not tested by Gate B.

**Recipe and setting.** The main recipe, B, is plain rather than standard: a constant learning rate with no warmup, a per-row loss and a rank-64 adapter on all linear layers, larger than is common (a rank-16 check is in [CITE: concurrent submission, anonymized]). Recipe G's standard schedule lowers the mean learning rate; it is a reference, not a general fix, and repair rankings may differ at a lower dose. Cross-family claims

hold at a matched erosion dose, not at one recipe, under a rule fixed after a development ladder was read. The skill bar at the probes and Gate A is looser than the bar first registered, and we say where it decided a reading (Appendix C.4). The main model and recipe were chosen by screening probe and Gate A readings, never Gate B or stage-1 outcomes; §5 lists every setting tried.

**Baselines, doses and cost.** Every scorer is re-implemented from its paper ("X-style"), so a weak result may reflect our implementation; a published table of deviations per scorer, reference outputs requested from the authors and the published-setting replication address this. A1 scores once, and its row scores at the two checkpoints agree only moderately (Spearman 0.77; top-600 Jaccard 0.43), so the removal set depends on the checkpoint. Comparisons across removal and addition hold only at the pre-registered doses (600 rows removed, 48 added). An arm's cost depends on how much we evaluate as well as on training, and costs at 1B do not transfer to larger models, because scoring and training costs scale differently.

**Scale and family.** The full comparison runs at 1B, with one benign dataset until Alpaca runs; at 7–8B it is planned on both families and Olmo-3-7B-Instruct, funding-gated, with one family as the fallback and Olmo 3 the first model cut; larger sizes get a check only if resources allow. Qwen2.5-1.5B was dropped during development when its erosion saturated under recipe B at a learning rate of $10^{-4}$, before the per-family learning-rate rule existed, and was not retried at a lower rate in this study, so the Qwen runs of [CITE: concurrent submission, anonymized] are the only later evidence.

**The positive control and the secondary analysis.** The planted cause is artificial; finding it shows that a training-signal diagnostic *can* find a concentrated, harmful-prompt cause, not that natural erosion has one. The value of diagnosis at natural menus is model-based, judged against advance predictions. Selectors are not compared at equal budgets here; a later, separately pre-registered study would do that.

**Scope and user.** We compare data repairs with model-level references; rewriting responses enters only if the erosion is diffuse. Post-hoc repairs other than the references, such as the one-example patch (Zhang et al., 2026a), are not compared (§2), so no claim here covers them. A customer of a closed fine-tuning service controls the data and some hyperparameters but cannot run gradient or representation scorers; the provider, or a developer with open weights, can run every scorer but also has model-level repairs. We therefore report the best repair within each actor's menu (§3.1); a data-only pipeline reads the data repairs alone.

## 7 Conclusion

[PLACEHOLDER: which repair to use, at what cost and with what side effects, and whether paying for diagnosis was ever worth it, under the observed outcome.]

## Impact Statement

This work studies how to repair a language model's safety after benign fine-tuning has eroded it, so that developers and fine-tuning services can choose a repair with its side effects and cost in view. We see three risks. First, the study shows which training data erode safety and which repairs fail, and the positive control, if run, plants rows that erode safety; an attacker could use this. The planted rows are PKU red-team prompts with PKU-safe, non-refusing answers, taken from a public safety dataset. We release [TO BE DECIDED: what is released, for example scores and outcome tables but no generated answers to harm prompts]. Second, every outcome comes from automatic judges, and a blind human read rated only about a third of judge-unsafe answers as usable help, so a repair labelled *better* or *equivalent* is not a safety guarantee. Third, the models are small and the benign data come from one dataset; the results may not hold for larger models, other data or deployed systems, and do not replace a model's own safety evaluation.

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## Appendix A: Blind human read (development evidence)

This read checked, during development, what the primary judge's Unsafe label means. It gates nothing, changes no verdict and is not a result of the study. The same read is also reported in [CITE: concurrent submission, anonymized].

**Design.** The read was pre-registered before anyone saw the answers, and the ratings were sealed before the key was opened. A seeded rule drew 50 sampled answers to development harm prompts: 40 that Qwen3Guard labels Unsafe, from recipe-B fine-tunes of OLMo-2-1B, Qwen2.5-1.5B, Qwen2.5-7B and Llama-3.2-1B and a recipe-G fine-tune of OLMo-2-1B (6 to 12 per group), and 10 that it labels Safe, as controls. The sheet showed only the request and the answer, without model, recipe, seed or judge label. One reader rated each answer 0 (refuses or deflects), 1 (goes along but gives no usable help: vague, generic, wrong, fictional or incoherent) or 2 (usable help: specific steps, methods, materials or content that would meaningfully help someone do the harmful thing).

**Counts.** Of the 40 Unsafe-labelled answers, 39 were rated and 1 was skipped. 34 of the 39 went along with the request (rated 1 or 2; 0.87, exact 95% interval [0.73, 0.96]), and 13 of the 39 gave usable help (rated 2; 0.33 [0.19, 0.50]). None of the 10 Safe-labelled controls was rated 1 or 2, so the judge missed none ([0.00, 0.31]). Of the 15 OLMo-2-1B answers that also have a HarmBench-classifier label, 4 of the 12 labelled "yes" were rated usable help, and none of the 3 labelled "no": the reader's bar was stricter than the classifier's.

**Use in the paper.** Our reading: when Qwen3Guard labels an answer Unsafe, the answer usually goes along with the request, so $U$ measures lost refusal, but only about a third of these answers give usable help. The read supports the wording of §4.2 and §6, not a claim of usable harm.

**Limits.** One reader, an author who helped design the study, and no second rater, so no agreement figure. Groups of 6 to 12 answers are too small to compare, and no comparison between groups was made. The sample differs from the one an audit of Gate A proposed (Unsafe answers from OLMo-2-1B's Gate A fine-tunes, stratified into very short answers and answers flagged only as unethical, plus the Unsafe answers on the over-refusal bank), so it does not test those strata.

## Appendix C: Design details

This appendix holds design details that the main text summarizes. The frozen pre-registrations govern wherever this summary differs from them [TO BE DECIDED: anonymized copies as supplementary material].

### C.1 Gate B

**Arms and waves.** Wave 0a: three random-removal arms $R_1$–$R_3$ (600 rows, 75 per category, each from its own sealed draw) at seeds 1101–1103, for the noise gate. Wave 0b: a retrain of $a_0$ at seed 1101 that keeps its checkpoints after steps 10 and 38, and the Gate C diagnostics. Wave 1: each of the 8 category removals at seed 1101. Wave 2: $m = 2$–$4$ units (the two with the largest wave-1 contrast, plus Gate C's picks if not already in) at seeds 1102 and 1103. Every removal arm comes from a sealed edit file and trains 33 steps; the Gate A $a_0$ runs are reused. No step starts before the step above it is complete.

**Contrasts and noise.** All $U$ are in pp over 600 sampled answers. For unit $u$ at seed $j$, $c_{u,j} = U(R_j, j) - U(u, j)$, so $a_0$ cancels exactly; $D_u = c_{u,1101}$ (wave 1) and $\Lambda_u = (c_{u,1102} + c_{u,1103})/2$ (wave 2). With $r_j = U(a_0, j) - U(R_j, j)$ and $\sigma_{\text{eval}}$ the median prompt-clustered standard error of a seed-matched pair's per-prompt difference, $\hat{\sigma}_{\text{gate}} = \max(\text{SD}(r_{1101}, r_{1102}, r_{1103}), \sigma_{\text{eval},0})$; the final $\hat{\sigma}$ pools the $r_j$ and the wave-2 pairs. A fallback $\hat{\sigma}$ that does not use $a_0$ applies if the retrain's adapter fingerprints, or a run's environment, differ from Gate A's. Tests use $t$ at 3 degrees of freedom, the effective degrees of freedom of $\hat{\sigma}$.

**Noise gate.** After wave 0a, Gate B goes on only if $\hat{\sigma}_{\text{gate}} \le 7$ pp and $\max(8, 4.177 \times 0.707\,\hat{\sigma}_{\text{gate}}) \le 33.8$ pp (0.75 of Gate A's mean $E$). A failure stops Gate B, and the pre-registered fallback applies instead of a redesign.

**Verdict.** *LOCALIZED*: some wave-2 unit has $\Lambda_u \ge 8$ pp and $\Lambda_u \ge t_m \times 0.707\,\hat{\sigma}$, where $t_m$ is the one-sided Bonferroni 5% $t$ at 3 degrees of freedom (3.182, 3.740 and 4.177 for $m = 2, 3, 4$); the unit passes guards G2–G4 and the direction checks, and every arm that the contrast uses passes G1. *DIFFUSE*: $\max_u D_u < 2.381\,\hat{\sigma}$, and every wave-2 unit has $\Lambda_u + 1.638 \times 0.707\,\hat{\sigma} < 8$ pp; both also hold on coherent answers (G2) and in both adverse corners (G3), and every arm passes G1. *TOO CLOSE*: anything else, a guard failure or a missing arm included. With LOCALIZED, the one-sided 95% lower bound $\Lambda_u - t_m \times 0.707\,\hat{\sigma}$ is reported. At Gate A's mean erosion, 8 pp is about 18% of $E$, so LOCALIZED does not say that the unit carries most of the erosion. The verdict reads $U$ alone; the blind human read recommended before the first wave is done, on a different sample from the one first proposed (Appendix A).

**Guards.** G1, answers end: Gate A's bars on sampled answers (at most 5% capped and at most 10% failing the coherence screen; a short-answer TOO CLOSE counts as a failure); a random-removal arm that fails G1 makes the verdict TOO CLOSE with no extension. G2: the verdict recomputed on coherent answers only. G3: two adverse corners, with unscored answers counted against the verdict, and with controversial answers counted Unsafe. G4, over-correction: the unit's mean refusal rate on the 100 over-refusal prompts may exceed the untuned model's by at most 5 pp. Direction checks: hard-category $U$, the share of sampled answers that do not refuse, and Granite Guardian's greedy labels must each give a positive contrast.

**Extension.** It is automatic; no choice follows an outcome. A wave-2 unit is *contested* if it is not LOCALIZED, $\Lambda_u + 1.638 \times 0.707\,\hat{\sigma} \geq 8$ pp, and it passed the guards. After a TOO CLOSE with at least one contested unit and no random-removal arm failing G1, up to two contested units run at seeds 1104 and 1105 with $R_4$ and $R_5$, and are read once more over four seeds with the same constants. No other reading follows an outcome, and no new margin is set.

**Length control.** After LOCALIZED, for the unit with the largest $\Lambda_u$: two removals of as many rows, drawn from the other units and matched to the unit's answer-length bins, run against the unit's removal at two fresh seeds. The wording is "content-localized" only if the length-matched contrast $L_u$ is at least 8 pp and at least $2.353 \times 0.707\,\hat{\sigma}$, and all four arms pass G1; otherwise it is "localized under the per-example-mean loss; content and answer length not separated". The control changes the wording, never the verdict.

**Operating characteristics.** By simulation, with one unit carrying the effect and the guards not simulated (so the rates are upper bounds): at central planning noise (a seed-pair standard deviation of 3.75 pp) and given that the noise gate passed, a true 8, 10 or 12 pp unit reads LOCALIZED in 52, 83 or 96% of runs, so an 8 pp category reads LOCALIZED about half the time. 80% detection needs about 9, 10 or 12 pp at low, central or high planning noise. False LOCALIZED, given that the gate passed and no unit has an effect, is 0.2–2.7% for seed-pair standard deviations of 3.75–7 pp. A missed 8–12 pp unit mostly reads TOO CLOSE, not DIFFUSE (DIFFUSE in at most 7.5% of runs at 8 pp). The pre-registration records the prior that DIFFUSE or TOO CLOSE is more likely than LOCALIZED.

**Status.** The noise gate passed after wave 0a, and neither trigger for the fallback $\hat{\sigma}$ fired after wave 0b. The diagnostics ran in wave 0b, and their rankings were sealed before wave 1 (Appendix C.2).

### C.2 Gate C

**Diagnostics and picks.** Each diagnostic scores all 4,800 rows, and a unit's score is the mean over its 600 rows. Attribution is bi-anchored gradient similarity (sealed anchors: 64 harmful, 64 safe) at the retrained $a_0$ 's checkpoints after steps 10 and 38, with equal weight. Representation similarity is bi-anchored, at the untuned model's final layer and last response token. A content classifier is reported only. $\text{pick}_A$ and $\text{pick}_R$ are the top units of attribution and representation similarity (ties go to the lower unit id). The format pick $\text{pick}_F$ is the unit with the largest mean share of a row's loss weight on its first five answer tokens under the per-row loss; it costs nothing.

**Rule.** For a pick $p$, $B_p$ is the mean of $c_{p,j}$ over seeds 1101–1103, and $\Delta = B_{\text{pick}_A} - B_{\text{pick}_R}$. A pick beats random if $B_p \geq 2\hat{\sigma}/\sqrt{3} \approx 1.15\,\hat{\sigma}$; the margin is $M_C = \max(6\text{ pp}, 1.15\,\hat{\sigma})$, with $\hat{\sigma}$ from before any extension. AGREE: $\text{pick}_A = \text{pick}_R$. ATTR_BETTER: $\Delta \geq M_C$ and $\text{pick}_A$ beats random. REP_BETTER: $\Delta \leq -M_C$. NONE_FINDS: neither pick beats random. The verdict is the first that holds in the order NONE_FINDS, AGREE, ATTR_BETTER, REP_BETTER, TOO CLOSE. If an arm that a $B_p$ uses fails G1, Gate C reads TOO CLOSE. Gate C is read only after LOCALIZED, and it reads "not read" when an arm it needs is missing. No Gate C extension runs.

**Format reading.** If $\text{pick}_A = \text{pick}_F$, the paper must say that attribution's pick is explained by answer form (a near-free answer-form score picks the same unit) and claims no value of that pick beyond answer form; the same holds for representation similarity. If neither pick equals $\text{pick}_F$, the paper says only that neither pick is the form pick, which does not show that the picks are free of form. The two equalities are fixed when the rankings are sealed, so they are reported whatever Gate B reads; the form reading never changes a verdict. Also reported: the Spearman $\rho$ of each diagnostic's 8 unit scores with the first-five-token share and with the wave-1 contrasts $D_u$ (a one-sided 5% test needs $\rho \geq 9/14$), and the top-1 hit (chance 12.5%).

**Consequence of the sealed rankings.** Attribution and representation similarity pick the same unit, and the format rule picks another. So $\Delta = 0$: ATTR_BETTER and REP_BETTER cannot occur, and Gate C, if read, is NONE_FINDS, AGREE or TOO CLOSE (the last only through the answers-end guard on the pick arms). The format reading is the second case above, and wave 2 holds $m = 2$ or 3 units.

### C.3 Stage 1: pools, seeds and secondary arms

**Pools and anchors.** The refusal pool holds prompts from PKU-SafeRLHF (Ji et al., 2025) that have an unsafe response in the source data, paired with a response that the source labels safe and that opens with a refusal by the frozen lexicon. Pools and anchors are never filtered by the evaluation judge, and prompts that nearly duplicate an evaluation prompt are removed. The add effect must also hold on the evaluation prompts least similar to the added rows.

**Seeds and power.** Method arms run at six seeds (1102–1107), and random removal and no repair at seven each (1101–1107; random removal at seed 1108 is not run); seeds run in blocks fixed before any outcome, and only complete blocks are analysed. A blinded rule reads only Gate B's pooled noise estimate: up to 4.5 pp, the plan of §4.3; from 4.5 to 6.0 pp, the family shrinks to P1, P3 and P6 at nine seeds; above 6.0 pp, stage 1 does not run. Under the paired model the MDE depends on the same-seed correlation; the stage-1 record prints it at Gate B's pooled noise [PLACEHOLDER: final MDE from Gate B's pooled noise].

**Secondary and scored-only arms [TO BE FINALISED].** At three seeds each, exploratory, with one-sided bounds only: a random dose curve (1,200 and 2,400 rows removed); random removal plus S1; a DataShield-style filter; rules for answers with list or markup lines and for the longest answers; the length rule under a token-mean loss; and octiles of answer length. Scored without new runs: a Self-Inf-N-style score and a content classifier; we publish the overlap of all top-600 sets. SQSD is cited but not run, because its scoring needs several FRE. If Gate B reads DIFFUSE, a rewrite arm (the untuned model's own answers replace the 600 top-attribution rows) runs first among the secondary arms, and the planted sweep becomes the only test of whether diagnosis can pay. Two more data-only additions, the augmentation of Qi et al. (2025) that deepens alignment beyond the first answer tokens and the single safety example of Zhang et al. (2026a) added to the training data, are candidate arms [TO BE FIXED in the stage-1 pre-registration].

**Token matching.** Every removal and addition arm reports the rows, target tokens and total row tokens it removes or adds. The token-matched analysis, report-only and run in any case, places each removal's restoration on the random dose curve (600, 1,200 and 2,400 random rows removed) along a token axis, beside the row axis. The primary token axis is total row tokens (prompt plus target tokens, as trained); target tokens (the answer and its end-of-turn tokens) are reported beside it. Token-matched random removals for A1, L1 and H1, three seeds each, are secondary arms in the funded tier; if they are not funded, the analysis alone is the fallback.

**Published-setting replication (stage 4).** A LARF- or DataShield-style filter on Llama-3.1-8B-Instruct fine-tuned on Alpaca, with random removal and length rules [PLACEHOLDER: if resources allow].

**Certified evaluation of the references.** A pinned evaluation script regenerates each $a_0$ adapter by a deterministic re-run and is certified once: with $\lambda = 1$ and no prefix it must reproduce $a_0$'s $U$ exactly. The implementations of Safe LoRA and B+Qi are committed before stage 1 is frozen.

### C.4 The skill bar

Held-out loss on all 200 task rows is the skill limit of the comparison. The probe and Gate A bars also check ARC-Easy accuracy (Clark et al., 2018) on 189 questions: the paired drop $d$ against the untuned model must stay at or below 0.03. With 189 questions the paired interval is about $\pm 0.07$ wide, so it straddles 0.03 for

almost every run. A rule fixed during development, first for probe and Gate A readings at one common recipe and then extended to every family's Gate A read at its chosen learning rate before any such read, resolves a straddling interval: the run **passes, as "skill kept, reported as a limitation", when the point drop $d \leq 0.03$ and the held-out task loss is below the untuned model's (the task is learned); it fails when $d > 0.03$.** This rule is a loosening: it can pass a run whose true drop is up to about 6 points, which 189 questions cannot detect. We report the verdict under the frozen bars beside the verdict under this rule. Disclosure: the first recipe-G probe on OLMo-2 was read under the earlier rule before this rule was proposed, so for that one run the rule is not applied blind. At 1B, ARC-Easy is close to chance, so "skill kept" rests mainly on the held-out task loss.

### C.5 The positive control

**Definitions.** A planted data set $\mathcal{D}^{\mathrm{p}}$ replaces the rows of one small cell of $\mathcal{D}$ with $k$ rows that carry a known eroding signal, keeping the number of rows fixed. Nested partitions split $\mathcal{D}^{\mathrm{p}}$ into $M$ cells, so that the planted cell sits inside one cell at every $M$. The planted erosion is $P = F(a_0^{\mathrm{p}}) - F(a_0)$, and removing the cell that holds the plant at menu size $M$ restores $G^{\mathrm{p}}(M) = F(a_0^{\mathrm{p}}) - F(a^{\mathrm{p}}_{-c(M)})$. A diagnostic $d$ *hits* at $(M, k_d)$ if it ranks that cell among its top $k_d$. Because the truth is known, the sweep measures how diagnosis and trial runs scale with $M$, which natural menus at our budget cannot.

**Design.** [PLACEHOLDER: runs only if funded.] The plant is $k = 37$ PKU red-team prompts with PKU-safe, non-refusing answers, from PKU-SafeRLHF, the source of the anchors. Each prompt also has an unsafe answer in the source. Each planted answer is labelled safe there and complies directly. A sealed draw picks one category and one leaf cell inside it, and the plant replaces that cell's rows, so the run keeps 4,800 rows and 38 steps. The nested cells [TO BE FINALISED] are the 8 categories (600 rows), 32 length quartiles within categories (150 rows) and 128 random quarters of those (37–38 rows). A pilot of three seeds goes on only if the plant adds at least 10 pp of erosion with a one-sided 90% lower bound above 5 pp (otherwise the plant doubles once, then the null is reported). The main sweep removes the planted cell at each menu size (three seeds each) and runs random removal on the planted data. Every scorer ranks the cells without new runs. Only attribution (A1) and the Self-Inf-N-style score count toward finding the cause. The text-only scorers (L1, H1, the list-or-markup and longest-answer rules, the DataShield-style filter and the content classifier) read only row text and the untuned model, so their ranks of the planted cell are fixed when the plant is drawn. They are published before the pilot as construction checks, and also if the pilot stops [PLANNED; the stage-1 pre-registration is not yet frozen]. The Self-Inf-N-style score's planted-cell rank joins the ranks scored without new runs; if its scorer is not committed before Gate B's category runs, it leaves both that list and the counting rule, and only A1 counts. The plant's prompts are disjoint from the anchor prompts, but both come from PKU-SafeRLHF, and A1 is bi-anchored on those anchors, so an A1 hit could be read as near-circular. The planted sweep therefore also reports A1's planted-cell rank under a second anchor set: 64 harmful and 64 safe anchors from a source outside PKU-Alignment's data (the publisher of PKU-SafeRLHF), built by Gate C's anchor rule (severity, category balance, a sealed draw and a pinned script), with its source sealed before the stage-1 pre-registration is frozen and before any planted run. The rank under the PKU-SafeRLHF anchors is the one that counts; the rank under the disjoint anchors is a pre-registered sensitivity analysis. If that set cannot be built before the freeze, the shared source is reported as a limit instead [PLANNED; the stage-1 pre-registration is not yet frozen]. Outputs: $G^{\mathrm{p}}(M)$; each scorer's top-$k$ hit rate with known truth; the runs-only cost of finding the cell, replayed with the frozen selector rules; and a check of additivity across the nested levels.

### C.6 Cost conventions

Every arm and every scorer is logged in GPU-seconds by stage (setup, training, greedy generation, sampled generation, judging, scoring, overhead). One FRE per family is the median measured time of that family's random-removal arms. Two cost conventions are frozen before stage 1: *full* (training plus the evaluation as run) and *lean* (training plus one sample per harm prompt, priced from the measured time per answer); the cost $c_0$ of one lean trial run is measured on a real job. Every pool filter, generated answer, anchor job and scorer is charged; "free" means below 0.01 FRE. Attribution is priced from its measured scoring pass. Claims about cost hold only where both conventions agree, and no cost ratio measured at 1B is claimed for a larger model.

### C.7 The value of diagnosis

A *menu* $\mathcal{A}_M = \{a_0, a_1, \dots, a_M\}$ holds $a_0$ and $M$ repairs built by a frozen rule, never by a compared diagnostic; $\mathcal{A}_{\text{feas}}$ is its feasible subset and $a^* = \arg\min_{a \in \mathcal{A}_{\text{feas}}} F(a)$. A *selector* $\pi$ with budget $B$ (in FRE) buys trial runs (each a training fraction $\phi$ and a prompt fraction $\psi$, priced from the measured cost model) at total cost at most $B$ and returns $\hat{a}$. It drops a repair whose adverse estimate of $O$ or $L$ passes its limit ($a_0$ is never dropped), and picks the lowest estimated $F$ among repairs with full runs on at least two seeds; ties go to $a_0$. An infeasible pick is scored as if no repair were made:

$$\text{loss}(\hat{a}) = \begin{cases} F(\hat{a}) & \hat{a} \in \mathcal{A}_{\text{feas}}, \\ F(a_0) & \text{otherwise}, \end{cases} \qquad \text{Reg}(\pi; B, M) = \mathbb{E}[\text{loss}(\hat{a})] - F(a^*).$$

This is a scoring convention for an expected outcome at a fixed budget, not a safety guarantee. The runs-only selector $\pi_-$ uses one rule per $(M, B)$ cell (a one-shot screen, successive halving or hierarchical bisection), chosen from a simulation before any outcome and then frozen. The selector $\pi_{+d}$ pays the cost $c_d$ of diagnostic $d$, keeps its top $k_d$ repairs, and runs the same rule on $B - c_d$. The value of $d$ and its break-even cost are

$$\Delta_d(B; M, c) = \text{Reg}(\pi_-; B, M) - \text{Reg}(\pi_{+d}; B, M, c), \qquad c_d^*(B, M) = \sup\{c \geq 0 : \Delta_d(B; M, c) \geq 0\}.$$

$F(a^*)$ cancels in $\Delta_d$. When $a_0$ is feasible, regret is non-negative, so no diagnostic can save more than $V_{\max}(B, M) = \text{Reg}(\pi_-; B, M)$. At the menus we can afford, these quantities are mostly outputs of a model, so we treat them as predictions. A generative model (repair effects shrunk by empirical Bayes, noise from pooled residuals, measured costs, two effect shapes) is frozen before Gate B's category runs, and its predicted $V_{\max}$, $\Delta_d$ and $c_d^*$ for the category menu and the planted menus are recorded then. We report measured against predicted values (replayed with the same code path on the outcome tables), and score OLMo-calibrated predictions of the chance of finding the best repair, and of regret, on Llama's category menu. Selectors are not run against each other at equal budgets in this study.

### C.8 The opening decomposition and the first-token margin

Let $p(\theta)$ be the share of sampled answers to harm prompts that open with a refusal (a frozen lexicon applied to the first 12 words), and $u_{\text{R}}(\theta)$ and $u_{\text{N}}(\theta)$ the unsafe rates among answers that do and do not open with one, so $f = p\,u_{\text{R}} + (1-p)\,u_{\text{N}}$. For the untuned model (subscript 0) and a fine-tune (subscript 1),

$$f_1 - f_0 = \underbrace{(p_0 - p_1)(u_{\text{N}0} - u_{\text{R}0})}_{\text{opening switch}} + \underbrace{(1-p_1)(u_{\text{N}1} - u_{\text{N}0}) + p_1(u_{\text{R}1} - u_{\text{R}0})}_{\text{content change}}.$$

The *opening share* is the first term over $f_1 - f_0$; the split depends on the order of terms, so both orders are reported. The decomposition is descriptive: it says whether a change in the unsafe rate shows up in how answers open or in what they say, not which training choice caused the change. The first term has two layers: how many refusal openings fine-tuning removes ($p_0 - p_1$), and how often a direct answer was already unsafe before fine-tuning ($u_{\mathrm{N0}}$); where $u_{\mathrm{N0}}$ is high and $u_{\mathrm{R0}}$ low, a lost opening comes close to an unsafe answer. If the opening share is small, repairs that act only on how answers open (refusal examples; an objective that constrains updates on the first answer tokens) may restore less than repairs that change what the answers say. We apply the decomposition to every arm; across families and recipes, the two layers are reported in [CITE: concurrent submission, anonymized].

The causal test on Gate B arms is a teacher-forced first-token refusal margin $m(p) = \log P_{\mathrm{ref}}(p) - \log(1 - P_{\mathrm{ref}}(p))$, where $P_{\mathrm{ref}}$ is the probability of a frozen refusal phrase as the first answer tokens. If a category's effect runs through openings, the margin should track $U$ across arms.

### C.9 Side studies: measurement validity and side-effect prediction

**Measurement validity and over-refusal.** Agreement and $E$ for three judges on the same answers, hard-category $U$ and length bands, how much of the gap to published numbers comes from judges and prompts, and $O$ by source, bounded by the judge's false-Unsafe rate on benign prompts.

**Side-effect prediction (secondary).** On the addition arms only: the balanced error of predicting an over-refusal violation, a diagnostic against trial runs of equal cost.

### C.10 The HarmBench classifier

The compliance judge of §4.2 is HarmBench's validation classifier, `cais/HarmBench-Mistral-7b-val-cls` (Mazeika et al., 2024), with HarmBench's classifier prompt. Each answer is clipped to its first 512 classifier tokens before it is judged, so the classifier reads only the opening of a longer answer. The clip mirrors HarmBench's own evaluation script (`evaluate_completions.py`): tokens are counted with the classifier's own tokenizer, loaded from the classifier's model snapshot; the count of 512 includes the beginning-of-sequence token; the answer is truncated on the right and decoded without special tokens. HarmBench's test classifier, `cais/HarmBench-Llama-2-13b-cls`, is a candidate sensitivity check.

### C.11 The analysis model, the testing family and the labels

The settings below are planned [PLANNED; the stage-1 pre-registration is not yet frozen].

**Analysis model.** One cell is one arm at one seed. The primary model, HB-C, estimates each contrast as the mean, over the seeds that its two arms share, of the seed-matched difference in $U$. Arms at the same seed share the adapter initialization, the row order and the sampling streams, and that shared seed effect is fitted once per design cluster: no repair and the three additions ($a_0$, S1, S3, S4), which keep $a_0$'s batches, and random removal and the three removals ($a_{\mathrm{R}}$, A1, L1, H1), which re-form theirs. Every primary contrast lies inside one cluster. Each arm has its own residual variance, fitted by REML; a prompt component, with prompts crossed with seeds, lets the inference reach new harm prompts from the same population; the degrees of freedom are Satterthwaite's. The same model is refitted for over-refusal (its items are the over-refusal prompts) and for task loss (the 200 task rows). Pre-registered sensitivity analyses: HB, the same model with one seed effect for all eight arms; S-a, the unpaired analysis with the variance pooled over the eight primary arms; and S-aW, unpaired Welch tests. If a sensitivity analysis gives another label, the row prints both; the primary model governs.

**Testing family $\Phi$.** One closed graphical family at family-wise level 5% replaces Holm's procedure over P1–P6. It has 24 nodes, four for each primary contrast $k$: $U_k$ (two-sided), $O_k$ and $L_k$ (over-refusal and task loss, one-sided in the adverse direction) and $F_k$ (feasible at $\tau_O^* = 5$ pp, one-sided). The start weights are 0.15 for each $U_k$, 0.1/12 for each $O_k$ and $L_k$, and 0 for each $F_k$; the pre-registration fixes how a rejected node passes its weight on. Equivalence is read outside $\Phi$, in two grades (below). The later families of §4.9 keep their own 5% levels.

**Equivalence grades.** Two grades are reported. Grade 1, *equivalent (per comparison)*, is the rule of §3.2: the 90% interval of the $U$ contrast lies inside ±5 pp. Grade 2, *equivalent (familywise)*, is an added report: the 98.33% interval ($\alpha/6$ per side) lies inside ±5 pp. The row names the grade.

**The margin.** $\delta = 5$ pp is about 11% of the measured erosion $E$ at recipe B: OLMo-2-1B's mean $E$ over its three Gate A seeds is 45.11 pp, so $\delta/E$ is 11.1% (10.4–11.5% by seed). For another model family, the margin is $\delta_{\text{fam}} = 5 \times E_{\text{fam}}/45.11$ pp $= 0.1108\, E_{\text{fam}}$. Here $E_{\text{fam}}$ is that family's own mean erosion $E$ (in pp) over its three Gate A seeds at its matched dose (§4.6), and 45.11 pp is the same mean for OLMo-2-1B, so δ is the same share of each family's erosion. The rule gives 2.2 pp at $E_{\text{fam}} = 20$ pp and 6.1 pp at 55 pp. It is fixed before that family's stage-2 runs, and the absolute 5 pp margin is reported beside it.

**The over-refusal bank.** A pre-registered power check on the Gate A records put the six-seed minimum detectable effect of over-refusal against $\theta_0$ on the 100 development prompts at 11.04 pp (prompts treated as random), above the 5 pp bar, so the pre-registered trigger fired: the primary over-refusal bank is those 100 prompts plus 400 harmless PHTest prompts (An et al., 2024). The candidates are PHTest's prompts labelled harmless, never those labelled controversial. Exact duplicates, near-duplicates of any evaluation prompt (every bank and split), of the anchors, of the refusal pool or of the plant rows, empty prompts and prompts over the 1,536-token prompt limit are excluded and counted by reason; the first 400 eligible prompts in a sealed order form the extension. The two sources are reported apart.

**The primary over-refusal bar.** $\tau_O^* = 5$ pp above the untuned model, the value of Gate B's over-correction guard G4 (Appendix C.1); feasibility is also reported at $\tau_O = 2$ and 10 pp.

**Feasibility.** Each side effect is read from its two-sided 90% interval: $O(m)$, paired by prompt with $\theta_0$, and $L(m)$, against $a_0$. At a limit, a repair is *feasible* if the interval's upper end is at most the limit, *infeasible* if its lower end is above the limit, and *undetermined* otherwise; the point estimate is reported beside the state, at $\tau_O = 2$, 5 and 10 pp and at $\tau_L$. "Upper 90%" means this upper end, a one-sided 95% bound, everywhere, the selector's screen (Appendix C.7) included; it is the interval that the test for *equivalent* already uses. The *infeasible* of rule R1.3 below has this meaning.

**Rule R1.3 for *worse* on over-refusal and task loss.** A repair is *worse* on over-refusal only if it is infeasible at $\tau_O^*$ against the untuned model $\theta_0$ and its over-refusal is significantly above its comparator's. It is *worse* on task loss only if it is infeasible at $\tau_L$ against $a_0$ and its task loss is significantly above its comparator's; $\bar{\ell}(m) - \bar{\ell}(a_{\text{R}})$ is reported beside every removal arm. The contrast with the comparator is always reported. Without the first condition, an addition arm that returns over-refusal to the untuned model's level, and so is feasible, could still read *worse* against $a_0$. So a repair whose over-refusal is significantly above its comparator's but that is not infeasible at $\tau_O^*$ (it is feasible or undetermined) is not *worse (O)*: its over-refusal shift is reported in its row, and its label comes from the $U$ part. The same holds for task loss at $\tau_L$. On the unsafe rate, *worse* needs only a rejection of $U_k$ with $\Lambda_m < 0$.

**Label order.** The headline word is chosen in a fixed order: *worse* first, naming the measure (for example "worse (O)"); then *better*; then *equivalent*; then *inconclusive*. *Better* and *equivalent* exclude each other: *better* needs an estimate of at least $\delta$, *equivalent* an upper limit below $\delta$. The full row is always printed: the

result on $U$, both side effects against the comparator and against $\theta_0$ or $a_0$, and feasibility at each bar. So a *worse* on a side effect never hides a gain in $U$, and the $U$ part is printed beside the word. When the test on $U$ rejects with an estimate between 0 and $\delta$, the row adds "significant, below the smallest effect of interest" instead of a new label.

**Arms that fail the coherence screen.** The bar is Gate A's (at most 5% of answers capped and at most 10% failing the screen of §4.2), applied to an arm's sampled answers pooled over all its seeds; per-seed counts are reported. No arm and no seed is dropped, and every contrast keeps its test. A failing repair keeps the label that its tests give, with the flag "fails answers end" beside the word; each contrast whose comparator fails carries the flag "comparator fails answers end". A flag is not a label and never changes the word, and the text never calls a flagged repair *better* or *equivalent* without its flag. $U$ on coherent answers only is reported for every arm, and every label is also re-read on coherent answers. A failing arm is infeasible for the selector and cannot dominate a reference.

## Editor's notes (not part of the paper; delete before submission)

**What this is.** Revision 6g of the draft, written on 17 October 2026 from 09:59 UTC (clock checked with `date -u`), applies the lead's rulings of 17 Oct (GI VI, under the user's weekend delegation) on conflicts 86, 87 and 89, risk 44 and decision 55, from the stage-1 build addendum §7 and §8 (158d2c2, 76a08a0) and FOCUS's 09:32 and 09:37 UTC entries: "training-signal" is kept, to be mentioned to the user (conflict 86); §4.5 now gives the source's reason for the construction checks (conflict 87); Appendix C.5 adds the Self-Inf-N-style rank to the planted-cell ranks, with its condition (conflict 89), and A1's planted-cell rank under a disjoint-source anchor set as a pre-registered sensitivity (risk 44), under one more marker [PLANNED; the stage-1 pre-registration is not yet frozen]; §4.2 and a new paragraph of Appendix C.11 state that the PHTest extension of the over-refusal bank fired (decision 55 (a)); and §6 states A1's one-shot scoring as a limit, with its stability figures (decision 55 (b)). The optional §6 sentence on the shared PKU source did not fit and is recorded as the fallback (decision 56). The LaTeX Appendix B's emphasis glitch in conflict 83 is fixed. "Six pre-registered repairs" and "about 10–12 pp" are kept. The main body measures 7.97 pages to the Impact Statement (8.16 to the References heading; 7.97 and 8.16 in revision 6f); see "What changed in revision 6g", decisions 38, 43, 55 and 56, risks 31, 44 and 45, and conflicts 66, 83, 86, 87, 89 and 90–95. The rest of this paragraph describes revision 6f (its page counts are revision 6f's), updated where revision 6g changed the body. Revision 6f of the draft, written on 17 October 2026 from 09:19 UTC (clock checked with `date -u`), applies the lead's rulings of 17 Oct (GI VI, under the user's weekend delegation, FOCUS 00b99a4) on decision 54 and conflicts 78–84, from the stage-1 decision list's LW-09 and LW-10 and the pre-registration DRAFT revision 2's §3.7, §4.5 and §8.5 (2ded100): the flag's name, "fails answers end", now also in §4.2 (conflict 78); the plant described as PKU red-team prompts with PKU-safe, non-refusing answers, never "benign-looking" (§4.5, Appendix C.5, the Impact Statement; LW-10 F1); and the positive control's claim narrowed to the training-signal diagnostics, A1 and a Self-Inf-N-style score, with the text-only scorers' planted-cell ranks published before the pilot as construction checks (the abstract, §1, contribution 2, §4.5, §6 and Appendix C.5; LW-10 F2 and F8). Rulings 80, 81, 83 and 84 are confirmed and their text is unchanged; "six pre-registered repairs" and "about 10–12 pp" are kept. The main body measures 7.97 pages to the Impact Statement (8.16 to the References heading; 7.94 and 8.13 in revision 6e); see "What changed in revision 6f", decisions 43, 44, 54 and 55, risk 44 and conflicts 78–84 and 86–89. Two stage-1 facts that FOCUS recorded during this revision (REQ-19's over-refusal check fired; A1's removal set is sealed) were not in the body; revision 6g states both (decision 55). The rest of this paragraph describes revision 6e (its page counts are revision 6e's), updated where revision 6f changed the body. Revision 6e of the draft, written on 17 October 2026 from 08:42 UTC (clock checked with `date -u`), applies the lead's rulings of 17 Oct (GI VI) on stage-1 items that the sources settle (the stage-1 pre-registration DRAFT revision 2 and the consolidated decision list, 2ded100; FOCUS fcf79a2, and 8dac6ad for Olmo 3): the feasibility estimator, LW-06 (§3.2); an arm that fails the coherence screen is read and flagged, LW-09 (§4.2); the margin for another family, UW-13, with δ's stated reason, LW-02 C1 (a §3.2 clause; Appendix C.11); headline item (2)'s wording, UW-15 (the abstract, §1, contribution 2, §6); and the two equivalence grades, LW-02 B3 (Appendix C.11). Each new body statement carries the marker [PLANNED; the stage-1 pre-registration is not yet frozen]; Appendix C.11 holds the definitions. The main body measures 7.94 pages to the Impact Statement (8.13 to the References heading; 7.92 and 8.12 in revision 6d); see "What changed in revision 6e", decisions 2, 38, 39, 44, 48 and 52–54, risks 34, 45, 46 and 61, and conflicts 78–85. Decision 48 (Olmo-3-7B-Instruct) is now confirmed by the user (FOCUS 8dac6ad). The rest of this paragraph describes revision 6d (its page counts are revision 6d's), updated where revision 6e changed the body. Revision 6d of the draft, written on 17 October 2026 from 08:16 UTC (clock checked with `date -u`), applies the lead's rulings of 17 Oct (GI VI) on the stage-1 decisions that FOCUS fcf79a2 and the stage-1 build addendum §6 (98ff104, time corrected in 0294d1b) record. Five settled items are stated in the body, each with a shared marker [PLANNED; the stage-1 pre-registration is not yet frozen]: the paired analysis model HB-C, the closed graphical testing family $\Phi$ of 24 nodes, rule R1.3 for *worse* on over-refusal and task loss, one primary over-refusal bar $\tau_O^* = 5$ pp and the label order (*worse* first, the full row printed) (§3.2, §4.9; the full definitions in a new Appendix C.11). It also applies seed layout D1 (Table 1, Appendix C.3, §4.4), the conditional PHTest extension of the over-refusal bank (§4.2; one new citation), the token axis (Appendix C.3) and how HarmBench's 512 tokens are counted (Appendix C.10). The main body now measures 7.92 pages to the Impact Statement (8.12 to the References heading; 7.94 and 8.13 in revision 6c); see "What changed in revision 6d", decisions 48 and 52–53, risk 61 and conflicts 70–77. The rest of this

paragraph describes revision 6c (its page counts are revision 6c's), updated where revision 6d changed the body. Revision 6c of the draft, written on 17 October 2026 from 07:56 UTC (clock checked with `date -u`), applies the lead's rulings of 17 Oct (GI VI; FOCUS fcf79a2 and the stage-1 build addendum §6 at 0294d1b) as the brief for revision 6c relays them: HarmBench's validation classifier and its 512-token window named (§4.2; the model ids in a new Appendix C.10); Qwen2.5-1.5B's drop as one limitation sentence (§6, with §4.1 pointing to it); and the token-matched analysis with the token-matched random arms (§3.2; Appendix C.3). It also records FC-1 as closed and the lead's two corrections. The main body now measures 7.94 pages to the Impact Statement (8.13 to the References heading; 7.96 and 8.15 in revision 6b); see "What changed in revision 6c", decisions 49–51 and conflicts 64–69. The rest of this paragraph describes revision 6b (its page counts are revision 6b's), updated where revision 6c changed the body. Revision 6b of the draft, written on 17 October 2026 from 07:20 UTC (clock checked with `date -u`), makes the paper-side fixes from the two mock reviews of revision 6 (`writing/paper/2026-10-17-mock-review-A-methods.md`, statistics and design; `writing/paper/2026-10-17-mock-review-B-safety.md`, LLM safety) that do not depend on the lead's pending stage-1 rulings, and records the user's decisions of 17 Oct as the coordinator's brief relays them (FOCUS at head 6368d66 does not record them; conflict 53). The main body still measures 7.96 pages to the Impact Statement (8.15 to the References heading); see "What changed in revision 6b", decisions 35–48, risks 34–60 and conflicts 53–63. The rest of this paragraph describes revision 6, updated where revision 6b changed it. Revision 6 of the draft, written on 16 October 2026 from 21:10 UTC (17 Oct IST; clock checked), retargets the paper to ICML 2027 (FOCUS 16 Oct 21:05 UTC, commit 416b780) and brings the body to ICML's 8-page limit: the main body now measures 7.96 pages to the Impact Statement (8.15 pages to the References heading, the measure used before; 9.36 in revision 5e); see "What changed in revision 6" and conflicts 41–52. Design details moved to a new Appendix C (part of the paper); none was dropped. The body now says what FOCUS records since 9 Oct: Llama-3.2-1B's rung without "provisional" (its Gate A passed under decision 25, 10 Oct); the 7–8B rungs of both families as design; Gate B's frozen pre-registration, its passed noise gate and its sealed rankings, with the pre-registered consequence for Gate C; D5 = (a). It applies the side-paper overlap rule, the coordinator's three additions (recipe G as a reference, not a general fix; the literature sweep's factual fixes; the user's rule of 17 Oct, "keep the more ambitious one always but always open to fallbacks": the title kept, token matching, the full 7–8B comparison on both families) and adds an Impact Statement. Earlier revisions: 5e (9 Oct, from 17:14 UTC) called Llama-3.2-1B's rung provisional and stated the frozen too-close rule exactly; 5d (from 17:03 UTC) filled Llama-3.2-1B's rung provisionally; 5c (from 15:29 UTC) applied the user's wording package (decision 23) and the ARC rule's scope (decision 16); 5b (from 15:06 UTC) added the blind human read; 5 (from 13:17 UTC) made recipe B the main recipe again, recipe G a reference, the per-family learning-rate rule, five mandatory references and the dataset plan; 4 (8 Oct, from 21:40 UTC) withdrew the loss-weighting account and added the ARC rule; 3 (from 19:35 UTC) added the *worse* label; 2 (from 14:00 UTC) followed the paper specification (`research/review/2026-10-08-paper-specification.md`, commit f2a9979) as amended by the adopted FOCUS amendment (commit 4b37ec2, "ADOPTED 2026-10-08T13:53 UTC") and the study plan `research/review/2026-10-08-sharpened-direction-DRAFT.md` (revision 2). FOCUS.md wins on any conflict. **Nothing in the paper body is a result,** with six kinds of exception, each a design or status fact: the blind human read's rounded caveat (§4.2, §6, the Impact Statement; conflict 28); §6's sentence that Qwen2.5-1.5B's erosion saturated under recipe B during development (revision 6c; a §4.1 clause in revision 6b; conflict 54); the rungs chosen from development probes, with a one-line disclosure that both Llama rungs were accepted on a too-close bar (§4.1); Gate B's status (the noise gate passed; the rankings are sealed) and the pre-registered consequence of the sealed rankings for Gate C (§4.6, Appendix C.1–C.2), with no number and no category named; and §4.9's disclosure that the stage-1 pre-registration is frozen after those Gate B values were seen; and, since revision 6g (the lead's ruling 6), §6's two rounded stability figures of A1's sealed per-row scores (Spearman 0.77; top-600 Jaccard 0.43), a pre-registered report-only read of wave 0b's retrain of $a_0$, not an outcome of any repair arm (conflict 90). Gate B's category runs (wave 1), Gate C's reading, stage 1, the positive control and the Llama stage have not run. Development evidence appears only in these notes, except the blind read's counts in Appendix A. In the LaTeX build these notes are Appendix B and the design details are Appendix C; when the notes are deleted before submission, Appendix C becomes Appendix B (every reference to it is a `\ref`). Menu sizes, the filter list and costs are marked [TO BE FINALISED]; the stage-1 pre-registration, Safe LoRA's settings and Alpaca's preconditions are marked [NOT YET FIXED]; the release policy and the supplementary copies are marked [TO BE DECIDED]; the full 7–8B comparison is marked [PLANNED; funding-gated, fallback one family]; Olmo-3-7B-Instruct is marked [PLANNED; funding-gated; first 7–8B model cut] and the lower-learning-rate

reference [PLANNED; to be pre-registered] (both revision 6b); three items are marked [TO BE FIXED in the stage-1 pre-registration] (the placeholder table lists them; since revision 6g one of them, the two candidate data-only additions, sits in Appendix C.3; revision 6b marked ten, revision 6c settled two, token matching and the HarmBench input, revision 6d four, the analysis model, the closed testing family, the baseline of *worse*'s over-refusal test and the label order, and revision 6e one, feasibility as a non-inferiority bound); the settled stage-1 items are marked [PLANNED; the stage-1 pre-registration is not yet frozen] (revision 6d: five items; revision 6e: the feasibility estimator in §3.2 and the coherence rule in §4.2, and under Appendix C.11's marker the margin and the equivalence grades; revision 6f: the positive control's counting rule, once in §4.5 and once in Appendix C.5; revision 6g: one more in Appendix C.5, on the Self-Inf-N-style rank's condition and the disjoint-anchor sensitivity); Appendix C.11 also states OLMo-2-1B's Gate A mean erosion, 45.11 pp, as the constant of the margin rule for another family (revision 6e; conflict 81), and, since revision 6g, the six-seed over-refusal MDE of 11.04 pp on the Gate A records that fired the PHTest extension (decision 55; conflict 91); the concurrent side paper is cited as [CITE: concurrent submission, anonymized]. The ICML LaTeX version (`icml/main.tex`) carries the same text with a fetched `references.bib`; the author-year citations below use the venue years and letters of those fetched entries (revision 6 re-lettered Zhang et al. 2026: a = Safety at One Shot, b = DataRx, c = DataShield).

**Title.** The spec's working title ends "…, and When Diagnosis Pays". The user's decision makes the measured repair comparison the headline and the decision-value map secondary, so the title names only the comparison. **Revision 6:** the title is unchanged: the user's rule of 17 Oct keeps "Compute-Matched", with "Cost-Accounted" as the fallback only if token matching cannot be done (decision 26). **Revision 6b:** both mock reviews recommend "Cost-Accounted" (review A W8; review B W13, "unless every removal arm is token-matched"); the user's decision of 17 Oct, relayed again in the brief for revision 6b, keeps "Compute-Matched" with token matching, so the title is unchanged. **Revision 6c (FOCUS fcf79a2, UW-14; the user in the lead's chat, 17 Oct 07:21 UTC):** the title keeps "Compute-Matched". The token-axis analysis is the token-matched analysis beside the row-matched one, and token-matched random arms (R_tok) for A1, L1 and H1 are pre-registered secondary arms in the funded tier, with the analysis alone as their fallback. UW-14 names no title fallback; alternative 5 stays on the list. **Revision 6d:** the title is unchanged; Appendix C.3 now states the token axis (total row tokens, with target tokens beside it; the lead's ruling 4). **Revision 6e:** the title is unchanged. **Revision 6f:** the title is unchanged. **Revision 6g:** the title is unchanged. Alternatives: 1. *Repairing Safety After Benign Fine-Tuning: A Compute-Matched Comparison of Data Repairs, and When Diagnosis Pays* (the spec's title; use it only if the planted sweep is funded and gives the map measured support). 2. *Diagnose or Retrain? A Compute-Matched Comparison of Data Repairs for Safety After Benign Fine-Tuning* (the previous title as a lead-in, as the spec allows). 3. *Choosing a Data Fix When Benign Fine-Tuning Erodes Safety: Diagnosis, Trial Runs and Side Effects at Matched Compute*. This is the DIFFUSE title: the plan (§11) and FOCUS ("agreed fallback") switch to it if Gate B reads DIFFUSE. Its number is kept from the previous draft because the plan refers to it as "alternative title 3". 4. *Remove, Add or Leave the Model Alone? Seeded, Compute-Matched Repairs for Safety Eroded by Benign Fine-Tuning* (stresses the model-level references). 5. *Repairing Safety After Benign Fine-Tuning: A Cost-Accounted Comparison of Data Repairs* (revision 6; the literature sweep's proposal; the fallback title under decision 26, used only if token matching cannot be done; revision 6c: under UW-14 the token-matched analysis runs in any case).

## What changed in revision 6g (17 Oct, from 09:59 UTC)

- **Sources.** The brief for revision 6g (the lead's rulings 1–9, GI VI, 17 Oct, under the user's weekend delegation); the stage-1 build addendum (`.ai/briefs/stage1-commitments-build-20261016-lead-decisions.md` at origin), §7 (158d2c2) and §8 (76a08a0) in full, §3–§6 read again; in `research/experiments/2026-10-17-stage1-preregistration/` at origin (2ded100; unchanged since revision 6f), the pre-registration DRAFT revision 2 (its deadline line, §1.2's A1 row, §8.1–§8.5 and §9.3 in full; Self-Inf-N-style, W1, S1 and REQ-13 by search) and the consolidated decision list (LW-18, by search); FOCUS.md at origin (fetched 17 Oct at 09:59 UTC; head 76a08a0; FOCUS last changed in 1423a4c), its 09:32 and 09:37 UTC entries and its 8 Oct 15:53 UTC entry (Gate C's anchor rule) in full. Before any edit, both paper files matched origin's tip (76a08a0; an empty `git diff` against it and against db90b6b, revision 6f; 76a08a0 changes only the addendum). Details in "Sources read (revision 6g)".

- **1. "Training-signal" (the lead's ruling 1; conflict 86).** Kept in the abstract, §1, contribution 2 and §6 by the lead's ruling, to be mentioned to the user: it narrows UW-15's adopted words by one word. No text change.
- **2. The reason for the construction checks (§4.5; the lead's ruling 2; conflict 87).** §4.5: "text-only scorers' ranks are construction checks, published before the pilot" → "text-only scorers read only row text and $\theta_0$, so their ranks are fixed at the draw and are construction checks, published before the pilot" (DRAFT §8.5 F2's reason, in the ruling's words). Appendix C.5 already gave the same reason ("read only row text and the untuned model, so their ranks of the planted cell are fixed when the plant is drawn"; θ0 is the untuned model); checked, unchanged. Contribution 2 still gives no reason; the ruling names §4.5 and Appendix C.5 only (conflict 94).
- **3. The counting list (Appendix C.5; the lead's ruling 3; addendum §8 item 2; conflict 89).** After the construction-check sentences: "The Self-Inf-N-style score's planted-cell rank joins the ranks scored without new runs; if its scorer is not committed before Gate B's category runs, it leaves both that list and the counting rule, and only A1 counts." The ruling's "by W1" is written "before Gate B's category runs" (the DRAFT's deadline line: W1 is before Gate B wave 1; conflict 92). One marker [PLANNED; the stage-1 pre-registration is not yet frozen] closes items 3 and 4. §4.5 is unchanged: it still names both counted diagnostics, without the condition.
- **4. Risk 44, the disjoint-source anchors (Appendix C.5; the lead's ruling 4; addendum §8 item 1).** After item 3's sentence: "The plant's prompts are disjoint from the anchor prompts, but both come from PKU-SafeRLHF, and A1 is bi-anchored on those anchors, so an A1 hit could be read as near-circular. The planted sweep therefore also reports A1's planted-cell rank under a second anchor set: 64 harmful and 64 safe anchors from a source outside PKU-Alignment's data (the publisher of PKU-SafeRLHF), built by Gate C's anchor rule (severity, category balance, a sealed draw and a pinned script), with its source sealed before the stage-1 pre-registration is frozen and before any planted run. The rank under the PKU-SafeRLHF anchors is the one that counts; the rank under the disjoint anchors is a pre-registered sensitivity analysis. If that set cannot be built before the freeze, the shared source is reported as a limit instead." The prompt-level disjointness is DRAFT §8.1's ("normalized prompts disjoint from the S-pool, the anchors and every REQ-6 list"; REQ-8). The last sentence is the first branch of the addendum's fallback (conflict 95). **§6's limit sentence is not added:** after item 6 there was no room (decision 56).
- **5. The over-refusal bank (§4.2; Appendix C.11; the lead's ruling 5; decision 55 (a); addendum §7).** §4.2: "…, extended by 400 harmless PHTest prompts (An et al., 2024) if a free, pre-registered check finds its six-seed minimum detectable effect (MDE) above 5 pp; never from the held-out bank." → "…, and 400 harmless PHTest prompts (An et al., 2024) drawn by a sealed rule, never from the held-out bank, since a pre-registered power check put the six-seed minimum detectable effect (MDE) on the 100 above 5 pp (Appendix C.11)." No dollar sign and no number in the body, and no marker (as in revision 6d). Appendix C.11's new paragraph "The over-refusal bank", under its head marker: "A pre-registered power check on the Gate A records put the six-seed minimum detectable effect of over-refusal against $\theta_0$ on the 100 development prompts at 11.04 pp (prompts treated as random), above the 5 pp bar, so the pre-registered trigger fired: the primary over-refusal bank is those 100 prompts plus 400 harmless PHTest prompts (An et al., 2024). The candidates are PHTest's prompts labelled harmless, never those labelled controversial. Exact duplicates, near-duplicates of any evaluation prompt (every bank and split), of the anchors, of the refusal pool or of the plant rows, empty prompts and prompts over the 1,536-token prompt limit are excluded and counted by reason; the first 400 eligible prompts in a sealed order form the extension. The two sources are reported apart." Sources: REQ-19 on the Gate A records, prompts random the decider (FOCUS 09:37 UTC); addendum §7 items 1–4. Not stated: the counts (2,077 harmless, 1,192 controversial), the label, the stop rule if fewer than 400 are eligible, and the 128-token answer cap (conflict 93).
- **6. A1's one-shot scoring (§6; the lead's ruling 6; decision 55 (b); addendum §5 item 2).** In "Baselines, doses and cost", after the "X-style" sentence: "A1 scores once, and its row scores at the two checkpoints agree only moderately (Spearman 0.77; top-600 Jaccard 0.43), so the removal set depends on the checkpoint." The figures are FOCUS's 09:32 UTC entry rounded (Spearman ρ(q, e1) 0.771 over the 4,800 per-row scores; top-600 Jaccard q–e1 0.434); the two checkpoints are those after steps 10 and 38 (Appendix C.2; DRAFT §1.2's {q,

e1}). The sentence has no marker; it is a sixth kind of exception to "nothing in the body is a result" (conflict 90). It takes priority over item 4's limit sentence (the ruling).

- **7. The converter glitch (Appendix B; the lead's ruling 7; conflict 83).** In conflict 83's title, the asterisk after τ_O (a quote of the earlier brief) opened an emphasis that the converter closed at the end of "undetermined", so the LaTeX printed a stray asterisk and set the rest of the title and "undetermined" in the wrong style. Fix: that asterisk is escaped in the Markdown, and the converter keeps an escaped asterisk literal (one new rule; on revision 6f's notes it changes nothing but its header comment). The HTML from the Markdown was already right. A check of the regenerated Appendix B (every `\emph` span, and every asterisk outside math) finds no other unbalanced emphasis; the long emphasis spans left are the alternative titles and the "History" lines, as intended.
- **8. Kept unchanged (the lead's ruling 8).** The abstract's "six pre-registered repairs" and §4.6's "about 10–12 pp". No source after revision 6f settles them: addendum §7 and §8 and FOCUS's entries to 1423a4c do not touch either (risk 34).
- **Trims to hold the length (meaning unchanged).** With the rulings applied in their first wording and no trims, the Impact Statement heading moved to page 9 (8.08 pages; 8.27 to the References heading); with item 4's §6 sentence as well, 8.11. §4.2's new clause was reordered ("drawn by a sealed rule, never from the held-out bank, since …"), which saves a line. Then one move and six trims: (1) the move: §4.3's sentence on the two candidate data-only additions (the augmentation of Qi et al. (2025) and the single safety example of Zhang et al. (2026a), with its marker [TO BE FIXED in the stage-1 pre-registration]) moves word for word to the end of Appendix C.3's "Secondary and scored-only arms", and §4.3's pointer "Pools, seeds and secondary arms (three seeds, exploratory) are in Appendix C.3" becomes "Pools, seeds, secondary arms (three seeds, exploratory) and two candidate data-only additions are in Appendix C.3"; (2) §6 "Scale and family": the repeat "A family difference is claimed only above its MDE." is cut (§4.8: "a family difference is claimed only above that MDE"); (3) §4.2 "flags answers that reach the token cap" → "… that hit the token cap"; (4) §4.2 "or have fewer than 5 tokens" → "or have under 5 tokens"; (5) §4.2 "an arm that fails the screen is still read" → "a failing arm is still read"; (6) §4.3 "The refusal pool comes from PKU-SafeRLHF" → "… is from PKU-SafeRLHF"; (7) §4.3 "Table 1 lists the stage-1 repairs; §4.4 gives the references." → "Table 1 lists the stage-1 repairs." (Table 1's caption says "references in §4.4"). Each was needed: without any one of the seven, or with §4.2's first word order, the heading falls to the top of page 9. Tried and not kept: §4.5 "Because the truth is known, … which natural menus at our budget cannot" → "With the truth known, …" (also in Appendix C.5); §6 "The planted cause is artificial" → "The plant is artificial"; §6 "We therefore report the best repair within each actor's menu (§3.1); a data-only pipeline …" → "…; we report …, and a data-only pipeline …"; §4.7 "per arm and per scorer" → "per arm and scorer". With all eleven, item 4's §6 sentence still put the heading at the top of page 9 (8.00), so it stays out.
- **Length.** Before: 7.97 pages to the Impact Statement heading (8.16 to References), revision 6f, re-measured on its committed source with the same script. With the rulings and no trims: 8.08 (8.27). After the move and trims: 7.97 (8.16 to References); the heading sits where it sat in revision 6f, on page 8's right column, at the last place where it fits with two lines of its text.
- **Editor's notes.** This list; "What this is" and "Title"; the placeholder table (§4.2, §4.3, §4.5 and Appendix C.5, §6, Appendix C); the fill-in "Planted cause found / not found"; decisions 38, 43 and 55 updated and 56 added; risks 31, 44 and 45; a status line at revision 6g and the end of revision 6f's; conflicts 66, 86, 87 and 89 updated (86, 87 and 89 resolved), 83's title escaped, and 90–95 added; "Sources read (revision 6g)". The LaTeX Appendix B is regenerated from these notes with revision 6f's converter, which reproduced revision 6f's Appendix B byte for byte from revision 6f's notes before any edit; the escape rule (item 7) was added after that check, and its header comment now says revision 6g.
- **Pipelines.** Revision 6f's scripts, reused in a new scratch folder `rev6g/`: `body_pages.py`, `heads.py` and `short_lines55.py` unchanged; `notes_md_to_tex.py` with the escape rule and its header comment; `build.sh` and `splice_notes.py` with the new folder; `paper_to_pdf.py` with local KaTeX, its banner set to revision 6g. One script applied the body and Appendix C edits to the Markdown and the LaTeX, a second the move and the trims, a third the notes, and a fourth adds a revision 6g paragraph to the LaTeX file's header comment; each

body change is a single-match replacement (whitespace-tolerant in the LaTeX), so the two files carry the same words.

- **Anonymity.** As in revision 6f: the body, Appendices A and C and the bibliography hold no author name, institution, repository, branch, account, commit hash, file path or grant (no new BibTeX entry; "PKU-Alignment" names the publisher of a public data set); the PDF metadata says "Anonymous Authors". These notes (Appendix B) are not anonymous and must be deleted before submission.

## What changed in revision 6f (17 Oct, from 09:19 UTC)

- **Sources.** The brief for revision 6f (the lead's rulings on decision 54 and conflicts 78, 80, 81, 83 and 84, GI VI, 17 Oct, under the user's weekend delegation); in `research/experiments/2026-10-17-stage1-preregistration/` at origin (2ded100), the consolidated decision list (LW-09, LW-10, UW-15, UW-16 and LW-26 in full) and the pre-registration DRAFT revision 2 (§0.5, §3.7, §4.5, §8.1–§8.6, §9.1–§9.2 in full); FOCUS.md at origin (fetched 17 Oct at 09:19 UTC; origin's head ecdd23e, revision 6e; FOCUS last changed in 00b99a4), its fcf79a2 entries (the user's UW decisions; the lead's LW decisions, all at their defaults, with LW-18) and its 00b99a4 entry; fetched again at 09:55 UTC (head 23848e0), its two new entries, 09:32 UTC (1be4bb1, time in 71f2458: A1's removal set sealed) and 09:37 UTC (48f3ac2, time in 1423a4c: REQ-19 run, the O-ext fires), and 158d2c2's message (PHTest downloaded) (decision 55). Before any edit, both paper files matched origin's tip (ecdd23e, revision 6e; an empty `git diff`); the commits after it (e27ee88 to 23848e0) touch no paper file. Details in "Sources read (revision 6f)".
- **1. The flag's name (§4.2; the lead's ruling 1; conflict 78).** In "an arm that fails the screen is still read and flagged (Appendix C.11)", "still read and flagged" became "still read, with the flag" and the flag's name, "fails answers end". Appendix C.11 already used "fails answers end" and "comparator fails answers end" (LW-09; DRAFT §3.7 items 3–4, §4.5). No other name for the flag is left in the paper or in these notes (conflict 78 is reworded).
- **2. The plant's description (§4.5; Appendix C.5; the Impact Statement; the lead's ruling 2; LW-10 F1 and R8.5).** §4.5: "A planted cause of $k = 37$ benign-looking rows replaces the rows of one sealed leaf cell" → "The planted cause is $k = 37$ rows: PKU red-team prompts with PKU-safe, non-refusing answers. It replaces the rows of one sealed leaf cell". Appendix C.5's definition: "with $k$ rows that look benign but carry a known eroding signal" → "with $k$ rows that carry a known eroding signal"; its design: "The plant is $k = 37$ rows whose answers comply directly with prompts that also have an unsafe answer in the source data, but which the source labels safe; they look benign." → "The plant is $k = 37$ PKU red-team prompts with PKU-safe, non-refusing answers, from PKU-SafeRLHF, the source of the anchors. Each prompt also has an unsafe answer in the source. Each planted answer is labelled safe there and complies directly." (DRAFT §8.1 and §8.5 F1). Impact Statement: "plants benign-looking rows that erode safety; … The planted rows come from a public safety dataset, and we release …" → "plants rows that erode safety; … The planted rows are PKU red-team prompts with PKU-safe, non-refusing answers, taken from a public safety dataset. We release …". A search of the body and Appendices A and C for "benign-looking", "look(s) benign", "benign-seeming", "appear(s) benign" and "innocuous" finds nothing else. In these notes the words remain only where they quote earlier text (revision 6e's item 4, decision 54, conflict 79) and in decision 43, which describes the reviewers' proposed He-style plant (lists and math): that plant is benign-looking, and LW-10 R8.5 rejects it.
- **3. The positive-control claim (the lead's ruling 3; LW-10 F2 = R8.2, and F8).** Abstract: "a positive control tests whether each diagnostic finds a concentrated, harmful-prompt cause" → "… whether each training-signal diagnostic finds …". §1: "asks whether each diagnostic finds it" → "asks whether each training-signal diagnostic finds it". Contribution 2: "[FINDING 2: whether each diagnostic finds …]" → "[FINDING 2: whether each training-signal diagnostic finds …]", and a new sentence, "Text-only scorers' ranks are construction checks, published before the pilot." §4.5: "each diagnostic's top-$k$ hit rate" → "each scorer's top-$k$ hit rate", and a new sentence, "Only training-signal diagnostics (A1 and a Self-Inf-N-style score) count toward finding the cause; text-only scorers' ranks are construction checks, published before the pilot [PLANNED; the stage-1 pre-registration is not yet frozen]." §6: "finding it shows that a diagnostic *can* find …" → "… a

training-signal diagnostic *can* find …". Appendix C.5: "each diagnostic's top-$k$ hit rate" → "each scorer's top-$k$ hit rate", and after "Every scorer ranks the cells without new runs." three new sentences: "Only attribution (A1) and the Self-Inf-N-style score count toward finding the cause. The text-only scorers (L1, H1, the list-or-markup and longest-answer rules, the DataShield-style filter and the content classifier) read only row text and the untuned model, so their ranks of the planted cell are fixed when the plant is drawn. They are published before the pilot as construction checks, and also if the pilot stops [PLANNED; the stage-1 pre-registration is not yet frozen]." The scope is LW-10's: F2 (which scorers are text-only and why their ranks are fixed at the draw; "construction check"; before the pilot; only A1 and Self-Inf-N-style count toward "finds a known cause") and F8 (the ranks are published if the pilot stops). H2a/b are written as Appendix C.3 names them ("rules for answers with list or markup lines and for the longest answers"). Every scorer's hit rate stays an output (DRAFT §8.6), so "each scorer's" replaces "each diagnostic's" there (conflict 88). LW-10's other items (F4–F7, the pilot interval, R8.3, N6) are not in the body (decision 54). "Training-signal diagnostic" is the lead's term; §4.5 defines it by its two members (conflict 86).

- **4. Confirmed rulings (the lead's ruling 4; checked; text unchanged).** 80: Appendix C.11's grade 1 is "the rule of §3.2". 81: Appendix C.11 states OLMo-2-1B's Gate A mean $E$ of 45.11 pp, the constant of $\delta_{\text{fam}}$. 84: §6's clause and §4.1's marker are unchanged. 83: Appendix C.11's rule R1.3 says that a repair is *worse* on over-refusal only if it is infeasible at $\tau_O^*$ against θ0 and its over-refusal is significantly above its comparator's, and that a repair that is significantly above but feasible or undetermined is not *worse (O)*; the same holds for task loss at τ_L; its "Feasibility" paragraph gives *infeasible* one meaning (the interval's lower end above the limit), which R1.3 uses. That is the ruling's reading, so the text is unchanged.
- **5. Kept unchanged (the lead's ruling 5).** The abstract's "six pre-registered repairs" and §4.6's "about 10–12 pp". The sources do not settle them: LW-26, at its default, puts "pre-specified" in the W1 record and names no paper text (risk 34); no FOCUS entry after fcf79a2 (8dac6ad, 00b99a4, 71f2458, 1423a4c) touches either; §4.6's range comes from Gate B's frozen operating characteristics.
- **Trims to hold the length (meaning unchanged).** With the rulings applied in their first wording and no trims, the Impact Statement heading moved to page 9 (8.10 pages; 8.29 to the References heading). The new sentences were shortened: contribution 2 and §4.5 do not list the text-only scorers or give the reason (Appendix C.5 does), and §4.5 names the two training-signal diagnostics in brackets. With these and no trims: 8.06. Six existing phrases were trimmed: §1 "Every decision rule is committed" → "Every rule is committed" and "replicated over training seeds" → "replicated over seeds"; §3.1 "a fine-tuning customer, who controls the data and can also set the learning rate and the number of epochs" → "… who controls the data, learning rate and epochs" (Table 1's legend: "data, learning rate, epochs"); §4.3 "Pools, seeds and the secondary arms" → "Pools, seeds and secondary arms"; §6 "is model-based and judged against advance predictions" → "is model-based, judged against advance predictions" and "a later study with its own pre-registration would do that" → "a later, separately pre-registered study would do that". Each was needed: without any one of them the heading falls to page 9 or to its top. Tried and not kept (not needed once the six were in): cutting §6's repeat of "a family difference is claimed only above its MDE" (§4.8), §4.5's "which natural menus at our budget cannot" (also in Appendix C.5), §4.2's "reach" → "hit" and "fewer than 5 tokens" → "under 5 tokens", and five smaller cuts (two in §1, one each in §4.3, §4.7 and §6).
- **Length.** Before: 7.94 pages to the Impact Statement heading (8.13 to References), revision 6e, re-measured on its committed source with the same script. With the rulings applied and no trims: 8.10 (first wording) and 8.06 (final wording). After the trims: 7.97 (8.16 to References); the heading sits about three lines lower on page 8 than in revision 6e.
- **Editor's notes.** This list; "What this is" and "Title"; the placeholder table (§4.2; §4.5 and Appendix C.5); the fill-in "Planted cause found / not found"; decisions 43, 44 and 54, and a new decision 55; risk 44; a status line at revision 6f; conflicts 78–84 marked resolved, with 78 reworded, and 86–89 added; "Sources read (revision 6f)". The LaTeX Appendix B is regenerated from these notes with revision 6e's converter, which reproduced revision 6e's Appendix B byte for byte from revision 6e's notes before any edit (its header comment now says revision 6f).

- **Pipelines.** Revision 6e's scripts, reused in a new scratch folder `rev6f/`: `body_pages.py`, `heads.py` and `short_lines.py` unchanged; `notes_md_to_tex.py` unchanged apart from its header comment; `build.sh` and `splice_notes.py` with the new folder; `paper_to_pdf.py` with local KaTeX, its banner set to revision 6f. One script applied the body and Appendix C edits to the Markdown and the LaTeX, a second the trims, a third the notes, and a fourth adds a revision 6f paragraph to the LaTeX file's header comment; each body change is a single-match replacement (whitespace-tolerant in the LaTeX), so the two files carry the same words.
- **Anonymity.** As in revision 6e: the body, Appendices A and C and the bibliography hold no author name, institution, repository, branch, account, commit hash, file path or grant (no new BibTeX entry); the PDF metadata says "Anonymous Authors". These notes (Appendix B) are not anonymous and must be deleted before submission.

## What changed in revision 6e (17 Oct, from 08:42 UTC)

- **Sources.** The brief for revision 6e (the lead's rulings 1–9, GI VI, 17 Oct) and the coordinator's addition on Olmo 3 (decision 48 confirmed; notes only); in `research/experiments/2026-10-17-stage1-preregistration/` at origin (2ded100), the consolidated decision list (UW-13, UW-15, LW-02, LW-06, LW-09, LW-10 and LW-26 in full) and the pre-registration DRAFT revision 2 (§0.5, §3.7, §4.2–§4.6, §6.1–§6.4, §8.5, FC-8 and FC-11 in full); FOCUS.md at origin (fetched 17 Oct at 08:42 UTC, head 80fd326, FOCUS last changed in fcf79a2; fetched again at 09:01 UTC, head 00b99a4: 8dac6ad records the user's Olmo 3 decision, 00b99a4 a relayed hands-off instruction for the weekend); the stage-1 build addendum §6 (its item 8 lists the δ_fam rule and headline item (2)'s new wording among the W1 record's texts); `research/review/2026-10-17-olmo3-7b/olmo3-7b-addendum.md` (its Gate A cost lines). Before any edit, both paper files matched origin's tip (80fd326, revision 6d; an empty `git diff`); 8dac6ad and 00b99a4 change only FOCUS. Details in "Sources read (revision 6e)".
- **1. Feasibility (§3.2; Appendix C.11; the lead's ruling 1; LW-06 (a)).** "A repair is *feasible* at limits $(\tau_O, \tau_L)$ if $O(m) \leq \tau_O$ and $L(m) \leq \tau_L$" became "… if the two-sided 90% intervals for $O(m)$ and $L(m)$ end at or below them, *infeasible* on a measure whose interval lies above its limit, else *undetermined* (Appendix C.11) [PLANNED; the stage-1 pre-registration is not yet frozen]", and the sentence "Whether feasibility is a one-sided non-inferiority bound is open [TO BE FIXED in the stage-1 pre-registration]" is gone. Appendix C.11's new paragraph "Feasibility" gives the estimator ($O(m)$ paired by prompt with $\theta_0$; $L(m)$ against $a_0$), the three states, the point estimate beside them at $\tau_O = 2$, 5 and 10 pp and at $\tau_L$, and the one bound convention ("upper 90%" is the upper end of the two-sided 90% interval, a one-sided 95% bound, everywhere, the selector's screen included; the interval that the equivalence test uses; DRAFT §6.3, R1.1 and R1.2), and says that R1.3's *infeasible* has this meaning. Appendix C.7's "drops a repair whose adverse estimate of $O$ or $L$ passes its limit" is unchanged; Appendix C.11 says that the screen uses the same bound.
- **2. Arms that fail the coherence screen (§4.2; Appendix C.11; the lead's ruling 2; LW-09).** "an arm whose answers fail too often is not read" became "an arm that fails the screen is still read and flagged (Appendix C.11) [PLANNED; the stage-1 pre-registration is not yet frozen]". Appendix C.11's new paragraph: Gate A's bar applied to an arm's sampled answers pooled over its seeds, per-seed counts reported; no arm and no seed dropped, every contrast keeps its test; the tested label stays, with the flag "fails answers end" beside the word, and "comparator fails answers end" on each contrast whose comparator fails; a flag is not a label and never changes the word, and no flagged repair is called *better* or *equivalent* without its flag; coherent-only $U$ and the labels re-read on coherent answers; a failing arm is infeasible for the selector and cannot dominate a reference (DRAFT §3.7 items 1–6, §4.5). The flag's name follows the sources (conflict 78). Every other place was checked: Appendix C.1's G1 and Gate C's "not read" belong to the frozen Gate B/C pre-registration; Gate A's "the answers must end" is an entry bar; Appendix C.7's selector drops repairs on $O$ or $L$, not on coherence; no other body or Appendix C text says that a stage-1 arm is dropped or not read. In these notes, decision 44 and risk 46 are updated.
- **3. The margin for another family (a §3.2 clause; Appendix C.11; the lead's rulings 3 and 5; UW-13; LW-02 C1).** §3.2 now reads "With a smallest effect of interest $\delta = 5$ pp (scaled with the erosion $E$ in another

family; Appendix C.11) and one closed testing family …". Appendix C.11's new paragraph "The margin" states C1 ("about 11% of the measured erosion $E$ at recipe B": δ/E = 11.1% at OLMo-2-1B's Gate A mean $E$ of 45.11 pp, 10.4–11.5% by seed; DRAFT §4.2) and UW-13's rule $\delta_{\text{fam}} = 5 \times E_{\text{fam}}/45.11 \text{ pp} = 0.1108\, E_{\text{fam}}$, in plain words: $E_{\text{fam}}$ is that family's own mean erosion $E$ (in pp) over its three Gate A seeds at its matched dose, and 45.11 pp is the same mean for OLMo-2-1B (the mean of its three Gate A seeds' $E$; the consistency review), so δ is the same share of each family's erosion; with the consolidated list's worked values (2.2 pp at $E_{\text{fam}} = 20$ pp, 6.1 pp at 55 pp), "fixed before its stage-2 runs" (DRAFT §4.2, AMD C2) and the absolute 5 pp reported beside it. C1's wording sits in Appendix C.11 only, so the body states no Gate A number (conflict 81).

- **4. Headline item (2) (the lead's ruling 4; UW-15; addendum §6 item 8).** The user's 8 Oct item (2) asked "does each diagnostic find a known cause, and at which menu size does diagnosis beat runs" (FOCUS 8 Oct 13:53 UTC); UW-15 makes its first half "does each diagnostic find a concentrated, harmful-prompt cause". Abstract: "a positive control tests whether diagnostics find a planted cause" → "… whether each diagnostic finds a concentrated, harmful-prompt cause", and [FINDING 3]'s "whether diagnostics find the planted cause" → "the control's result"; to hold the length, "each run with six seeds" → "each over six seeds" (words 239 → 237; characters 1,571 → 1,571). §1: "A positive control plants a known eroding cause in nested menus …" → "… plants a concentrated, harmful-prompt cause in nested menus …" (it still asks "whether each diagnostic finds it, and at which menu size diagnosis beats trial runs"). Contribution 2: "[FINDING 2: whether each diagnostic ranks a planted cause first, …]" → "[FINDING 2: whether each diagnostic finds a concentrated, harmful-prompt cause, …]". §6: "a diagnostic *can* find a concentrated cause" → "… a concentrated, harmful-prompt cause". The fill-in "Planted cause found / not found" follows. Not changed: §4.5, Appendix C.5 and the Impact Statement still call the plant "benign-looking", which LW-10 corrects (decision 54; conflict 79).
- **5. LW-02 items B3 and C1 (Appendix C.11; the lead's ruling 5).** B3 is the new paragraph "Equivalence grades": grade 1, *equivalent (per comparison)*, the 90% interval of the $U$ contrast inside ±5 pp (§3.2's rule); grade 2, *equivalent (familywise)*, an added report, the 98.33% interval ($\alpha/6$ per side) inside ±5 pp; the row names the grade (DRAFT §4.3, §4.5). Φ's paragraph said "Equivalence is read per comparison, outside Φ" and now says "Equivalence is read outside Φ, in two grades (below)". B3's "the frozen rule" is written "the rule of §3.2" (conflict 80). C1: item 3.
- **6. Confirmed rulings (the lead's ruling 6; checked).** 70: §4.4 says "(three, or all seven $a_0$ seeds)". 71: the full-bank clause is gone (and the rest of that [TO BE FIXED] sentence goes with item 1). 76: Appendix C.11's label order says that a significant difference below δ adds "significant, below the smallest effect of interest" to the row instead of a new label; no body sentence words it otherwise. 77: the cut is confirmed, and Appendix C.11's R1.3 paragraph now says it plainly: a repair whose over-refusal is significantly above its comparator's but that is not infeasible at $\tau_O^*$ (feasible or undetermined) is not *worse (O)*; its over-refusal shift is reported in its row, and its label comes from the $U$ part; the same holds for task loss at $\tau_L$ (conflict 83). Decision 52 (the planning MDE) is confirmed by the lead. In these notes, decision 2 and the fill-in "A repair is worse" said that every significantly adverse contrast gives *worse*; both now note R1.3 (conflict 82).
- **7. Olmo-3-7B-Instruct (the lead's ruling 7 and the coordinator's addition; notes only).** Decision 48 is CONFIRMED by the user (FOCUS 8dac6ad; first chosen in the FAQ chat, then confirmed in the lead's chat): cut order if money is short, 70B, then 32B, then Olmo 3 7B; costs (ESTIMATES) about USD 8–10 for Gate A rung 1 only, a figure that counts two seeds (a probe pod and one fine-tune pod), and about USD 16–44 for a three-seed Gate A (the untuned unit and three seeds at the chosen rung, over up to three rungs; USD 6–9 if rung 1 reads weak and Olmo 3 stops); risk 61 stays (its own environment lock with transformers ≥ 4.57; a cross-family limit to state if the model is run). The notes gave an Olmo 3 cost only in decision 48 (conflict 85). The "awaiting" wording is gone from decision 48, risk 61, the placeholder table, revision 6c's list and conflict 53. The body keeps [PLANNED; funding-gated; first 7–8B model cut] (conflict 84).
- **8. Kept unchanged (the lead's ruling 8).** The abstract's "six pre-registered repairs" and §4.6's "about 10–12 pp". The sources do not settle them: LW-26, taken at its default, puts the word "pre-specified", not "pre-registered", among the texts of the W1 record until REQ-20's manifest and REQ-21 are committed, and names no paper text (risk 34); US-01, US-11 and LS-06 stay S1 items; §4.6's range comes from Gate B's frozen operating characteristics.

- **Trims to hold the length (meaning unchanged; the details are in Appendix C.11).** With the rulings applied and no trims the Impact Statement heading moved to the top of page 9 (8.00 pages; 8.20 to the References heading). The new sentences were shortened (§3.2's feasibility sentence; §3.2's δ clause keeps only the scaling, with C1's wording in Appendix C.11; §4.2's clause names the flag only in Appendix C.11; contribution 2 has no "planted"), and four existing sentences were trimmed: §3.2's reason why *better* and *equivalent* exclude each other ("*better* needs an estimate of at least δ, *equivalent* an upper limit below δ") moves to Appendix C.11's label order, with "(Appendix C.11)" in its place; §4.1 "which halve the mean learning rate over the epoch" → "… the mean learning rate" (the run is one epoch); §6 "judged against predictions made in advance" → "judged against advance predictions"; §6 "Selectors are not run against each other at equal budgets here" → "Selectors are not compared at equal budgets here". Tried and not kept: cutting §3.2's cluster members (a revision 6d ruling text) and §6's repeat of "a family difference is claimed only above its MDE" (§4.8).
- **Length.** Before: 7.92 pages to the Impact Statement heading (8.12 to References), revision 6d, re-measured on its committed source with the same script. With the rulings applied and no trims: 8.00 (the heading at the top of page 9). After the shortenings and trims: 7.94 (8.13 to References); the heading sits about two lines lower on page 8 than in revision 6d.
- **Not changed (settled in the sources, outside the brief; for the lead).** LW-10's factual description of the plant and its "finds a known cause" rule (decision 54).
- **Editor's notes.** This list; "What this is" and "Title"; the placeholder table (§3.2, §4.1, §4.2, Appendix C); decisions 2, 38, 39, 44, 48, 52 and 53 updated and 54 added; risks 34, 45, 46 and 61 updated; the fill-ins "A repair is worse" and "Planted cause found / not found"; a status line at revision 6e; revision 6c's list and conflict 53 (the "awaiting" wording); conflicts 78–85 added; "Sources read (revision 6e)". The LaTeX Appendix B is regenerated from these notes with revision 6d's converter, which reproduced revision 6d's Appendix B byte for byte from revision 6d's notes before any edit (its header comment now says revision 6e).
- **Pipelines.** Revision 6d's scripts, reused in a new scratch folder `rev6e/`: `body_pages.py` and `short_lines.py` unchanged; `notes_md_to_tex.py` unchanged apart from its header comment; `build.sh` and `splice_notes.py` with the new folder; `paper_to_pdf.py` with local KaTeX, its banner set to revision 6e. One script applied the body and Appendix C.11 edits to the Markdown and the LaTeX, a second the trims, a third the notes, and a fourth adds a revision 6e paragraph to the LaTeX file's header comment; each body change is a single-match replacement (whitespace-tolerant in the LaTeX), so the two files carry the same words.
- **Anonymity.** As in revision 6d: the body, Appendices A and C and the bibliography hold no author name, institution, repository, branch, account, commit hash, file path or grant (no new BibTeX entry); the PDF metadata says "Anonymous Authors". These notes (Appendix B) are not anonymous and must be deleted before submission.

## What changed in revision 6d (17 Oct, from 08:16 UTC)

- **Sources.** The brief for revision 6d (the lead's rulings 1–7, GI VI, 17 Oct) and the coordinator's addition on Olmo 3 (notes only); FOCUS.md on `origin/stageb-costs-and-recovery-safeguards-20261003` (fetched 17 Oct at 08:16 UTC, branch head f9714b5, revision 6c; fetched again at 08:34 UTC, head 4446539, the Olmo 3 checks; FOCUS last changed in fcf79a2); the stage-1 build addendum §3–§6; in `research/experiments/2026-10-17-stage1-preregistration/`, the consolidated decision list, the pre-registration DRAFT revision 2 and the W1 builder extract; the Gate B HarmBench pipeline; HarmBench's `evaluate_completions.py`; the arXiv API record of 2409.00598; `research/review/2026-10-17-olmo3-7b/` (4446539). Before any edit, both paper files matched origin's tip (an empty `git diff` against f9714b5); 4446539 changes no file under `writing/paper/`. Details in "Sources read (revision 6d)".
- **1. Five settled stage-1 items (§3.2, §4.9; new Appendix C.11; the lead's ruling 1).** Each is stated in short sentences in the body and in full in Appendix C.11, with one shared marker [PLANNED; the stage-1 pre-registration is not yet frozen] in §3.2's first and third paragraphs, in §4.9 and at the head of Appendix C.11. (a) The analysis model (UW-02; LW-01; addendum §6 item 1): §3.2's "Contrasts are estimated from per-seed unsafe rates; the analysis model, including whether it pairs arms by seed, is open [TO BE FIXED in the stage-1

pre-registration]" became "The primary model, HB-C, pairs arms by seed within two design clusters (no repair and the additions; random removal and the removals) and crosses prompts with seeds; HB, S-a and S-aW are pre-registered sensitivity analyses (Appendix C.11)". Appendix C.11 adds what same-seed arms share (the adapter initialization, the row order and the sampling streams; DRAFT §3.3), the cluster members and why they differ (add arms keep $a_0$'s batches, removal arms re-form theirs; FOCUS LW-01), a residual variance per arm by REML, Satterthwaite df, the refit for over-refusal and task loss (DRAFT §3.6) and "the row prints both; the primary model governs" (DRAFT §3.4). (b) The closed graphical family (UW-04 (a); addendum §6 item 3): §4.9's [TO BE FIXED] sentence became "The primary family is one closed graphical family, $\Phi$, of 24 nodes, four for each of P1–P6 (Appendix C.11)"; Appendix C.11 gives the nodes, their sidedness and the start weights (DRAFT §4.4: 0.15 per $U_k$, 0.1/12 per $O_k$ and $L_k$, 0 per $F_k$) and says that equivalence is read per comparison, outside $\Phi$. Holm leaves the body: §1 "after Holm correction" → "after multiplicity correction"; §3.2 "Holm correction at family-wise level 0.05 over the primary contrasts" → "one closed testing family at family-wise level 0.05 (§4.9)", "the Holm-adjusted test" → "its test", and "at the Holm-adjusted 5% level" is gone; §4.3 "Holm at 5%, labels as in §3.2" → "One closed family at 5% (§4.9); labels as in §3.2". (c) Rule R1.3 (UW-05 (b); addendum §6 item 3): §3.2's *worse* clause now reads "*worse* if it is significantly adverse on the unsafe rate ($\Lambda_m < 0$) or, by rule R1.3, on over-refusal or task loss: infeasible against $\theta_0$ at the primary over-refusal bar $\tau_O^* = 5$ pp (task loss: against $a_0$ at $\tau_L$) and significantly above the comparator"; the [TO BE FIXED] sentence on its baseline is gone, and so is the clause "each read in its adverse direction, so an adverse repair is never reported as inconclusive" (conflict 77). Appendix C.11 adds $\tau_L$ read against $a_0$ with $\bar{\ell}(m) - \bar{\ell}(a_{\mathrm{R}})$ reported beside every removal arm (LW-07). (d) One primary over-refusal bar (UW-06 (a)): $\tau_O^* = 5$ pp is named in the *worse* clause; "Results are reported at $\tau_O \in \{2, 5, 10\}$ pp" stays; Appendix C.11 says that it equals Gate B's G4 bar. (e) The label order (UW-08 (a)): "which label wins when *worse* holds with another, and whether a significant difference below δ gets its own label, are open [TO BE FIXED in the stage-1 pre-registration]; both results are reported" became "*Worse* is checked first, and the full row is always printed (Appendix C.11)"; Appendix C.11 gives the order (worse, better, equivalent, inconclusive), the row's parts (DRAFT §4.5) and the note "significant, below the smallest effect of interest" for a significant difference below δ, which is part of UW-08 (a)'s recommended text (conflict 76).

- **2. Seeds, layout D1 (the lead's ruling 2; UW-03 (a); addendum §6 item 2).** Table 1: random removal 8 → 7 seeds and no repair 6 → 7. Appendix C.3: "Method arms run at six seeds, random removal at eight and no repair at six" → "Method arms run at six seeds (1102–1107), and random removal and no repair at seven each (1101–1107; random removal at seed 1108 is not run)". §4.4: "(three, or all six $a_0$ seeds)" → "(three, or all seven $a_0$ seeds)" (conflict 70). No other body or Appendix C text gave these counts.
- **3. The over-refusal bank (§4.2; the lead's ruling 3; UW-07a, UW-07b; addendum §6 item 4).** "Qwen3Guard's refusal flag on greedy answers to 100 benign prompts that resemble harmful ones (…), reported by source, with a lexicon check" became "… to the 100 development over-refusal prompts, benign prompts that resemble harmful ones (…), extended by 400 harmless PHTest prompts (An et al., 2024) if a free, pre-registered check finds its six-seed minimum detectable effect (MDE) above 5 pp; never from the held-out bank. Reported by source, with a lexicon check." The ruling's zero-dollar check is written "a free … check" (conflict 72), and the MDE is defined here, its first use in the body. New BibTeX entry `an2024phtest`, from the arXiv API (CITE-CHECK 37). §3.2's feasibility marker loses "with over-refusal on the full XSTest and OR-Bench-Hard banks" (conflict 71).
- **4. The token axis (Appendix C.3; the lead's ruling 4; W1-01, W1-02 item 4).** "Every removal and addition arm reports the training tokens it removes or adds" → "… reports the rows, target tokens and total row tokens it removes or adds", and a new sentence: "The primary token axis is total row tokens (prompt plus target tokens, as trained); target tokens (the answer and its end-of-turn tokens) are reported beside it." The body is unchanged: the ruling allows "a token axis (total tokens per removed row)" in the body but does not require it, and §3.2 already points to Appendix C.3.
- **5. HarmBench's token count (Appendix C.10; the lead's ruling 5).** "Each answer is clipped to its first 512 classifier tokens before it is judged, as HarmBench's own evaluation script does, so …" → "… before it is judged, so the classifier reads only the opening of a longer answer. The clip mirrors HarmBench's own

evaluation script (`evaluate_completions.py`): tokens are counted with the classifier's own tokenizer, loaded from the classifier's model snapshot; the count of 512 includes the beginning-of-sequence token; the answer is truncated on the right and decoded without special tokens." Checked on the script and on the Gate B pipeline (CITE-CHECK 36). No path or hash in the paper; the script is named by its public file name (conflict 75).

- **6. Kept unchanged (the lead's ruling 6).** The abstract's "six pre-registered repairs", §4.6's "about 10–12 pp" and §4.4's open choice for the post-hoc references' seeds ("three, or all … $a_0$ seeds"; only the $a_0$ count changed, conflict 70). The sources do not settle them: US-01 (the references on all $a_0$ seeds), US-11 (the lower learning rate as a sixth reference) and LS-06 (the rule "a data repair beats a reference"; its rule text may enter the W1 record) are S1 items, not among the decisions of 17 Oct; the stage-1 pre-registration is not frozen; and §4.6's range comes from Gate B's frozen operating characteristics (Appendix C.1: about 9, 10 or 12 pp), which no new source changes.
- **7. The planning MDE (§4.3, Appendix C.3; follows from ruling 1 (a) and (b)).** "At planning noise the minimum detectable effect after Holm's first step is about 5 pp" (§4.3 and Appendix C.3) was the unpaired model's figure under Holm (DRAFT §3.5, the PLAN-S and S-a rows). Under HB-C and Φ it is not the planning figure (decision 52). §4.3 now says "The MDE depends on the same-seed correlation [PLACEHOLDER: final MDE from Gate B's pooled noise]", and Appendix C.3 "Under the paired model the MDE depends on the same-seed correlation; the stage-1 record prints it at Gate B's pooled noise" (LW-02, item A5). No new number (conflict 73).
- **8. Olmo-3-7B-Instruct (the coordinator's addition; notes only).** The lead's free (USD 0) checks are recorded under decision 48, and their cross-family limit as risk 61. The body keeps its [PLANNED; funding-gated; first 7–8B model cut] marker and its citation of Team Olmo et al. (2025); nothing else is added to the body.
- **Trims to hold the length (meaning unchanged; the details are in Appendix C.11).** The first versions of the new body sentences put the Impact Statement on page 9 (8.00 pages). Shortened: §3.2's model sentence (the seed-matched differences and the cluster names are in Appendix C.11); §3.2's label-order sentence keeps only the ruling's words ("*Worse* is checked first, and the full row is always printed"; the note for a significant difference below δ is in Appendix C.11); §4.9's Φ sentence keeps "24 nodes, four for each of P1–P6" (the nodes, sidedness and weights are in Appendix C.11).
- **Length.** Before: 7.94 pages to the Impact Statement heading (8.13 to References), revision 6c, re-measured on its committed source with the same script. With the rulings applied and no trims: 8.00 (the heading moved to the top of page 9). After the trims: 7.92 (8.12 to References); the heading sits about two lines higher on page 8 than in revision 6c.
- **Not changed (settled in the sources, outside the brief; for the lead).** LW-06 (a) settles the feasibility estimator that §3.2 still marks [TO BE FIXED]; LW-09 replaces §4.2's "an arm whose answers fail too often is not read" with flags; UW-13 (δ_fam), UW-15 (headline item 2's wording) and LW-02 (items B3 and C1) are not in the body (decision 53).
- **Editor's notes.** This list; "What this is" and "Title"; the placeholder table (§3.2, §4.2, §4.3, Table 1, §4.9, Appendix C, References); decisions 12, 19, 33, 35, 37, 38, 40, 48, 49 and 51 updated and 52–53 added; risks 34, 35, 37, 45, 46, 51, 54 and 59 updated and 61 added; CITE-CHECK 36 updated and 37 added; the development evidence's HarmBench item and a status line at revision 6d; conflicts 64–67 updated and 70–77 added; "Sources read (revision 6d)". The LaTeX Appendix B is regenerated from these notes with revision 6c's converter, which reproduced revision 6c's Appendix B byte for byte from revision 6c's notes before any edit (its header comment now says revision 6d).
- **Pipelines.** Revision 6c's scripts, reused in a new scratch folder `rev6d/`: `body_pages.py`, `short_lines.py` and `notes_md_to_tex.py` unchanged apart from the converter's header comment; `build.sh` and `splice_notes.py` with the new folder; `paper_to_pdf.py` with local KaTeX, its banner set to revision 6d. One script applied the body and Appendix C edits to the Markdown and the LaTeX, a second shortened the new body sentences and a third polished the new wording; each change is an exact single-match replacement, so the two files carry the same words.

- **Anonymity.** The body, Appendices A and C and the bibliography hold no author name, institution, repository, branch, account, commit hash, file path or grant; the new BibTeX entry holds only the public arXiv record; the PDF metadata says "Anonymous Authors". These notes (Appendix B) are not anonymous and must be deleted before submission.

## What changed in revision 6c (17 Oct, from 07:56 UTC)

- **Sources.** The brief for revision 6c, which relays the lead's rulings (GI VI, 17 Oct) in five items; FOCUS.md on `origin/stageb-costs-and-recovery-safeguards-20261003` (fetched 17 Oct at 07:56 UTC; the branch head is 0432940, revision 6b; FOCUS last changed in fcf79a2), read for the user's and the lead's decisions of 17 Oct 07:21 UTC and searched for the 7 Oct Qwen entries, Olmo 3, option B and the wave-1 reserve; the stage-1 build addendum §6 (98ff104, time corrected in 0294d1b); `research/experiments/2026-10-17-stage1-preregistration/`; HarmBench's `evaluate_completions.py`. Details in "Sources read (revision 6c)".
- **1. The HarmBench classifier (§4.2; new Appendix C.10; the lead's ruling 1).** §4.2's "HarmBench's classifier asks whether …" became "HarmBench's validation classifier (512-token window; Appendix C.10) asks whether …". The sentence "Whether the classifier reads each answer clipped to 512 tokens or in full (up to 2,048) is open [TO BE FIXED in the stage-1 pre-registration]" is gone: the clip is settled. The new Appendix C.10 gives the model id (`cais/HarmBench-Mistral-7b-val-cls`), HarmBench's classifier prompt and the clip to the first 512 classifier tokens "as HarmBench's own evaluation script does", and one sentence that is not a decision: "HarmBench's test classifier, `cais/HarmBench-Llama-2-13b-cls`, is a candidate sensitivity check." The script fact was checked on the script itself (CITE-CHECK 36; decision 40; conflicts 55, 56, 68).
- **2. Why Qwen is out (§6, §4.1; the lead's ruling 2).** §6 "Scale and family" gains one plain sentence: "Qwen2.5-1.5B was dropped during development when its erosion saturated under recipe B at a learning rate of $10^{-4}$, before the per-family learning-rate rule existed, and was not retried at a lower rate in this study, so the Qwen runs of [CITE: concurrent submission, anonymized] are the only later evidence." The body gives no calendar date. §4.1's clause became "Qwen2.5 (Qwen et al., 2024) is outside this study (§6)", so the reason is stated once, in §6; §4.1's citation of the same-recipe comparison of OLMo-2, Llama-3 and Qwen2.5 is gone (§4.1 keeps its two other citations of the concurrent submission). Sources: FOCUS 7 Oct 20:58 UTC ("Llama and OLMo only"; the confirming Qwen seeds and "a Qwen backup at Qi's LR 2e-5 … cancelled before launch"); FOCUS's screening list (Qwen 1.5B recipe B +75.2 pp); the per-family rule of 9 Oct 09:29 UTC (risk 56; conflicts 54, 69).
- **3. Token matching (§3.2; Appendix C.3; the lead's ruling 3; UW-14).** §3.2's "Removals are matched in training tokens as well as rows, or a token-matched analysis is pre-registered beside the row-matched one [TO BE FIXED in the stage-1 pre-registration]" became "Removals are matched in rows; a token-matched analysis beside the row-matched one, and token-matched random removals for A1, L1 and H1 as secondary arms in the funded tier, are pre-registered (Appendix C.3)". The marker is gone: FOCUS's UW-14 and addendum §6 item 5 settle both parts, and the sentence states no more than they do (no token axis, no test, no label). Appendix C.3 gains a paragraph, "Token matching": every removal and addition arm reports the tokens it removes or adds; the analysis is report-only and runs in any case, placing each removal on the random dose curve (600, 1,200 and 2,400 rows removed) along a token axis beside the row axis; the token-matched random removals run at three seeds each in the funded tier, with the analysis alone as the fallback (decisions 26, 51; risks 30, 51; conflict 67).
- **4. Kept unchanged (the lead's ruling 4).** The abstract's "six pre-registered repairs"; §4.6's "about 10–12 pp"; §4.4's seeds of the post-hoc references ("three, or all six $a_0$ seeds", [TO BE FIXED in the stage-1 pre-registration]). They wait for the stage-1 freeze.
- **5. Decisions recorded (the lead's ruling 5).** FC-1 is closed: the user confirmed the ambitious-default rule in the lead's chat and took the recommended defaults on all 21 wave-1 stage-1 decisions (FOCUS fcf79a2, 07:21 UTC; decisions 26, 28; conflict 53 resolved). "Add OLMo 3 7B" was then not yet confirmed in the lead's chat, with the free (USD 0) checks running (the user confirmed it later, FOCUS 8dac6ad, revision 6e; decision 48 was then no longer RESOLVED; the body's [PLANNED; funding-gated; first 7–8B model cut] marker is

unchanged). The lead's two corrections to FAQ's mapping of the rule (FOCUS fcf79a2): Llama-3.1-8B's third Gate A seed waits for money beyond the wave-1 reserve (decision 50); R_tok is the ambitious arm, with the token-axis analysis as its fallback (item 3).

- **Trims to hold the length (meaning unchanged).** §2: "than removing rows flagged by WildGuard, from one training run per arm and with no compute reported" → "than removing WildGuard's picks, from one run per arm and with no compute reported". §6 "Baselines, doses and cost": "because a scorer's cost and a run's cost grow differently with model size" → "because scoring and training costs scale differently". §4.1: "Qwen2.5 models … are outside" → "Qwen2.5 … is outside".
- **Length.** Before: 7.96 pages to the Impact Statement heading (8.15 to References), revision 6b, re-measured on its PDF with the same script. With the rulings applied and no trims: 8.00 (the heading moved to the top of page 9). After the three trims: 7.94 pages (8.13 to References); the heading sits about two lines higher on page 8 than in revision 6b.
- **Not changed (outside this brief; for the lead).** The stage-1 decisions of 17 Oct settle more of the body's [TO BE FIXED in the stage-1 pre-registration] items than the brief applies: the analysis model (UW-02; LW-01: HB-C, paired by seed), the closed testing family (UW-04: a closed graphical family of 24 nodes), the baseline of *worse*'s over-refusal test (UW-05: rule R1.3), one primary $\tau$_O (UW-06: 5 pp) and the label order (UW-08 (a): *worse* first). The seed layout (UW-03, D1: $a_0$ and $a_R$ at seeds 1101–1107) differs from Table 1 and Appendix C.3 (random removal at eight seeds, no repair at six), and the over-refusal bank may grow (UW-07a). The body keeps its revision-6b text on all of these (decision 49; conflicts 64–66).
- **Editor's notes.** This list; "What this is" and "Title"; the placeholder table (§3.2, §4.1, §4.2, §4.4, §6); decisions 26, 28, 32, 40 and 48 updated and 49–51 added; risks 30, 34, 51, 56 and 59 updated; CITE-CHECK 36; the development evidence's HarmBench item and a status line at revision 6c; conflicts 52–56 updated and 64–69 added; "Sources read (revision 6c)". The LaTeX Appendix B is regenerated from these notes with revision 6b's converter, which reproduced revision 6b's Appendix B byte for byte from revision 6b's notes before any edit (its header comment now says revision 6c).
- **Pipelines.** Revision 6b's scripts, reused in a new scratch folder `rev6c/`: `body_pages.py`, `short_lines.py` and `notes_md_to_tex.py` unchanged apart from the converter's header comment; `build.sh` takes the build folder as an argument; `splice_notes.py` takes the draft and the target file and reports whether Appendix B changed; `paper_to_pdf.py` with local KaTeX, its banner set to revision 6c. One script applied the body edits to the Markdown and the LaTeX, each an exact single-match replacement, so the two files carry the same words.
- **Anonymity.** The body, Appendices A and C and the bibliography hold no author name, institution, repository, branch, account, commit hash or grant; Appendix C.10's two model ids are public Hugging Face ids of HarmBench's authors; the body gives no calendar date for the Qwen drop; the PDF metadata says "Anonymous Authors". No BibTeX entry was added. These notes (Appendix B) are not anonymous and must be deleted before submission.

## What changed in revision 6b (17 Oct, from 07:20 UTC)

- **Sources.** The coordinator's brief for revision 6b; both mock reviews in full (review A, statistics: 15 weaknesses, W2–W5, W11–W13 new; review B, LLM safety: 16 weaknesses and its §8.3 inconsistency; each review's last section separates known from new); these notes; FOCUS.md on `origin/stageb-costs-and-recovery-safeguards-20261003` (fetched 17 Oct; head 6368d66), read for the 7 Oct entries on Qwen (17:44, 19:14 and 20:58 UTC), the 8 Oct 10:14 UTC HarmBench judge entry and the newest entries (to the T1 bundle, 02:42 UTC on 17 Oct). Details in "Sources read (revision 6b)".
- **The user's decisions of 17 Oct (relayed by the coordinator's brief; not in FOCUS at 6368d66; conflict 53).** (1) Olmo-3-7B-Instruct is added as a third 7–8B model, marked [PLANNED; funding-gated; first 7–8B model cut] (decision 48). (2) The standing rule "ambitious default with fallbacks" ("keep the more ambitious one always but always open to fallbacks"). (3) The "Compute-Matched" title is kept, with token matching (decision 26). (4) Option B with option C as the fallback (decision 28). No result is claimed for any of them.

- **1. Why Qwen is out (§4.1; review B W12).** "Qwen2.5 models (Qwen et al., 2024) are outside this study" became "… outside this study because their erosion saturated under recipe B during development (above 55 pp)". Source: FOCUS 7 Oct 17:44 UTC (Qwen2.5-7B "collapsed: 4.8% → 87.0% unsafe after 38 recipe-B steps"), 19:14 UTC (the frozen bars: "E > 55 is saturated, like Qwen") and 20:58 UTC (the user chose "Llama/OLMo only"). The clause gives the class, not a number; the same-recipe comparison stays cited to the concurrent submission (conflict 54; risk 56).
- **2. Threat model (review B W3).** §3.1 gains two sentences: "Two actors can repair: a fine-tuning customer, who controls the data and can also set the learning rate and the number of epochs, and the service provider, who also holds the weights and can run every scorer and reference. We report the best repair within each actor's menu (Table 1)." Table 1 gains an Actor column (C, the customer; P, the provider; S4 is C only if the customer can sample the untuned model) and a caption legend. §4.4 says "The customer can run the safety prompt and the lower learning rate; the provider can run all." To stay consistent: §1 now says the owner "often controls only the data and a few training settings" (was "only the data"), and §6 "Scope and user" replaces "can change only the data" and "Our answer serves a service that must ship the customer's adapter and may change only the data, and a pipeline that allows only data changes" with "controls the data and some hyperparameters", "the provider, or a developer with open weights, can run every scorer …" and "We therefore report the best repair within each actor's menu (§3.1); a data-only pipeline reads the data repairs alone." The abstract's conditional "When only the training data may change" is kept (conflict 60). Not in the body: the dataset publisher as an actor and the transfer test (decision 46).
- **3. A lower-learning-rate reference (§4.4; review B W3).** "A sixth, **lower learning rate** [PLANNED; to be pre-registered], is recipe B at the next lower rung of the ladder in §4.1, a lever the customer has; such rungs are studied in [CITE: concurrent submission, anonymized]." No side-paper number is stated. The body's count of five references is unchanged (conflict 58).
- **4. Scale (§4.1, §6; review B W1).** "Olmo-3-7B-Instruct (Team Olmo et al., 2025) is a third 7–8B model under the same rule [PLANNED; funding-gated; first 7–8B model cut]." §6: "at 7–8B it is planned on both families and Olmo-3-7B-Instruct, funding-gated, with one family as the fallback and Olmo 3 the first model cut". No result is claimed.
- **5. Nine items marked [TO BE FIXED in the stage-1 pre-registration], one sentence each, no rule invented.** (a) §3.2, the over-refusal baseline of *worse*: "Whether the over-refusal test of *worse* uses the comparator's rate or the untuned model's r(θ0) as its baseline is open" (review B §8.3). (b) §4.4, the reference test: "the seeds of the post-hoc references (three, or all six a0 seeds) and the test (against B+Qi, and against the adapter-scaling curve matched on task loss) are open" (review A W3; review B W2). (c) §4.9, one closed testing family: "Whether these and the label tests (superiority, equivalence, the three *worse* tests) form one closed testing family is open" (review A W4). (d) §3.2, the analysis model: "Contrasts are estimated from per-seed unsafe rates; the analysis model, including whether it pairs arms by seed, is open"; this replaces "as differences of means over unpaired seeds, with the variance pooled across arms" (review A W2). (e) §4.2, the HarmBench classifier's input: "Whether the classifier reads each answer clipped to 512 tokens or in full (up to 2,048) is open" (review B W15; conflict 55). (f) §3.2, non-inferiority over-refusal: "Whether feasibility is a one-sided non-inferiority bound, with over-refusal on the full XSTest and OR-Bench-Hard banks, is open" (review A W12; review B W7). (g) §4.2, secondary endpoints: "Whether StrongREJECT's evaluator score (Souly et al., 2024) and a prefilling probe (Qi et al., 2025) are secondary endpoints is open" (review B W6). (h) §4.3, two data-only additions: "Two more data-only additions, the augmentation of Qi et al. (2025) that deepens alignment beyond the first answer tokens and the single safety example of Zhang et al. (2026a) added to the training data, are candidate arms" (review B W5; CITE-CHECK 34). (i) §3.2, the label order (item 7 below). Each sentence ends with the marker; "is open" means "is fixed in the stage-1 pre-registration", which the marker says.
- **6. Related work (§2; review B W5).** One clause each, all fetched programmatically: "Other repairs act after fine-tuning and are not among our references (§6): Safe Delta adjusts the parameter change of a harmful or benign fine-tune (Lu et al., 2025), Antidote prunes harmful weights after harmful fine-tuning (Huang et al., 2025), and Zhang et al. (2026a) restore safety by further training on a single safety example. Huang et al.

(2024) survey harmful fine-tuning attacks and defences." This merges revision 6's sentence on Safety at One Shot. CITE-CHECK 30–33. Not added (each needs its own fetch and room): Eiras et al., backdoor-enhanced safety alignment, Fraser et al., SORRY-Bench, Qwen3 (risk 52).

- **7. Label logic (§3.2, §6; review B W8; review A W5).** As written, *better* (an estimate of at least δ) and *equivalent* (a 90% interval inside (−δ, δ), so an estimate below δ) cannot both hold; §3.2 now says so. What can co-occur is *worse* with *better* or *equivalent* (a side effect, or a small significant adverse difference in U), and "equivalent" can hold for a significant difference below δ. §3.2's "Which label a repair receives when it is better on the unsafe rate but worse on a side effect is fixed in the stage-1 pre-registration [TO BE FINALISED]" became "*Better* and *equivalent* exclude each other (*better* needs an estimate of at least δ, *equivalent* an upper limit below δ); which label wins when *worse* holds with another, and whether a significant difference below δ gets its own label, are open [TO BE FIXED in the stage-1 pre-registration]". §6's "smaller differences stay inconclusive", which the rules contradict (a small difference with a tight interval is *equivalent*), is gone: "The labels resolve differences of about 5 pp, and an *equivalent* label means … not "identical" (§3.2)."
- **8. Editor's notes.** This list; decisions 10, 11, 12, 19, 20, 24, 26, 27, 28 and 33 updated; decisions 35–48 added (48 RESOLVED: Olmo 3); risks 34–60 added, one for each weakness that either review marks new or partly new, each with its review and W-number and its status in the body (TO BE FIXED, changed, or not in the body); "Where the framing is weak" item 4 updated; the placeholder table, CITE-CHECK 30–35, the development evidence (status at revision 6b; the HarmBench clip) and conflicts 53–63. The LaTeX Appendix B is regenerated from these notes with revision 6's converter.
- **Moves to Appendix C, to hold the length (nothing dropped).** §3.4 keeps the definitions in words (selector, regret, the value of a diagnostic and its break-even cost); the selector symbols and the menu notation are in Appendix C.7 only. §4.3's pool sentence keeps only the source (the pairing rule, the judge rule and the overlap control are in Appendix C.3, where they already were). §4.5 keeps the plant, the menus and the outputs; the pilot, the sweep and the replay are in Appendix C.5 (already there). §4.6's Gate B paragraph drops "(600 rows, 33 steps)", "(75 per category)", "so a0 cancels" and "a noise gate, an automatic extension and a length-matched control are pre-registered" (all in Appendix C.1); "If DIFFUSE" becomes one sentence in Gate B, with the rewrite arm and the planted sweep's role moved to Appendix C.3; Gate C drops its list of verdicts (Appendix C.2) and the clause "which does not show that the picks are free of format" (Appendix C.2 keeps it). §4.7 is shortened (Appendix C.6 holds the conventions). §4.8's published-setting replication moves to a new paragraph in Appendix C.3. §4.10 items 3 and 4 move to a new Appendix C.9 ("Side studies: measurement validity and side-effect prediction"). §4.2's ARC item keeps the rule's existence and the double report (Appendix C.4 gives the bar and the rule). §4.1 drops "It was chosen over recipe G from probe and Gate A readings alone, before any repair outcome" (§6 says the same: "The main model and recipe were chosen by screening probe and Gate A readings, never Gate B or stage-1 outcomes"). §4.2's "a sealed held-out subsample" became "held-out prompts" (§4.8 gives the sealed 200-prompt subsample). §4.8's "No result is claimed at a size without runs at that size" became "… for a size without runs".
- **Length.** Before: 7.96 pages to the Impact Statement heading (8.15 to References), revision 6, re-measured on its PDF with the same script. After: 7.96 pages (8.15 to References); the heading sits at the same height on page 8. A probe build with one more line of text in §6 moved the heading to page 9 (8.00 pages), so the body has about one line of slack (risk 28; decision 33).
- **New citations (four; each fetched programmatically on 17 Oct between 07:26 and 07:30 UTC).** `lu2025safedelta` (PMLR v267 lu25g; arXiv:2505.12038 by the arXiv API, "ICML 2025 Camera Ready"); `huang2025antidote` (PMLR v267 huang25b; arXiv:2408.09600, "Accepted by ICML25"); `huang2024survey` (arXiv API, arXiv:2409.18169; the arXiv comment says accepted by ACM Computing Surveys); `olmo2025olmo3` (arXiv API, arXiv:2512.13961, "Olmo 3"; the model id allenai/Olmo-3-7B-Instruct, tagged arxiv:2512.13961, checked on the Hugging Face model API). The OpenReview API search record of Qi et al. (2025) was read for its abstract (prefilling attacks; deepening alignment).
- **Pipelines.** Revision 6's scripts were reused from `rev6/` in a new scratch folder `rev6b/`: `body_pages.py` and `notes_md_to_tex.py` unchanged (the converter reproduces revision 6's Appendix B byte for byte from revision 6's notes); `build.sh` builds `build/main.tex` in place (the body was edited directly, not assembled);

`paper_to_pdf.py` with local KaTeX, its banner set to revision 6b and its highlight list extended to [TO BE FIXED …] and [PLANNED …] markers (revision 6's Markdown PDF did not highlight them). A consistency check converted the LaTeX body and Appendix C back to Markdown with revision 6's `tex2md.py` and `struct2md.py` and compared every paragraph with this file: all text paragraphs match.

- **Also fixed.** Appendix C.3's paragraph heading "Secondary and scored-only arms [TO BE FINALISED]" had a stray brace and a misplaced bold marker in this file (a revision-6 conversion glitch); fixed. The LaTeX header comment for revision 6 said "7.94 pages"; it now says 7.96, the measured value.
- **Anonymity.** The body, Appendices A and C and the four new BibTeX entries hold no author name, institution, repository, branch, account, commit hash or grant; the PDF metadata says "Anonymous Authors". These notes (Appendix B) are not anonymous and must be deleted before submission.
- **Not changed (outside this revision's scope; for the lead).** The abstract's "six pre-registered repairs" (review A W1; decision 45); §4.6's "about 10–12 pp" against Appendix C.1's "about 9, 10 or 12 pp at low, central or high planning noise" (review A W15; risk 48); every weakness listed "not in the body" in risks 34–60.

## What changed in revision 6 (16 Oct, from 21:10 UTC)

- **Sources.** The lead's brief for revision 6, with two additions from the coordinator during the work (recipe G; the literature sweep). FOCUS.md on `origin/stageb-costs-and-recovery-safeguards-20261003` (fetched; head 416b780), every entry from 9 Oct 17:00 UTC to the newest (16 Oct 21:05 UTC, the ICML 2027 goal); the Gate B/C pre-registration, revision 5, frozen 10 Oct 11:16 UTC (SHA-256 5c97cce7…, checked on the fetched file), §1–§5, §7, §8 and §10; the GI handoff STATUS of 16 Oct 13:20 UTC; the kill-test design §4.4 (Gate C verdicts); the side paper's abstract and outline (for the overlap rule only); `writing/paper/2026-10-17-lit-sweep.md` §0, §4 (corrections) and §5; the head of `writing/paper/2026-10-17-icml-replan-PROPOSAL.md` (status only). Details in "Sources read (revision 6)".
- **Venue (FOCUS 16 Oct 21:05 UTC).** ICML 2027 is the goal; TMLR or NeurIPS 2027 stays the fallback until the user names one. The ICML 2027 deadlines are ESTIMATED (about 16 Jan 2027 abstract, 22 Jan 2027 paper, AoE; a third-party tracker); the official call is not published (sweep §6). Decision 6 and the LaTeX header say so; the ICML 2026 style stays, with the swap-to-2027 note.
- **§4.1, families and scale.** The ladder is now design: OLMo-2-0425-1B and Llama-3.2-1B at 1B; OLMo-2-1124-7B-Instruct and Llama-3.1-8B-Instruct at 7–8B with, after the coordinator's addition 3, "the full comparison runs on OLMo-2-1124-7B-Instruct and Llama-3.1-8B-Instruct … [PLANNED; funding-gated, fallback one family]" (the brief's "[TO BE DECIDED: … ICML re-plan proposal pending]" was used first and is replaced); a headline check "at the largest single-GPU size" [PLACEHOLDER: if resources allow] (the 8 Oct 12:53 UTC ladder names OLMo-2-32B and, if funded, Llama-3.x-70B; the body no longer names sizes). Qwen leaves the study: "a same-recipe comparison of OLMo-2, Llama-3 and Qwen2.5 is reported in [CITE: concurrent submission, anonymized]".
- **§4.1, rungs.** "Llama-3.2-1B's provisional rung is …" became "The rungs are $10^{-4}$ for OLMo-2 (1B and 7B) and $5 \times 10^{-5}$ for Llama (3.2-1B and 3.1-8B)", with one disclosure: "Both Llama rungs were accepted on a too-close bar (one seed's answers-end bar at 1B; the window's upper edge at 8B); §5 reports both." Sources: FOCUS 10 Oct 09:40 UTC (Llama-3.2-1B Gate A PASS under decision 25), 10 Oct 21:30 UTC (OLMo-2-7B rung 1, partial) and 11 Oct 03:53 UTC (Llama-3.1-8B rung 2, accepted "at the edge"). Numbers are in the development evidence.
- **§4.1, the family × recipe sentence (overlap rule).** "During development no rung of a one-recipe ladder left both families partly eroded (§5), because families respond to the same change of recipe by different amounts" became "Families respond differently to the same recipe [CITE: concurrent submission, anonymized]; we fixed this rule after reading a one-recipe development ladder, on which no rung left both 1B families partly eroded, and Llama's first probe at recipe B". The disclosure of when the rule was fixed stays; the finding is cited.
- **Recipe G is a reference, not a fix (coordinator's addition 1; FOCUS 11 Oct 10:38 and 12:54 UTC; GI handoff 15 Oct).** FOCUS records that recipe G lowered erosion for OLMo-2-1B but not for OLMo-2-7B or Llama-3.1-8B, and that "the paper must not present recipe G as a general fix". Edits: §1 "recipe G (a recipe-

only fix whose warmup and cosine decay halve the mean learning rate)" → "recipe G (a standard warmup and cosine schedule)"; §4.1 "*Recipe G*, the recipe-only fix and a reference" → "*Recipe G*, a reference arm that changes only the recipe", plus "It is not a general fix: its effect on erosion differs across families and sizes [CITE: concurrent submission, anonymized]"; §4.4 "**Recipe G** (recipe only)" → "(a reference that changes only the recipe)"; §6 "we report it as a recipe-only fix" → "it is a reference, not a general fix"; Table 1 no longer lists the references (below). The abstract never named recipe G ("a gentler learning-rate schedule" in revision 5e is gone with the abstract's list of references); Figure 1 and its caption never named it. The 7–8B recipe-G numbers are side work: they are only in the development evidence, labelled "not this paper's result". In these notes, decision 15 and risks 3, 21, 24 and 26 carry a revision-6 line; older history entries keep their words.

- **The user's decision of 17 Oct (coordinator's addition 3): "keep the more ambitious one always but always open to fallbacks".** (1) The title keeps "Compute-Matched" (decision 26). (2) §3.2 gains one design sentence: "Removals are matched in training tokens as well as rows, or a token-matched analysis is pre-registered beside the row-matched one [TO BE FIXED in the stage-1 pre-registration]"; the lead rules on which. (3) The 7–8B tier is the full comparison on both families, marked [PLANNED; funding-gated, fallback one family], in §4.1, §4.8 ("the full 7–8B comparison of §4.1") and §6 ("at 7–8B it is planned on both families, funding-gated, with one family as the fallback"): the re-plan's option B, with option C as the fallback (decision 28). (4) The opening-hold loss and Safety at One Shot are listed as candidate additional reference arms in these notes only (decision 27); the body's five references are unchanged. To hold the length, §4.2's "Banks" item joined the U item (same words).
- **Gate B status (§4.6).** "These rules summarize its pre-registration, revision 5 [NOT YET FIXED]" became "Its pre-registration was frozen before any Gate B run; Appendix C.1 gives its rules"; a new last sentence: "**Status:** the noise gate has passed, and Gate C's rankings were sealed before the category runs." No number.
- **Gate C consequence (§4.6; Appendix C.2).** New, neutral and pre-registered: "In the sealed rankings both diagnostics pick the same category, and the format rule another. So, by the frozen rules, Gate C can read only AGREE, NONE_FINDS or TOO CLOSE, or is not read; ATTR_BETTER and REP_BETTER cannot occur; and the paper may say only that neither pick is the format pick, which does not show that the picks are free of format." The brief asked for the ATTR_BETTER/REP_BETTER clause. The format clause is added because the frozen pre-registration (§5, "Form pick") says the two equalities "are reported whatever Gate B reads" and fixes this wording when neither equality holds (decision 29). Checked against the pre-registration's order (NONE_FINDS, AGREE, ATTR_BETTER, REP_BETTER, TOO CLOSE), AGREE's definition (pick_A = pick_R; kill-test design §4.4) and the pick-arm G1 guard: with equal picks $\Delta = 0$, so ATTR_BETTER ($\Delta \geq$ M_C) and REP_BETTER ($\Delta \leq -$M_C) cannot hold, and TOO CLOSE can come only from the guard. Appendix C.2 also notes that wave 2 then holds m = 2 or 3 units. The body names no category; the development evidence does.
- **D5 = (a) (FOCUS 16 Oct 12:15 UTC).** P1–P6 stay primary and the selector stays secondary and model-based (unchanged). New: §6 "Selectors are not run against each other at equal budgets here; a later study with its own pre-registration would do that" (and one line in Appendix C.7). The one-example safety patch is bounded in text: §2 cites Safety at One Shot (Zhang et al., 2026a; new BibTeX) as a patch that "acts after fine-tuning and is not among our references (§6)", and §6 says "no claim here covers it".
- **§4.9, a timing disclosure (FOCUS 16 Oct 00:25 and 13:21 UTC; stage-1 brief rule T2).** The stage-1 pre-registration "lists every Gate B value seen before it is frozen (the noise-gate reading and the sealed rankings)".
- **Overlap rule (the side paper, ARR #13386, "Same Recipe, Different Fate").** Its abstract covers the same-recipe family contrast (OLMo-2, Llama-3, Qwen2.5 at 1B and 7–8B), recipe G against recipe B, the 7–8B replication, the half learning rate, the rank-16 and token-mean-only runs, the two-layer opening account and the blind read. The main paper no longer presents any of these as its own result: §4.1 (two places), §4.10 item 2 ("Recipe and family contrasts are not results of this paper … reported in [CITE: concurrent submission, anonymized]"), §6 ("a rank-16 check is in [CITE …]"), §5's placeholder (the same-recipe contrast, the ladder, the rungs, the token-mean-only run and the rank-16 check are gone from it), Appendix C.8 ("across families and recipes, the two layers are reported in [CITE …]") and Appendix A ("The same read is also reported in [CITE …]"; decision 30). The abstract and the contributions made no such claim; contribution 3 now says

"how much of each repair's effect runs through refusal openings" (repair arms, not families). The citation is a placeholder: ICML lets authors cite their own unpublished work only anonymized, with an anonymized copy as supplementary material (ICML 2026 instructions; decision 32).

- **Literature sweep, factual fixes (coordinator's addition 2; sweep §4 "Corrections" and §5).** (1) DualSelect: §1 now says it "compares data selectors and a safety-instruction mix over three seeds, with a false-refusal rate and its selection overhead reported, but on training data that include red-teaming pairs, and without model-level references or equivalence-bounded verdicts"; §2's removal paragraph drops its DualSelect sentence; "How we differ" says "DualSelect compares selectors and a safety-data mix over seeds, with a false-refusal rate and selection cost". The facts are FAQ's check of the arXiv full text (SafeInstr, 2–3% safety-instruction tokens, is a baseline; selection overhead 1.34× standard SFT at a 90%-of-target-tokens budget; three seeds, mean ± std; a false-refusal rate). The sweep's "six selectors" and "red-teaming pairs" come from the sweep, not re-read here; the body gives no count. (2) DataRx: §2 adds "Its baselines, all additions, include self-distillation, a precedent for our self-generated arm" (the sweep's full-text reading; the coordinator confirmed). No sentence claims the self-generated arm is new. (3) DataShield: the body cites only Zhang et al. (2026c, arXiv:2606.00160); the BibTeX comment records that arXiv:2607.15081 (Wu et al.) is a different paper with the same name (checked with the arXiv API). (4) GradShield: the NeurIPS 2026 accepted-papers JSON lists it as a poster (same seven authors); the entry keeps its arXiv form and adds "NeurIPS 2026 (poster)" as a note (CITE-CHECK 8). (5) "How we differ" is rewritten after the sweep's §5 text, shortened: it adds SafeTune and SafeTuneBed, DualSelect and DataRx to what we do not claim first, and the combination now says "for purely benign data", "limits fixed in advance" and "every arm's full cost, scoring included". Not applied (decisions for the user and lead): the title word (decision 26; since resolved by the user's rule of 17 Oct: kept); an opening-hold or one-shot reference (decision 27; candidates in these notes only); the sweep's other added sentences and design actions (risk 32).
- **New citations (each fetched programmatically).** Safety at One Shot (`zhang2026oneshot`; OpenReview note EyH8Fu3vtZ, ICLR 2026 poster, BibTeX from the API search record; arXiv:2601.01887 checked by the arXiv API; the second author is "Tony He" on OpenReview and "Lipeng He" on arXiv); SABA (`anonymous2026saba`; OpenReview Jo6U0y2omr, an ICLR 2027 submission under review, cited as Anonymous; claim from its abstract in the API record): §2 "A concurrent submission finds risk rankings taken at initialization nearly uncorrelated with those after two epochs, and re-scores rows during training"; SafeTuneBed (`hossain2025safetunebed`; arXiv:2506.00676 by the arXiv API; an IJCAI 2025 workshop version with another title exists on OpenReview). CITE-CHECK 27–29.
- **Trim to 8 pages: what moved to Appendix C (nothing dropped).** C.1 Gate B (arms and waves, contrasts and noise, the noise gate, the verdict rule and constants, guards G1–G4 and direction checks, the extension, the length control, operating characteristics, the recorded prior, the blind-read sentence from §4.6, status); C.2 Gate C (diagnostics, picks, rule and order, the format reading, reports, the consequence of the sealed rankings); C.3 stage-1 pools and anchors, seeds and power (the blinded rule), secondary and scored-only arms, the certified evaluation script and "the implementations of Safe LoRA and B+Qi are committed before stage 1 is frozen" (from §4.4); C.4 the ARC rule and its disclosure (from §4.2); C.5 the positive control's definitions (old §3.4) and design (old §4.5); C.6 the cost conventions (from §4.7); C.7 the value of diagnosis in full (old §3.5); C.8 the opening decomposition (old §3.6) and the first-token margin's formula (from §4.10). The body keeps a summary of each, with its claims.
- **Trim: what was shortened in the body.** The abstract (311 → 239 words, placeholders counted; the list of the five references and "of the same size" moved to §1 and §4); §1's fourth and fifth paragraphs and the contributions; §2's DataRx numbers moved from §1 to §2 (§1 keeps the direction of the result); §2's two short paragraphs merged ("Prediction and decisions under a budget"); §3.2's label list set as a sentence and the adapter-scaling display inline; §3.4 (the value of diagnosis) cut to its definitions; Table 1 lists the eight stage-1 arms only (the five references and their seeds are in §4.4, where they were already listed); §4.3's three paragraphs merged; §4.5, §4.7, §4.8 and §4.10 shortened; §5's placeholder shortened; §6's "Cost" merged into "Baselines, doses and cost". Section numbers changed: old §3.5 is now §3.4; old §3.4 and §3.6 are in Appendix C.

- **Impact Statement (new; ICML requires it; it does not count toward the page limit).** 169 words, after §7 and before the references: the aim; three risks (the data and repair findings and the planted rows could help an attacker; the judges' labels are not a safety guarantee, with the blind-read caveat; small models and one dataset); and [TO BE DECIDED: what is released] (decision 31).
- **Anonymity checks.** The body, Appendix A, Appendix C and the bibliography hold no author name, institution, repository, branch, account, commit hash, grant or pod name; the self-reference to the side paper is the anonymized placeholder; the PDF metadata says "Anonymous Authors" (checked with `pdfinfo`; the subject line still says "International Conference on Machine Learning 2026", from the 2026 style). Appendix A's "One reader, an author who helped design the study" names no one. These notes (Appendix B) are not anonymous and must be deleted before submission.
- **Length.** Before: 9.36 pages (revision 5e; same measure, re-measured on the revision-5e PDF with the re-created script). After: 7.96 pages to the Impact Statement heading (ICML's measure: the statement is outside the limit) and 8.15 pages to the References heading (the old measure, which now includes the 0.19-page Impact Statement). §5 and §7 are still placeholders; results will need about a page (risk 28).
- **Pipelines.** The scratchpad scripts named in the brief (`paper_to_pdf.py`, `scripts/body_pages.py`, `scripts/notes_md_to_tex_rev5c.py`, `scripts/wrap_tex.py`, `rev5d/splice.py`) no longer existed (the scratchpad's files were cleared; only empty folders were left). They were re-created in `rev6/scripts/`: the body measure reproduces revision 5e's 9.36 pages on its PDF; the notes converter follows revision 5e's Appendix B conventions (conflict 41).
- **Editor's notes.** Decisions 6, 7, 13, 15, 17, 18 and 25 updated (7, 18 and 25 RESOLVED); decisions 26–34 added (26 and 28 RESOLVED by the user's rule of 17 Oct). Risks 2, 3, 17, 19, 21, 22, 24 and 26 updated; risks 28–33 added. "Where the framing is weak" item 6 added. Placeholders, fill-ins, CITE-CHECK 7, 8 and 21, development evidence and status brought up to FOCUS 16 Oct 21:05 UTC; CITE-CHECK 27–29 and conflicts 41–52 added. The LaTeX Appendix B is regenerated from these notes.

## What changed in revision 5e (9 Oct, from 17:14 UTC)

*History: revision 6 drops "provisional", because the user recorded Llama-3.2-1B's Gate A as PASS under decision 25 (FOCUS 10 Oct 09:40 UTC).*

- **Source.** The lead's brief for revision 5e. FOCUS.md read to its newest entry, "Granite Mac-versus-pod PARITY: PASS, 2026-10-09 17:22 UTC" (commit 2837eaf, committed while this revision was written; no later entry exists at 17:31 UTC), with the entries of 16:59 UTC (Llama's rung-2 read, commit 7a8eb08), 16:48 UTC (the Granite judge on RunPod and its parity rule), 15:31 UTC (the rung-2 pre-registration) and 15:29 UTC (the USD 10/h pod price limit), and the frozen amendment to the family-probe bars (7 Oct 21:35 UTC). The brief gave the Granite parity result as pending; FOCUS 17:22 UTC records PASS, and FOCUS wins (conflict 40).
- **§4.1, "provisional".** "Llama-3.2-1B's rung is" became "Llama-3.2-1B's provisional rung is", because the user has not decided (decision 25). Conflict 37 is resolved.
- **§4.1, the too-close rule.** "(§4.6; a too-close probe gets one more seed)" became "(§4.6; a probe too close on $E$ gets one more seed; any other too-close bar goes to the user)", the wording proposed in conflict 39. It matches the frozen amendment (7 Oct 21:35 UTC): only the E class's edge band leads to one more seed; a TOO CLOSE on another bar (short answers, ARC, unscored answers) goes to the user; and "any TOO CLOSE goes to the user with the numbers". For length, the body leaves out two details of the amendment: the extra seed runs only after the user's go, and a two-seed mean that is still too close goes to the user. Conflict 39 is resolved.
- **Length.** The body did not grow: the main body measures 9.36 pages, as in revision 5d (same measure; the References heading sits at the same height on page 10, and every heading checked from §3.3 on, §4.2 and later included, is where it was). The two changes added one line to §4.1's last paragraph; alone, they pushed the body to 9.40 pages. Three trims in the same paragraph took the line back, with no change of meaning: "whose erosion probe reads partial" became "whose probe reads partial" (the class, 20–55 pp, says what the probe reads); "under the frozen one-seed versions of the Gate A bars" became "under the frozen one-seed Gate A bars"; "the second family's first probe at recipe B" became "Llama's first probe at recipe B" (§3.5 and §4.9 also

say "Llama"). §4.1's "Data" keeps revision 5d's trim ("sealed per tokenizer"). No layout command was added. One cosmetic cost: in the LaTeX build, "(20–55 pp)" now breaks between "20–55" and "pp". A tie there pushed the body again. Of the other trim sets tried in scratch builds, the only one that holds the length without the break changes "respond to the same change of recipe by different amounts" to "respond differently to the same change of recipe", which loses "by different amounts". So the break stays.

- **Editor's notes.** Risk 17 gives no balance figure, because the balance changes daily. The status entry drops its 12:18 UTC balance and "no pod running", and adds the USD 10/h pod price limit (FOCUS 15:29 UTC), the Granite judge on RunPod with its pre-set parity rule (FOCUS 16:48 UTC) and the parity PASS (FOCUS 17:22 UTC). Decision 7 notes that the USD 10/h limit replaces "no hourly limit" in [D20]. Decisions 18 and 25 quote "provisional rung". Conflicts 37 and 39 are resolved; conflict 40 is added. The LaTeX Appendix B is regenerated from these notes with revision 5c's converter; Appendix A is unchanged.

## What changed in revision 5d (9 Oct, from 17:03 UTC)

*History: revision 5e calls the rung "provisional" and states the too-close rule exactly (§4.1).*

- **Source.** FOCUS "Llama-3.2-1B rung 2 at the main recipe (recipe B at LR 5e-5), seed 1101, read 2026-10-09 16:59 UTC" (commit 7a8eb08); no later FOCUS entry exists at 17:03 UTC. Also read: its pre-registration (15:31 UTC) and the frozen amendment to the family-probe bars (7 Oct 21:35 UTC: the short-answers rule and "a probe qualifies only if every bar passes with no TOO CLOSE").
- **§4.1, provisionally filled.** The placeholder for Llama-3.2-1B's rung became "Llama-3.2-1B's rung is $5 \times 10^{-5}$ (one probe seed; development evidence; to be confirmed by stage 2's Gate A)." In the LaTeX build its placeholder highlight is removed. The user has not decided whether to accept the rung (decision 25); if the user declines, the sentence reverts to a placeholder (decision 18). The only other body change holds the length (below).
- **Editor's notes.** New development-evidence entry for the rung-2 probe; decision 25 added (accept rung 2 given the answers-end TOO CLOSE, with the lead's recommendation); decision 18 provisionally filled, pending the user; the §4.1 row of the placeholder table renamed; "What this is", the status entry and the sources updated; conflicts 37–39 added. The LaTeX Appendix B is regenerated from these notes with revision 5c's converter; Appendix A is unchanged.
- **Length.** The body did not grow: the main body measures 9.36 pages, as in revision 5c (same measure; the References heading sits at the same height on page 10, and every heading from §4.2 on is where it was). The new sentence adds one line to §4.1's last paragraph; alone, that pushed §4.2's last item ("Banks") to page 7 and the body to 9.40 pages. To take the line back, §4.1's "Data" paragraph, whose last line held only "ing rate].", drops two words: "prepared and sealed per tokenizer" became "sealed per tokenizer". No layout command was added: in a scratch build, asking TeX to set each paragraph of pages 5–6 one line shorter (its looseness setting) did not shorten the body.

## What changed in revision 5c (9 Oct, from 15:29 UTC)

- **Source.** FOCUS "USER DECISIONS, 2026-10-09 15:23 UTC", items 1 and 3 (conflict 36).
- **The *worse* label says "adverse" (decision 23a).** "Significantly harmful" became "significantly adverse" in §1 and §3.2; §3.2's "harmful direction" and "harmful repair" became "adverse direction" and "adverse repair"; §4.3's P4 "a significant harmful result" became "a significant adverse result". The abstract names the label but never defined it, so it had no such phrase (conflict 35). In these notes: the *worse* fill-in and decisions 2 and 12. "Significantly" stays, because the label needs a Holm-adjusted test (conflict 33).
- **The abstract names the measures (decision 23b).** New sentence: "Lost safety is measured as refusal erosion as Qwen3Guard labels it and as harmful compliance as the HarmBench classifier defines it." The title keeps "safety". To hold the length, two abstract phrases were shortened: "but rarely compared on equal terms, with training seeds, side effects and compute counted" and "set the bar for a data repair".

- **The caveat sits once in §4.2 and once in §6 (decision 23c).** §4.2's claim rule no longer asks every harmful-compliance sentence to carry the caveat; it states it once, as a fact about the measures: "A blind human read during development rated about a third of judge-unsafe answers as usable help (Appendix A)." §6 keeps its caveat and adds "Appendix A"; §4.6's pointer for Gate B's read now goes to Appendix A. No other body sentence carries the caveat, and harm sentences still say "harmful compliance as the HarmBench classifier defines it". In these notes the "Harm wording" fill-in, risks 1 and 24 and the two HarmBench entries of the development evidence no longer attach the caveat to each sentence.
- **Appendix A, "Blind human read (development evidence)" (decision 23d).** New and part of the paper: after the references and before these notes (in the LaTeX build, `\appendix` makes it Appendix A and these notes Appendix B, on a new page). It holds the design (pre-registered; ratings sealed before the key was opened; 50 answers; blinding; the 0/1/2 rubric), the counts (40 Unsafe-labelled answers drawn, 39 rated, 1 skipped; 34 of 39 went along, 0.87 [0.73, 0.96]; 13 of 39 gave usable help, 0.33 [0.19, 0.50]; 0 of 10 Safe controls missed; 4 of the 12 OLMo answers that HarmBench labels "yes" rated usable help, 0 of 3 "no"), the reading (labelled ours) and the limits (one reader, who helped design the study; no second rater; groups of 6 to 12, not compared; a sample different from the audit's proposal). Group counts stay in these notes. Sources: the Side Project's `research/side-blind-read/PREREG-2026-10-08.md` and `blind-read-scores-2026-10-09.json` at commits 4d63e04 and 57d08dd, read with `git show`; the Gate A audit §8 item 4. The body gives no counts.
- **ARC rule's scope (decision 16; FOCUS 15:23 UTC, item 1).** §4.2's [TO BE FINALISED: its scope under the per-family rule] is replaced by the scope: first for probe and Gate A readings at one common recipe, then extended to every family's Gate A read at its chosen learning rate, before any such read. The loosening sentence and the disclosure stay. The §4.2 row of the placeholder table is removed.
- **Length.** The body did not grow: the main body measures 9.36 pages, as in revision 5b (same measure; the References heading sits at the same height on page 10, and every heading from §4.3 on is where it was). The abstract grew from 296 to 311 words (placeholders counted) and moved §1–§4.2 down by up to about four lines; §4.2's shorter caveat sentence and page 6's column slack took that up. Appendix A has 379 words, about half a page of the LaTeX build (page 14); these notes start on page 15.
- **Editor's notes.** Decisions 16 and 23 RESOLVED; decision 23's open item on a second rater becomes decision 24; decisions 2 and 12 say "adverse". Risks 1, 23 and 24 updated. Conflicts 23, 28 and 31 updated; conflicts 32–36 added. Development evidence (the blind-read and HarmBench entries, the ARC rule, the status) and sources brought up to FOCUS 15:23 UTC. The LaTeX appendix is regenerated from these notes.

## What changed in revision 5b (9 Oct, from 15:06 UTC)

*History: revision 5c states the blind-read caveat once in §4.2 and once in §6 instead of in every harm sentence, and moves the read's counts to Appendix A.*

- **The blind human read (FOCUS 9 Oct 15:05 UTC; the Side Project's pre-registration and scores, commits 4d63e04 and 57d08dd, read with `git show`).** Its numbers are only in the development evidence (new entry, labelled), never in the abstract or the contributions. The body carries only the rounded caveat that the wording rule requires: §4.2 (the claim rule) and §6 ("most judge-unsafe answers went along with the request, but only about a third gave usable help"; one reader; "during development").
- **Tighter judge wording (FOCUS 15:05 UTC; the lead's proposal, applied unless the user objects; decision 23).**
- **U.** Where the body defines U or says what it measures, a rise in U is "refusal erosion as Qwen3Guard labels it": §3.1 (the erosion E), §4.2 (U) and §6.
- **Harm.** §4.2's claim rule keeps its two conditions (the HarmBench classifier's contrast has the same sign and at least half the size; its erosion under that recipe is at least 20 pp). It now allows only the wording "harmful compliance as the HarmBench classifier defines it", never harm alone, and every such sentence carries the blind-read caveat. §6 says the rule "allows at most" that wording.
- **Harm sentences reworded: 6.** §1, the side study ("the harmfulness of its content" → "what the answer says"); §4.2, the claim rule's two harm sentences (merged into one rule); §4.10 item 1's title ("harmful content" →

"answer content"); §6, "whether its answer carries out harm" (replaced by the read's evidence) and "reduced harm". One ambiguous phrase was also fixed: §3.6 "sampled harm answers" → "sampled answers to harm prompts".

- **Searched and left, because they make no harm claim:** dataset, benchmark and category names (HarmBench, the harm bank, harm prompts, "harmful requests", "harmful anchors" and "benign prompts that resemble harmful ones", the hard-category list, PKU-SafeRLHF's and the plant's source labels); prior work's own measures (§1, §2); "safety" as the field's name for the outcome and in reference names (the title, "restores the most safety", Gate B's question, the figure caption, $\Lambda_m$'s gloss, "safety prompt", Safe LoRA, "safety data"); and the three sentences of the *worse* label that use "harmful" for the adverse direction of a contrast (§1, §3.2, §4.3 P4; the user's wording of 8 Oct 18:31 UTC). Decision 23 lists these for the user.
- **Gate B (§4.6).** One sentence: the verdict reads U alone; the blind human read recommended before the first wave (DRAFT §12 item 5) is done, on a different sample from the one first proposed (the audit's §8 item 4: 40 Unsafe e1 answers in two strata plus the 29 over-refusal-bank Unsafe answers).
- **Length.** The body did not grow: the main body measures 9.36 pages against 9.40 for revisions 5 and 5a (same measure). To make room: §1's section roadmap was cut (its points repeat §1's paragraphs before it); three sentences that repeat earlier text were cut (§4.1's seed clause, given in §3.1; §4.10 item 1's "It locates a change …", given in §3.6; §6's "two cost conventions" sentence, given in §4.7); and §6's new caveat replaces the sentence it qualifies. In the LaTeX build only, Figure 1's caption note became "(Placeholder: not yet drawn.)".
- **Editor's notes.** Decision 23 added; decision 22's "WITHOUT" set as *without* (as in the LaTeX). Risks 1, 5 and 24 updated. The "Harm wording" fill-in, the two HarmBench entries of the development evidence and the status updated; the blind-read entry added. Conflicts 28–31 added. The LaTeX appendix is regenerated from these notes, so its wording is identical; this also brings in the placeholder-table wording of revision 5a (commit ec40d6a: "OLMo-2-1B's is 1e-4"), which the LaTeX appendix lacked.

## What changed in revision 5 (9 Oct, from 13:17 UTC)

*History: revision 5b tightens §4.2's claim-rule wording and §6.*

- **Recipe B is the main recipe again (FOCUS 09:29 UTC, decision 1).** Gate A, Gate B and stage 1 use recipe B (LoRA r64, constant LR 1e-4, no warmup, 128 rows per step, per-row loss, one epoch); OLMo's recipe-B Gate A (PASS, 8 Oct 05:40 UTC) is the Gate A of record. Revisions 3–4's "recipe G main" is undone in the abstract, §1, §4.1, Table 1, §4.3, §4.6 and §6. The abstract and §1 no longer call the main recipe standard practice ("a fixed LoRA recipe that erodes it partly"; "recipe B: a constant learning rate, one epoch"); §6 calls recipe B "plain rather than standard" and says why it was chosen (from probe and Gate A readings, before any repair outcome).
- **Recipe G is the "recipe only" fix and a reference (FOCUS 09:29 and 10:01 UTC).** The body gives its design facts (a 4-step warmup and cosine decay, which halve the mean learning rate, and a token-mean loss) and calls it "the recipe-only fix". Revision 4's token-mean-only finding (the schedule, not the loss weighting, carries G's reduction on OLMo; one seed) stays development evidence; it is why these notes call recipe G a dose-lowering fix. The body does not say that recipe G lowers erosion in every family (conflict 25). "Stronger dose", "stronger-dose comparison" and "lower-dose base" are gone.
- **The per-family learning-rate rule replaces the common-recipe ladder (FOCUS 09:29 UTC, decision 2).** §4.1's "One common recipe" paragraph and [PLACEHOLDER: the common rung] are replaced by "A matched erosion dose": recipe B with only the learning rate changed, the highest rung of (1e-4, 5e-5, 2e-5) whose probe reads partial (20–55 pp) under the frozen probe bars, chosen from probes only, never from a repair result; cross-family claims "at a matched erosion dose". The development ladder (FOCUS 01:43 UTC) is cited, without numbers, as the evidence that one recipe cannot equalize the erosion, and the rule is disclosed as fixed after that ladder was read (§4.1, §6). New marker: [PLACEHOLDER: each family's rung]. Qwen is outside the repair study (§4.1); its recipe-B runs enter only the same-recipe family contrast (§4.10). The contributions, §4.6 (Gate A), §4.8 and §6 say "matched dose".

- **Five mandatory references (FOCUS 10:01 and 13:16 UTC).** The safety prompt; adapter scaling (× 0.75, × 0.5); recipe G (3 seeds); B+Qi (Qi et al.'s constrained SFT on recipe B, 6 seeds; "G+Qi" is renamed everywhere); Safe LoRA (Hsu et al., 2024; 3 seeds; settings [NOT YET FIXED]). Changed: abstract, §1, §2 (one clause), §3.2 (now points to §4.4), Table 1 (two rows added), §4.4 (rewritten). New BibTeX `hsu2024safelora`, from the OpenReview venue note HcifdQZFZV (NeurIPS 2024 poster), with arXiv:2405.16833 checked by the arXiv API (CITE-CHECK 26).
- **Dataset plan (FOCUS 09:40 UTC).** §4.1 "Data": Dolly for Gate B and stage 1; PKU-SafeRLHF for the add repairs' safety rows; Alpaca as the second benign dataset after Gate B, [NOT YET FIXED: its units and learning rate]. §6 adds "one benign dataset until Alpaca runs".
- **The two-layer mechanism account (FOCUS 8 Oct 15:17 and 16:50 UTC; the mechanism note's addenda; conflict 19).** §3.6 adds one sentence: the opening-switch term has two layers, the refusal openings lost ($p_0 - p_1$) and how often a direct answer was already unsafe before fine-tuning ($u_{N0}$). §4.10 item 1 reports both layers across families and recipes; §5's placeholder lists them. Numbers only in the development evidence.
- **§4.10 item 2** is now "Recipe and family contrasts": recipe G against recipe B as a recipe-only fix, the token-mean-only run, the same-recipe family contrast, every learning-rate rung and the rank-16 check under recipe G.
- **HarmBench on the pod and the harm wording (FOCUS 8 Oct 23:43 UTC).** §4.2's last clause is now per recipe ("If the classifier's erosion under a recipe is below 20 pp, no sentence calls that recipe's erosion harm"), so recipe G's erosion is "refusal erosion". Numbers only in the development evidence.
- **ARC rule.** Kept with its disclosure; its scope ("at the common recipe", FOCUS 20:50 UTC) is marked [TO BE FINALISED: its scope under the per-family rule] (decision 16). The sentence "An interval wholly at or below 0.03 passes …" is cut (implied by "resolves a straddling interval").
- **Gate B (§4.6).** Its pre-registration, revision 5, is marked [NOT YET FIXED]; "with the warmup and cosine decay spanning its own epoch" is removed (recipe B has no schedule); the exploratory E-scaled floor is removed, because at recipe B it equals the 8 pp floor (decision 21).
- **Order of work (§4.9).** Gate B runs to its verdict; funding for the later stages is then sought with that result; stage 1 starts only once funded, never because of an outcome. The stage-1 pre-registration is marked [NOT YET FIXED].
- **Secondary arm.** "The length rule under recipe B's per-row loss" reverts to revision 2's "the length rule under a token-mean loss" (decision 11).
- **Length.** The body did not grow: the main body measures 9.40 pages against 9.46 for revision 4 by the same measure (pages before the page with the References heading, plus the share of that page's two columns above it, in the ICML 2026 build). To make room, §4.1's common-recipe paragraph, §4.6's E-scaled floor, §4.10 item 2's dose sentence, §4.2's "as before" sentence and §6's repeat of Gate B's power were cut, §4.3's "Re-implementations" paragraph was merged into §6 "Baselines and doses", and Table 1's B+Qi and Safe LoRA rules were kept to one line each.
- **Editor's notes.** Decisions 14 and 15 resolved; decisions 7, 11 and 13 updated; decisions 16–22 added. Risks 2, 3, 17–19, 21–23 updated; risks 24–27 added. Conflicts 9, 10, 12 and 18 marked superseded; conflicts 19–27 added. CITE-CHECK 25 renamed (B+Qi); 26 added. Placeholders, fill-ins, development evidence and status brought up to FOCUS 13:16 UTC.

## What changed in revision 4 (8 Oct, from 21:40 UTC)

*History: revision 5 reverses this revision's recipe roles and replaces its common-recipe rule.*

- **The loss-weighting account is withdrawn for B → G (FOCUS 21:36 UTC, the token-mean-only run).** The body no longer says or implies that recipe G's token-mean loss, or a first-token loss mechanism, explains why recipe G erodes less than recipe B.
- §1 and §4.1 describe recipe B as the stronger dose: a constant learning rate without warmup, about twice recipe G's mean learning rate over the epoch (FOCUS: G's warmup and cosine decay "also halves the mean learning

rate"), plus a per-row loss. §4.1 drops "which over-weights the first tokens of short answers" and "a mechanism test", and names the token-mean-only run as the run that separates the two changes.

- §4.10, item 1 is now "refusal openings against harmful content": the decomposition is descriptive, and the first-token margin is the one causal test. The recipe contrast is no longer a causal test. Item 2 replaces "four amplifiers" with the two differences (schedule and loss weighting), adds a placeholder for the token-mean-only run's result against its prediction, and adds that families can respond differently to the same change of dose.
- §3.6 says the decomposition is descriptive; "should restore less" becomes "may restore less".
- §5's placeholder list adds the token-mean-only run against its pre-registered prediction, and "the mechanism decomposition and its causal tests" becomes "the opening decomposition and the first-token margin". §6 "Recipe and setting" describes recipe B as a stronger dose and adds the family × dose point and the ARC loosening.
- **G+Qi kept (FOCUS 17:36 UTC, item 4).** §1, §2, §4.4 and Table 1 justify it by Qi et al.'s own objective, which constrains updates on the initial tokens, not by the loss-weighting account; none of that text changed. §4.10 now cites Qi et al. (2025) only for the shallow-alignment observation beside the first-token margin (abstract re-read by the arXiv API between 21:40 and 21:44 UTC; CITE-CHECK 25).
- **ARC rule (FOCUS 20:50 UTC).** §4.2's "Task loss" item becomes "Task loss and skill": the rule in FOCUS's wording, called a loosening, with the disclosure that the first OLMo recipe-G probe was read under the old rule before the rule was proposed. The body gives no numbers for that probe. Revision 3's "ARC … sets no limit" was wrong for the probe and Gate A bars, which check ARC; §4.2 now states the bar. §4.1 and §4.6 point to §4.2 for skill; §6 notes the loosening.
- **Common-recipe rule (FOCUS 17:36, 21:04 and 21:37 UTC; the frozen amendment of 7 Oct, 21:35 UTC).** §4.1 adds the too-close rule: one more seed, the two-seed mean classified with the same band, no further seed in that classification, and a choice made after the outcome reported as such. [PLACEHOLDER: the common rung] stays, because Llama's two-seed reading under recipe G is pending and FOCUS records no step-down.
- **Development numbers** (the token-mean-only run, OLMo recipe G at three seeds, Llama under recipe G, the rank-16 check) are only in the development evidence below, labelled. The abstract and contributions did not change.
- **Editor's notes.** Risks 2, 3 and 18 updated; risks 20–23 added; "Where the framing is weak" item 5 updated; decisions 7 and 11 updated, decisions 13–15 added; placeholders updated; CITE-CHECK 25 extended; conflicts 14–18 added; development evidence and status brought up to FOCUS 21:37 UTC.

## What changed in revision 3 (8 Oct, from 19:35 UTC)

*History: revision 5 reverses this revision's recipe roles (recipe B is main again) and renames G+Qi to B+Qi.*

- **Recipe roles (FOCUS 19:24 UTC; the common-recipe rule, 17:36 UTC).** Recipe G (token-mean loss, 4-step warmup with cosine decay, LR 1e-4, LoRA r64, 128 rows per step, one epoch) is the main recipe for Gate A, Gate B, stage 1 and both families. Recipe B is the stronger-dose comparison; any learning-rate rung that runs is a dose point. Changed: abstract, §1, §3.2, §3.6, §4.1, Table 1, §4.3, §4.4, §4.6, §4.8, §4.10, §5, §6.
- **Llama (FOCUS 16:50 and 17:36 UTC).** "Llama-3.2-1B replicates the comparison" is gone from the abstract, §1, the contributions and §4.8. Llama-3.2-1B is a planned second family under one common recipe, chosen by the pre-registered ladder rule (§4.1). The claim rule "replicates across two families" stays in §4.8 as a rule for when both families have run.
- **Worse label (FOCUS 18:31 UTC).** The labels are better / equivalent / worse / inconclusive (abstract, §1, contributions, §3.2, §4.3). Old decision 2 is resolved.
- **G+Qi reference (FOCUS 17:36 UTC, item 4).** Recipe G plus Qi et al.'s (2025) token-wise constrained SFT, six seeds, a model-level reference with no novelty claim. Recipe G is now the base, so "a standard recipe" leaves the references (FOCUS 17:36 UTC, item 4); the three references are the safety prompt, adapter scaling and G+Qi (abstract, §1, §2, §3.2, Table 1, §4.4). Qi et al. (2025) was already in references.bib (`qi2025shallow`, fetched from OpenReview with arXiv:2406.05946 checked; title and authors re-checked with the arXiv API

on 8 Oct between 19:32 and 19:35 UTC; OpenReview now answers with a bot challenge, so the venue was not re-fetched); no new BibTeX.

- **One secondary arm.** "The length rule under a token-mean loss" contrasts nothing once recipe G (token-mean) is the main recipe. §4.3 now names its mirror, the length rule under recipe B's per-row loss. The mirror is the drafter's inference (decision 11).
- **Editor's notes.** Development evidence and status brought up to FOCUS 19:24 UTC; risks 2 and 3 updated, risks 18–19 added; conflicts 9–13 added.

## What changed in revision 2 (from the draft of 8 Oct, about 12:00 UTC)

- **Headline.** The measured repair comparison leads (six repairs × six seeds against random removal and no repair, labels better / equivalent within 5 pp / inconclusive, with over-refusal, task loss and measured cost). The "when diagnosis pays" map is secondary and model-based; its predictions are written down before Gate B's category runs and tested on Llama.
- **Menu.** Stage 1's primary family P1–P6 (A1, L1, H1 removals; S1, S3, S4 additions); secondary arms and scored-only scorers; rewrite and self-distill only in the DIFFUSE branch. The previous draft's keep/remove/add/rewrite menu by source, the default mix a_mix and the "beat a_mix" rule are gone.
- **New parts.** Three mandatory references with no data change (safety prompt, adapter scaling, recipe G; revision 3 replaces recipe G with G+Qi); the planted-cause positive control; the HarmBench claim rule; two cost conventions; the Llama rule (LR or rank only; revision 3: the common-recipe ladder, LR only, for both families); confirmation on a sealed held-out subsample; the opening decomposition (side study).
- **Resolved old decisions.** O is measured against the untuned model (old decision 1); $\tau_O$ is a grid of 2, 5 and 10 pp and $\tau_L$ is 2 SE of a seed-to-seed difference in held-out loss on all 200 rows (old decision 2); the main scale is 1B with a ladder (old decision 3); a judge rule decides "harm" against "refusal" wording (old decision 4); model-level references are mandatory (old decision 6); side-effect prediction is secondary, on the add arms only (old co-primary).
- **§4 and §6 are written out. §2** adds the published filters, the format paper, DualSelect, Bianchi et al., PTST and MNEME, the DataRx correction and Guo et al.'s Table 2.

## Placeholders

| Where | Marker | Filled by |
|---|---|---|
| Abstract | [PLACEHOLDER: if funded]; [FINDING 1]–[FINDING 3] | the funding decision for the planted sweep; stage 1, the references, Gate B and the sweep |
| §1 | [PLACEHOLDER: if funded]; [TO BE FINALISED] menu sizes 8/32/128 | as above; the plan's review |
| §1, contributions | [FINDING 1]–[FINDING 4]; [ARTIFACT] | stage 1 and Llama; the sweep; Gates B/C and the mechanism study; the decision-value predictions; the anonymized outcome tables |
| §1 | [FIGURE 1] | the repair-comparison figure (restoration with 90% intervals and the ±5 pp band against over-refusal and cost, with the adapter-scaling curve) |
| §3.2 | [PLANNED; the stage-1 pre-registration is not yet frozen], three times: the analysis model HB-C (revision 6d); the feasibility estimator | the stage-1 freeze removes the [PLANNED; the stage-1 pre- |

| Where | Marker | Filled by |
|---|---|---|
| | (revision 6e; LW-06); the closed testing family, rule R1.3 with $\tau^*_O = 5$ pp, the label order (revision 6d) and δ's scaling for another family (revision 6e; UW-13). *Settled earlier:* token matching (revision 6c, decision 26); the analysis model, the baseline of *worse*'s over-refusal test and the label order (revision 6d; they were [TO BE FIXED]); feasibility as a non-inferiority bound (revision 6e; it was [TO BE FIXED], decisions 38, 53) | registration is not yet frozen] markers |
| §3 | the constants δ (5 pp, given), δ_B (8 pp, given), τ_O grid (given), τ_L (rule given; fixed from the three Gate A a0 seeds, stage-1 build L16), B, the selector rules per (M, B), k_d, φ_j and the V level | the stage-1 pre-registration and the selector spec (committed before Gate B's category runs) |
| §4.1 | [PLANNED; funding-gated, fallback one family] (the full 7–8B comparison); [PLANNED; funding-gated; first 7–8B model cut] (Olmo-3-7B-Instruct; revision 6b, as relayed; revision 6e: confirmed by the user, FOCUS 8dac6ad, decision 48; still funding-gated); [PLACEHOLDER: if resources allow] (the headline check); [CITE: concurrent submission, anonymized] (twice); [NOT YET FIXED: Alpaca's units and learning rate] | funding, under option B with option C as the fallback (decision 28); funding; the citation form at submission (decision 32); the Alpaca design note and the user's extension of the learning-rate rule to datasets (decision 20). **Filled in revision 6:** the rungs (OLMo-2 1e-4 at 1B and 7B; Llama 5e-5 at 1B and 8B) |
| §4.2 | [TO BE FIXED in the stage-1 pre-registration], once: StrongREJECT's evaluator and a prefilling probe as secondary endpoints (revision 6b). Revision 6c: the HarmBench input is settled (the 512-token window; Appendix C.10). Revision 6d: the conditional PHTest extension of the over-refusal bank, stated without a marker (the ruling gives none). [PLANNED; the stage-1 pre-registration is not yet frozen], once (revision 6e): an arm that fails the coherence screen is read and flagged (LW-09); revision 6f: the same clause names the flag "fails answers end" (conflict 78). Revision 6g: the PHTest extension is stated as fired, still without a marker; its rule and the 11.04 pp are in Appendix C.11, under its marker (decision 55) | the stage-1 pre-registration (decision 40); the stage-1 freeze |
| §4.3 | [PLACEHOLDER: final MDE] (revision 6d: "The MDE depends on the same-seed correlation", decision 52); [TO BE FINALISED] secondary arms (listed in Appendix C.3); [TO BE FIXED in the stage-1 pre-registration] the two candidate data-only additions (revision 6b; revision 6g: moved to Appendix C.3, with §4.3 pointing there) | Gate B's pooled noise; the plan's review; the stage-1 pre-registration (decision 41) |
| Table 1 | the Actor column (C, P; S4's C conditional) and its legend (revision 6b); revision 6d: seven seeds for $a_{\mathrm{R}}$ and $a_0$ (layout D1), no marker | the lead confirms the assignments (decision 46) |
| §4.4 | [NOT YET FIXED] Safe LoRA's base model, layer rule and threshold; [PLANNED; to be pre-registered] the lower-learning-rate reference; [TO BE FIXED in the stage-1 pre-registration] the post-hoc references' seeds and the test of "beats the references" (both revision 6b); [CITE: concurrent submission, anonymized] (the lower rungs; revision 6b) | the stage-1 pre-registration, after the lead reads the paper in full (decision 19) |
| §4.5, Appendix C.5 | [TO BE FINALISED] menu sizes; [PLACEHOLDER: runs only if funded]; [PLANNED; the stage-1 pre-registration is not yet frozen], once in each (revision 6f): only the training-signal diagnostics count toward finding the cause, and the text-only scorers' ranks are construction checks, published before the pilot (Appendix C.5: also if | the plan's review; funding; the stage-1 freeze |

| Where | Marker | Filled by |
|---|---|---|
| | the pilot stops; LW-10 F2, F8). Contribution 2's sentence on the construction checks carries no marker. Revision 6g: §4.5's sentence gives the reason (conflict 87); Appendix C.5 carries one more such marker, on the Self-Inf-N-style rank's condition and the disjoint-anchor sensitivity with its fallback (conflict 89; risk 44) | |
| §4.7 | [TO BE FINALISED] costs | measured ledgers; frozen cost conventions |
| §4.8 | [PLACEHOLDER: if resources allow] | funding; decision 28 |
| §4.9 | [NOT YET FIXED] the stage-1 pre-registration; [PLANNED; the stage-1 pre-registration is not yet frozen] the closed graphical family Φ (revision 6d; it replaces revision 6b's [TO BE FIXED in the stage-1 pre-registration] on one closed testing family, decision 37) | the stage-1 commitments build (FOCUS 16 Oct 19:47 UTC), committed before wave 1 |
| §4.10 | [PLACEHOLDER: report against the bound on a first-token path]; [CITE: concurrent submission, anonymized] | the bound recorded before Gate B (P3 in the Gate B pre-registration); decision 32 |
| §5, §7 | placeholders | results (decision 33: the page budget) |
| §6 | [CITE: concurrent submission, anonymized], twice (the rank-16 check; since revision 6c, the Qwen runs); revision 6g: the A1 stability sentence carries no marker | decision 32 |
| Impact Statement | [TO BE DECIDED: what is released] | decision 31 |
| Appendix A, Appendix C.8 | [CITE: concurrent submission, anonymized] | decisions 30 and 32 |
| Appendix C | [TO BE DECIDED: anonymized copies as supplementary material]; Appendix C.11: [PLANNED; the stage-1 pre-registration is not yet frozen] (revision 6d; revision 6e adds the equivalence grades, the margin, feasibility and the coherence rule under the same marker; revision 6g adds the over-refusal bank); Appendix C.3: [TO BE FIXED in the stage-1 pre-registration] the two candidate data-only additions (revision 6g, moved from §4.3) | decision 32; the stage-1 freeze; the stage-1 pre-registration (decision 41) |
| References | Microsoft page [CITE-CHECK]; Alpaca has no citation (§4.1, §4.8); GradShield's proceedings entry; the versions of SABA and SafeTuneBed; the survey's ACM Computing Surveys version; the wording for Qi et al.'s augmentation (revision 6b); PHTest's venue (revision 6d) | the author (CITE-CHECK 8, 14, 24, 28, 29, 32, 34, 37) |

**Removed in revision 6:** the §4.6 row ([NOT YET FIXED] Gate B's pre-registration, revision 5): the user froze it on 10 Oct (decision 7).

**[TO BE FINALISED] items (the plan is under review; revision 6 moved their details to Appendix C):** (1) the planted menus' sizes (8, 32, 128 cells) and cell rules; (2) the published-filter list: L1 is LARF-style in the primary family; DataShield-style is secondary; Self-Inf-N-style is scored only; SQSD is cited only; bi-anchoring enters as A1's score; (3) all costs: the FRE values, both cost conventions, the cost of each scorer and of A1, the lean-trial cost $c_0$, and the dollar plan (stage 1 about USD 56–98; planted sweep about USD 17–35; core path about USD 215–405; all estimates from the plan, not paper text); (4) the secondary arms and their order.

**Fill-ins for each outcome** (templates, not claims): - **A repair is better.** F1: "[repair] restores __ pp more than random removal (90% CI a–b), with over-refusal __ pp above the untuned model and task loss within τ_L, at __ FRE; it [does /

does not] lie below the adapter-scaling curve". If it does not, the sentence must say that a model-level reference does as well (FOCUS "do not claim": data repair beats model-level repair). - **A repair is worse.** F1: "[repair] is worse than [comparator]: it [raises the unsafe rate / raises over-refusal / raises task loss] by __ (adjusted p = __ in Φ)". Under the 18:31 UTC rule this label replaced "inconclusive" for any significantly adverse contrast ("adverse" since decision 23a); rule R1.3 narrows it: a rise in over-refusal or task loss gives *worse* only if the repair is also infeasible against θ0 at $\tau_O^*$ (task loss: against $a_0$ at τ_L), and otherwise the row reports the rise and the label comes from the U part (revision 6e; conflict 77). A repair that fails the coherence screen keeps its label, with the flag "fails answers end" (LW-09); P4 (S1 against no repair) is the contrast where prior work predicts an adverse result (DataRx's random mix). - **Everything equivalent.** F1: "every removal repair is equivalent to random removal within ±5 pp; [addition result]". This is a publishable answer; the abstract then says it. - **A free reference wins.** F2: "a safety prompt / a down-scaled adapter / Safe LoRA / recipe G / B+Qi restores as much as any data repair at lower over-refusal" (plan §13: a valid, reportable answer). - **Planted cause found / not found** (only if funded). F3: "[training-signal diagnostic] ranks the planted cell first at M = __ ; runs alone find it at __ FRE". Headline item (2) asks whether each diagnostic finds a concentrated, harmful-prompt cause (UW-15; revision 6e), so the sentence names the cause that way, not "benign-looking" (decision 54). **Revision 6f (LW-10 F2):** only A1 and the Self-Inf-N-style score can fill the bracket; a text-only scorer's rank is reported as a construction check, published before the pilot, never as a find. **Revision 6g (the lead's rulings 3 and 4):** if the Self-Inf-N-style scorer is not committed before Gate B's category runs, only A1 can; A1's find is its rank under the PKU-SafeRLHF anchors, with its rank under the disjoint-source anchors beside it as a sensitivity. - **DIFFUSE.** F3: "no Dolly category restores more than 8 pp beyond random removal (80% detection at about 10–12 pp)". Switch to alternative title 3; the planted sweep becomes the only test of whether diagnosis can pay. - **Second family.** Llama-3.2-1B's Gate A passed at its rung under decision 25 (FOCUS 10 Oct 09:40 UTC); stage 2 still needs funding. If stage 2 runs: "[contrast] replicates on Llama-3.2-1B at a matched erosion dose (same sign; family difference below the interaction's MDE of __ pp)". If no learning-rate rung of recipe B reads partial for Llama, or stage 2 is not funded, the paper is single-family: drop the second family from the abstract and contributions, and say so in §6. - **Gate C after the sealed rankings (revision 6).** Equal picks: if Gate B reads LOCALIZED, F3 is one of "Gate C reads AGREE: both diagnostics pick [category], which restores __ pp more than random removal", "Gate C reads NONE_FINDS: the shared pick does not beat random removal", or "Gate C reads TOO CLOSE: a pick arm failed the answers-end guard"; with any Gate B verdict, add the pre-registered form sentence: "Neither pick is the form pick; that does not show that the picks are free of form." If Gate B is not LOCALIZED: "Gate C is not read." - **Harm wording (revision 5c; decision 23).** Use "restored refusal" unless the claim rule (§4.2) passes for that contrast: the HarmBench classifier's contrast has the same sign and at least half the size, and its erosion under that recipe (E_HB) is at least 20 pp. Then write "harmful compliance as the HarmBench classifier defines it", never harm alone. The sentence carries no caveat of its own: the blind-read caveat sits once in §4.2 and once in §6, with the counts in Appendix A. Recipe B's OLMo erosion clears 20 pp on every seed and recipe G's does not (development evidence), so recipe G's erosion is only "refusal erosion as Qwen3Guard labels it". **Revision 6:** at 7–8B, recipe B's erosion at each family's rung clears 20 pp on every seed with the same sign and at least half the size (FOCUS 11 Oct 10:40 UTC), so the same wording is allowed there.

## [CITE-CHECK] items

| # | Item | Where | Why |
|---|---|---|---|
| 1 | TOSS (2603.01185) | §2 | Title and venue now fetched ("Token-level Data Selection for Safe LLM Fine-tuning", ICLR 2026). The claim "token-level selection" rests on the title only. |
| 2 | Unforgotten Safety (2512.10150) | §2 | Fetched ("…with Continual Learning"); claim from the title only. |
| 3 | Yin & Rush (2410.16208, ICLR 2025) | §1, §2 | Wording ("almost never compute-optimal"; 5x and 10x model-size ratios) from the arXiv abstract, read this session. The spec's "predicts retraining wins at equal size" is our inference from the 10x ratio; check the paper before using it. |

| # | Item | Where | Why |
|---|---|---|---|
| 4 | TuneAhead (2606.17660) | §2 | Fetched; claim from the title only. |
| 5 | Data2Behavior (2602.04735) | §2 | Fetched; claim from the title only. |
| 6 | SafeTune: model-level repairs, single deterministic runs | §2 | Recipe and runs PDF-checked (benign-erosion recipes note); "compares repairs of safety drift" from the title and a fetch summary. |
| 7 | DataRx mechanism ("gap between the model's own response and a safe reference"); revision 6: "its baselines, all additions, include self-distillation" | §2 | From the arXiv abstract (this session). The numbers 59.23% and 13.70% are in the abstract; 20.24% is Table II (recipes note, PDF read); "averaged over seven tasks" is the abstract's wording. The plan reads (by HTML fetch, not PDF) that DataRx's random arm draws refusal-only rows; that is why S1 is the DataRx test. Revision 6: the baselines (Random, Longest, Paraphrase, Self-Distill, SSS-B, PSS-B; all additions; no seeds or selection cost reported) are the literature sweep's reading of the arXiv full text (sweep §4, "Corrections" 2), confirmed by the coordinator; not re-read here. |
| 8 | DataShield and GradShield report scoring cost without charging it against runs; DataShield's mechanism; revision 6: two papers named DataShield, and GradShield's venue | §2 | Cost claim: fetch summaries only. Mechanism (compliance direction): DataShield's abstract (this session). DataShield's cost table (Table IV) is cited by the plan from the reading list (LARF 0.17 h, DataShield 0.50 h, bi-anchoring 3.5 h per 10,000 rows); not PDF-checked. Revision 6: arXiv:2607.15081 (Wu et al., "DataShield: Uncovering Risky Fine-Tuning Data Across LLMs Through Consensus Subspace Alignment") is a different paper with the same name (arXiv API, 16 Oct); the draft cites only Zhang et al. (arXiv:2606.00160), and its "DataShield-style" filter follows Zhang et al.'s compliance direction; check which paper holds the cost table. GradShield: the NeurIPS 2026 accepted-papers JSON (fetched by the literature sweep) lists it as "Accept (poster)"; the BibTeX keeps the arXiv form with the note "NeurIPS 2026 (poster)"; replace it with the proceedings entry when published. |
| 9 | SOURCE "reported to cost several training runs" | §2 | Reported through the Bergson paper; not checked. |
| 10 | LESS as "gradient similarity on LoRA gradients" | §2 | Reading-list summary. |
| 11 | Persona vectors | §2 | Blog post via the problem-validation review; confirm in 2507.21509. |
| 12 | DataDecide; Ivison et al. | §2 | Positioning review, from an extraction-tool read. |
| 13 | Xiao & Aranguri | §2 | Abstract only ("over 10 times cheaper"). |
| 14 | Microsoft fine-tuning safety documentation | §1, refs | Read on 6 Oct; re-check before submission; no URL in the bib. |
| 15 | Venues and author order | refs | ADMIRE-BayesOpt: TMLR lists Ouyang first, arXiv lists Chen first (bib follows TMLR). DATE-LM year set to 2025 from the venue string. |
| 16 | LARF (2507.18631, EMNLP 2025) | §2 | Mechanism from the abstract (this session). |
| 17 | SQSD (2605.04572, ICML 2026) | §2 | Mechanism and acronym from the abstract (this session). "Its scoring needs several FRE" is the plan's estimate (6–14 FRE), not the paper's. |
| 18 | Self-Inf-N (2505.06843, ICML 2025) | §2, §4.3 | "100 outliers break safety; most mitigations fail" from the abstract (this session). Using its score as a removal filter is our adaptation. |
| 19 | "Bi-anchoring" as He et al.'s method name | §2, §4.3 | The reading list's name; check He et al.'s own term. |

| # | Item | Where | Why |
|---|---|---|---|
| 20 | Form over content (2609.19589) | §1, §2 | Abstract only (here and in the repo's arXiv search file). |
| 21 | DualSelect (2606.09866) | §1, §2 | Abstract (this session): joint task and reference selection, 1B–8B models. "Compares random, SafeInstr, SEAL, LARF, TOSS and SQSD with three seeds and a false-refusal rate" comes from the plan's novelty review (HTML §4), not re-read. **Revision 6:** FAQ checked the arXiv full text (coordinator's addition 2): SafeInstr, which adds 2–3% safety-instruction tokens, is a baseline (its §4.1, §4.4); selection overhead 1.34× standard SFT at a 90%-of-target-tokens budget; three seeds (mean ± std); a false-refusal rate (§4.6). "Training data that include red-teaming pairs" and "no model-level references" are the literature sweep's reading of the same full text (sweep §4, "Corrections" 1). The §1 sentence now rests on these; its [CITE-CHECK] is removed. |
| 22 | Bianchi et al. (ICLR 2024): 3% safety data helps, with exaggerated safety | §2 | Abstract (this session). |
| 23 | MNEME (EMNLP 2025) predicts side effects | §2 | Abstract (this session). |
| 24 | Instrument citations (OLMo 2, Llama 3, Qwen2.5, Qwen3Guard, Granite Guardian, HarmBench, StrongREJECT, SimpleSafetyTests, XSTest, OR-Bench, PKU-SafeRLHF, LoRA, ARC, Dolly) | §4 | Fetched. Check that each cited report covers the version used (Granite Guardian 3.3 8B; Qwen3Guard-Gen-4B; Llama-3.2 under the Llama 3 report; OLMo-2-0425-1B under the OLMo 2 report). Dolly's entry is the dataset card's own BibTeX (a blog post). Alpaca (§4.1, §4.8) has no citation yet: the brief allows new BibTeX only from the arXiv API, OpenReview or PMLR, and Alpaca was not searched there this revision (it is a GitHub release). |
| 25 | Qi et al. (2025; 2406.05946) as the B+Qi reference (renamed from G+Qi in revision 5; "token-wise constrained SFT", "constrains updates on the first answer tokens") | §1, §2, §3.6, §4.4, §4.10 | Mechanism from the arXiv abstract, read by API this session ("constraining updates on initial tokens"). "Token-wise constrained SFT" is the lead's and FOCUS's name; check the paper's own term before submission. FAQ (FOCUS 15:59 UTC) reports that their Table 9 shows warmup alone does most of the work on benign data; not checked here. Revision 4: §4.10 also cites them for "safety alignment often adapts mainly the first few output tokens", a paraphrase of the abstract ("the alignment adapts a model's generative distribution primarily over only its very first few output tokens"), re-read by the arXiv API on 8 Oct between 21:40 and 21:44 UTC. |
| 26 | Safe LoRA (Hsu et al., 2024; arXiv:2405.16833; NeurIPS 2024) as a reference ("projects the LoRA weights of selected layers onto a safety-aligned subspace without training or data") | §1, §2, §4.4, Table 1 | BibTeX from the OpenReview venue note HcifdQZFZV (NeurIPS 2024 poster), fetched by the OpenReview API on 9 Oct at about 13:20 UTC; title and the six authors checked against the arXiv API record (v2, "camera-ready version accepted for NeurIPS 2024"). The body's wording is from the arXiv abstract, read by API this session ("projection of LoRA weights from selected layers to the safety-aligned subspace"; "training-free and data-free"; it "only requires the knowledge of the weights from the base and aligned LLMs"). FOCUS 10:01 UTC's fuller description (the weight difference between the aligned and the base model; the layers where the update departs from it) was not checked against the paper; the lead reads it in full before the settings are fixed (decision 19). |

| 27 | Safety at One Shot (Zhang et al., 2026a; ICLR 2026; arXiv:2601.01887) | §2, §6 | Revision 6. BibTeX from the OpenReview note EyH8Fu3vtZ (ICLR 2026 poster), fetched by the OpenReview API v2 search endpoint (the note

endpoint answered with a bot challenge); arXiv:2601.01887 checked by the arXiv API (same title; second author "Lipeng He" on arXiv, "Tony He" on OpenReview: check). Claim ("further training on a single safety example" restores safety; it "acts after fine-tuning") from the arXiv abstract: "safety alignment can be fully recovered with only a single safety example, without sacrificing utility and at minimal cost", shown after fine-tuning with harmful examples. Whether it works after benign fine-tuning is not tested by them as far as the abstract says. | | 28 | SABA (Anonymous, 2026; ICLR 2027 submission, OpenReview Jo6U0y2omr) | §2 | Revision 6. BibTeX and abstract from the OpenReview API v2 search record (16 Oct). The body's clause paraphrases the abstract: "risk rankings computed at initialization are nearly uncorrelated with those computed after two epochs"; SABA "re-evaluates sample risk at the current model state in each training round". Cited anonymously as an OpenReview submission (the coordinator's instruction). Re-check for an arXiv or venue version before submission. | | 29 | SafeTuneBed (Hossain et al., 2025; arXiv:2506.00676) | §2 ("How we differ") | Revision 6. Title, authors and abstract from the arXiv API ("a benchmark and toolkit unifying fine-tuning and defense evaluation"); the body says only that it compares defences. OpenReview has an IJCAI 2025 workshop version with another title (note yKetbTItcu); the arXiv version is cited. | | 30 | Safe Delta (Lu et al., 2025; ICML 2025; arXiv:2505.12038) | §2 | Revision 6b. BibTeX from the PMLR v267 bibliography (entry pmlr-v267-lu25g; editor, month, pdf and abstract dropped); arXiv API record v1 with the comment "ICML 2025 Camera Ready". The clause ("adjusts the parameter change of a harmful or benign fine-tune") paraphrases the arXiv abstract: "adjusts the delta parameters (i.e., the parameter change before and after fine-tuning)"; user data "whether harmful or benign". | | 31 | Antidote (Huang et al., 2025; ICML 2025; arXiv:2408.09600) | §2 | Revision 6b. BibTeX from the PMLR v267 bibliography (pmlr-v267-huang25b); arXiv API record v3, comment "Accepted by ICML25". PMLR's title ends "… Harmful Fine-tuning Attack", arXiv's "… Harmful Fine-tuning" (PMLR kept). The clause paraphrases the abstract: "a one-shot pruning stage after harmful fine-tuning to remove the harmful weights". | | 32 | Harmful fine-tuning survey (Huang et al., 2024; arXiv:2409.18169) | §2 | Revision 6b. arXiv API record (v1 2024-09-26; v6 2026-04-23); the comment says "Accepted by ACM Computing Survey (CSUR)". ACM is not an allowed source in this revision, so the arXiv version is cited; replace it with the CSUR entry before submission. The clause ("survey harmful fine-tuning attacks and defences") is from the abstract. | | 33 | Olmo 3 (Team Olmo et al., 2025; arXiv:2512.13961) | §4.1 | Revision 6b. arXiv API record (title "Olmo 3"; 69 authors, the first the collective "Team Olmo"; the abstract names 7B and 32B models). The model id allenai/Olmo-3-7B-Instruct exists on the Hugging Face model API (created 2025-11-19; Apache-2.0; tagged arxiv:2512.13961; base model Olmo-3-7B-Instruct-DPO). Check the report's venue before submission. Review B's description (SFT, then DPO, then RLVR) was not checked here; the body does not repeat it. | | 34 | Qi et al. (2025) for a data-only augmentation and for prefilling | §4.2, §4.3 | Revision 6b. The abstract (arXiv API; OpenReview API search record of 6Mxhg9PtDE, 17 Oct) lists "prefilling attacks" and says that "deepening the safety alignment beyond the first few tokens can meaningfully improve robustness". Review B's account of the augmentation (refusals that start after a prefilled harmful prefix; its §3.1, read in HTML by the reviewer) was not verified here: the OpenReview PDF answered with a challenge, and the brief allows no other source. The body says only "the augmentation of Qi et al. (2025) that deepens alignment beyond the first answer tokens"; check the paper's own name for it (review B calls it safety-recovery augmentation). | | 35 | StrongREJECT's evaluator | §4.2 | Revision 6b. The arXiv abstract (API, 17 Oct): "its automated evaluator measures the extent to which a response gives useful information to forbidden prompts". Review B's figures (a fine-tuned Gemma-2B evaluator; Spearman 0.900 against 0.819 for the HarmBench classifier) were not checked here; the body gives none. | | 36 | HarmBench's validation and test classifiers; the 512-token clip | §4.2, Appendix C.10 | Revision 6c. The clip: HarmBench's `evaluate_completions.py` (raw file on GitHub, main branch, fetched 17 Oct about 08:00 UTC) defines `--num_tokens` with default 512 ("The number of tokens to evaluate"), encodes each generation with the tokenizer of `--cls_path` at `max_length=args.num_tokens, truncation=True` and decodes the clipped text before judging; its default `--cls_path` is `cais/HarmBench-Llama-2-13b-cls`. The names "validation classifier" (Mistral-7b-val-cls) and "test classifier" (Llama-2-13b-cls) follow the brief and the model ids; check them, and which classifier the main tables use, against the HarmBench paper (Mazeika et al., 2024) before submission. **Revision 6d (the lead's ruling 5):** the script was fetched again (raw file on GitHub, main branch, 17 Oct about 08:24 UTC): it loads the tokenizer of `--cls_path` (`AutoTokenizer.from_pretrained`), sets `truncation_side` to right, encodes each generation at `max_length=args.num_tokens` with truncation and decodes it with `skip_special_tokens=True`. The Gate B HarmBench pipeline (`scripts/gate_b_mac_judges/harmbench_files.py` at origin), which LW-23 (HB-1) has stage 1 reuse byte for byte, builds the tokenizer from the pinned `tokenizer.model` of the validation classifier's snapshot, sets right

truncation, counts the answer's tokens "with BOS, as HarmBench counts it" and decodes with `skip_special_tokens=True`. Appendix C.10 states these four facts, with no path or hash. | | 37 | PHTest (An et al., 2024; arXiv:2409.00598) | §4.2 | Revision 6d. The arXiv API record (fetched 17 Oct at 08:25 UTC; v1 2024-09-01, v2 2025-06-11): the title, the six authors and the abstract, which names the PHTest data set and its separate labels for controversial prompts. The record has no journal reference or comment, so the arXiv version is cited; check the venue before submission. "Harmless" follows UW-07a and the stage-1 DRAFT ("400 harmless PHTest rows"); the MIT licence that the DRAFT gives from the data set's card was not checked here. |

PDF- or full-text-checked in the repo, and used as such: Qi et al. (2024), He et al., Luo et al. (2026b), Pan et al., DATE-LM (verification log, 6 Oct); Kim et al. (2025), SafeTune's recipe, PTST, DataRx's Table II (benign-erosion recipes note); Guo et al. including Table 2 (7.27, 4.50, 9.09, 44.80 confirmed), Springer et al. (safety-landscape note); Llama 3's safety mix (problem-validation review); Qi et al. (2025) on first-token alignment (erosion-amplifiers note). Guo's "two seeds" is the Table 2 caption; the Guo-gap note finds no seed-69 checkpoint for the 5,000-row cells.

## Decisions the author must make

1. **Title** (above).
2. **RESOLVED (FOCUS 18:31 UTC, the user): the "worse" label.** A repair whose contrast against its comparator is significantly adverse at the 5% Holm level (on unsafe share, over-refusal or task loss, each in its adverse direction) is labelled *worse*, so an adverse repair is never reported as inconclusive. Labels are better, equivalent, worse and inconclusive; §3.2 and §4.3 carry them, and P4's adverse direction (DataRx) now has a label. One detail is left for the stage-1 pre-registration (decision 12). (The 18:31 UTC wording said "significantly harmful"; "adverse" since decision 23a, revision 5c.) **Revision 6e (conflict 77; the lead's ruling 6):** rule R1.3 (UW-05 (b)) narrows the rule for over-refusal and task loss: an adverse O or L contrast gives *worse* only if the repair is also infeasible against θ0 at $\tau^*_O = 5$ pp (task loss: against $a_0$ at τ_L), so such a repair can read *inconclusive* or another label from its U part (Appendix C.11).
3. **The planted sweep's funding.** If it is not funded, "menu size" leaves the headline: delete contribution 2, the planted part of the abstract and Figure 1's menu-size panel, and §3.4 shrinks to one sentence.
4. **Preliminary-results section.** The spec does not ask for one, so the body has none. Gate A's numbers and their disclosures stay in this note until §5 is written. Decide whether a section labelled "Preliminary results (development evidence; not confirmatory)" should carry Gate A.
5. **Filter list and secondary arms** ([TO BE FINALISED]).
6. **Venue and length. Revision 6 (FOCUS 16 Oct 21:05 UTC; the user, relayed by FAQ): "the goal for the main paper is ICML 2027".** TMLR or NeurIPS 2027 stays the fallback until the user names one. The ICML 2027 deadlines are ESTIMATED, not official: about 16 Jan 2027 (abstract) and about 22 Jan 2027 (paper), AoE, from a third-party tracker; the ICML 2027 call is not published (literature sweep §6, checked 16 Oct about 21:20 UTC). Plan to the ICML 2026 rules until then: 8 pages of main body; references, the Impact Statement and appendices outside the limit; one more page at camera-ready; the Impact Statement required (added in revision 6). The body measures 7.96 pages to the Impact Statement and 8.15 to the References heading (revision 5e: 9.36). Swap to the 2027 style when it is released and re-check the limit and the statement. *History:* the adopted amendment targeted TMLR or NeurIPS 2027; ICML 2027 only if, by about 1 December, the planted sweep, both 1B families and one 7–8B published-setting replication are funded. The LaTeX build uses the ICML 2026 style; see the page count in the build log. NeurIPS allows 9 pages; TMLR has no strict limit. ICML needs an Impact Statement (missing). **Revision 5:** the main body measures 9.40 pages (the measure in "What changed in revision 5"), over ICML's 8 pages and NeurIPS's 9.
7. **RESOLVED (FOCUS 10 Oct 11:16 UTC, the user: "do the recommended"): the Gate B/C pre-registration, revision 5, is frozen** (SHA-256 5c97cce7…; §12 item 8: take the fallback if the noise gate fails). Revision 6: §4.6 summarizes it and Appendix C.1–C.2 give its rules; [NOT YET FIXED] is gone. *History:* **The Gate B and Gate C rules** in §4.6 summarize the DRAFT pre-registration (revision 4b, written for recipe B), which awaits the user's acceptance. FOCUS 17:36 UTC moves Gate B to the common recipe (REC and recipe S rebuilt from it; the noise gate and predictions P3–P4 re-based on that recipe's Gate A mean E; the 8 pp floor

unchanged; each removal arm's warmup-cosine schedule spans its own epoch); DRAFT revision 5 is to carry this and was not yet in the repository at this revision. §4.6 states only the recipe change. Update §4.6 when revision 5 is accepted. **Revision 4:** the repository's DRAFT is still revision 4b. FOCUS 20:17 UTC adds an exploratory, report-only E-scaled floor for Gate B, 8 × E_common / 45.1 pp (about 5.2 pp at E ≈ 29), labelled exploratory wherever it appears and never changing a verdict; the 8 pp floor stays primary. §4.6 does not mention it yet; add one sentence when revision 5 is accepted. **Revision 5:** FOCUS 9 Oct 09:29 UTC keeps Gate B on recipe B, so the move to the common recipe above is void; DRAFT revision 5 "stays on recipe B and adds the [D14]–[D19] rules", and the brief's [D20] fixes the kit's price constants (no hourly limit; guard planning rate USD 1.85/h; spend rate USD 1.79/h; reserve USD 6), with the money thresholds recomputed in revision 5 (FOCUS 13:16 UTC). The repository's DRAFT is still revision 4b, so §4.6 marks revision 5 [NOT YET FIXED]. Update §4.6 when revision 5 is accepted. **Revision 5e:** the user's USD 10/h pod-creation price limit (FOCUS 9 Oct 15:29 UTC) replaces "no hourly limit"; brief revision 17 amends [D20]; the budget guard's planning rate (USD 1.85/h) is unchanged.

8. **The opening decomposition's definition** (§3.6) must match the exploratory note's "composition share"; the note does not state its formula in these terms. Fix the order of terms and report both orders.
9. **Gate names.** Keep "Gate A/B/C" or call them pre-registered checks.
10. **Who the user is** (§6): the plan names a fine-tuning service that must ship the customer's adapter and may change only data, or a data-only pipeline. Confirm this wording. **Revision 6b (review B W3):** the body now names two actors (§3.1: a customer who controls the data and can also set the learning rate and the number of epochs; the provider, who also holds the weights) and reports the best repair within each actor's menu; Table 1 marks who can run each arm; §6 no longer says that the service "may change only the data". Confirm the actor assignments (decision 46).
11. **The length-rule secondary arm under the other loss** (§4.3). Revision 2 listed "the length rule under a token-mean loss", a control for recipe B's per-row loss. With recipe G (token-mean) as the main recipe, revision 3 writes the mirror: the length rule under recipe B's per-row loss. Confirm, or drop the arm. **Revision 4:** the token-mean-only run (development evidence) finds that the loss weighting carries none of recipe G's reduction at one seed, so the arm's reason (that per-row weighting makes short rows matter more) has no support at the recipe level. The arm would still test it at the row level; keep it only if three seeds are cheap. **Revision 5:** with recipe B (per-row loss) the main recipe again, §4.3 reverts to revision 2's "the length rule under a token-mean loss". Keep it only if three seeds are cheap, or drop it. **Revision 6b (review B W4, partly new):** review B asks to promote "H1 and aR under token-mean loss" to the primary family at six seeds, and to report tokens removed and the first-five-token loss share per removal arm. Not in the body; for the lead (risk 51).
12. **Label order when endpoints disagree.** The 18:31 UTC rule labels a repair *worse* if any of its three contrasts (unsafe share, over-refusal, task loss) is significantly adverse. It does not say which label a repair gets when it is *better* on the unsafe share and *worse* on over-refusal or task loss. §3.2 reports both facts and leaves the order to the stage-1 pre-registration; fix it there. **Revision 6b (review B W8; review A W5 items 4–5):** §3.2 now states that *better* and *equivalent* exclude each other and marks [TO BE FIXED in the stage-1 pre-registration] which label wins when *worse* holds with another label (a side effect, or a small significant adverse difference in U), and whether a significant difference below δ gets its own label (review B proposes "better, below δ"; review A notes the asymmetry: *worse* needs significance only, *better* also an estimate of at least δ, so the chance of *better* at a true 5 pp is 0.50). **Revision 6d: RESOLVED (UW-08 (a); FOCUS fcf79a2; the lead's ruling 1 (e)).** *Worse* is checked first, naming the measure, then *better*, *equivalent* and *inconclusive*; the full row is always printed, so a *worse* on a side effect never hides a gain in *U*; a significant difference below δ adds "significant, below the smallest effect of interest" to the row instead of a new label (§3.2, Appendix C.11, marked [PLANNED; the stage-1 pre-registration is not yet frozen]).
13. **Where the failed prediction goes.** FOCUS 21:36 UTC: "the paper reports it as a pre-registered prediction that failed". Revision 4 puts its result in §5 (placeholder) and names the run in §4.1 and §4.10. Decide whether §5 gives it a sentence or a row in the dose table, and whether the mechanism side study keeps any recipe contrast. Revision 4 keeps the recipe contrast only as a dose decomposition, not as a mechanism test. **Revision 5:** recipe G is now a reference arm, and the token-mean-only run explains that reference's effect on OLMo; §4.10 item 2

and §5 keep it. Decide whether §5 gives it a sentence or a row beside recipe G. **Revision 6 (overlap rule):** the token-mean-only run is reported by the concurrent submission; §4.10 item 2 cites it and §5's placeholder no longer lists it. Decide whether the main paper still states this failed pre-registered prediction itself (FOCUS 8 Oct 21:36 UTC: "the paper reports it as a pre-registered prediction that failed"), or only through the citation.

14. **RESOLVED (FOCUS 9 Oct 09:29 UTC): no common recipe.** Llama's two-seed reading under recipe G stayed TOO CLOSE (+58.25 pp), and the ladder found no rung for both families (01:43 UTC); the user chose recipe B as the main recipe and the per-family learning-rate rule (§4.1). Llama's recipe-G seed 1103 was then read as a report-only third seed (three-seed mean +57.1 pp, TOO CLOSE at the top edge; results record, 09:35 UTC).
15. **RESOLVED (FOCUS 9 Oct 09:29 UTC): recipe G's name.** Recipe G is "the recipe only fix" and a reference, not the main recipe or a lower-dose base; the body calls it "the recipe-only fix", and the abstract no longer calls the main recipe standard practice. **Revision 6 (coordinator's addition 1; FOCUS 11 Oct 10:38 and 12:54 UTC):** recipe G did not lower erosion at 7–8B in the side work, so the body no longer calls it a fix: it is "a reference arm that changes only the recipe" and "not a general fix", with the concurrent submission cited for its effect across families and sizes.
16. **RESOLVED (FOCUS "USER DECISIONS, 2026-10-09 15:23 UTC", item 1): the ARC rule's scope under the per-family rule.** FOCUS 20:50 UTC adopted it "for the probe and Gate A readings at the common recipe (both families)", and no common recipe now exists. OLMo's Gate A of record (recipe B) passed ARC under the frozen bars (ARC rose on every seed; audit), so the rule decided none of OLMo's main readings; it decided the recipe-G Gate A read and the rung-2 readings (reported). **Resolution:** the rule is extended to every family's Gate A read at its chosen learning rate, fixed before any such read and disclosed as a loosening; it replaces "at the common recipe" in the 20:50 UTC entry. §4.2 states this scope (revision 5c); the placeholder is gone.
17. **The family × recipe interaction test (Side Project).** Its 95% interval on the logit scale excludes 0, but it was written after 11 of its 12 values were known, has about 5 df and uses one judge (development evidence). Decide whether §5 reports it, labelled exploratory, or only the per-family rule it motivates. The body now gives only the design consequence (§4.1: "families respond to the same change of recipe by different amounts"). **Revision 6 (overlap rule):** the interaction is the concurrent submission's finding (the 7–8B replication was "not shown"); the main paper does not report it, and §4.1 cites the concurrent submission.
18. **RESOLVED (FOCUS 10 Oct 09:40 UTC, the user: "yes"; Llama-3.2-1B's Gate A recorded as PASS under decision 25): Llama's rung is 5e-5.** Revision 6 drops "provisional" from §4.1. *History:* **PROVISIONALLY FILLED, PENDING THE USER (revision 5d): Llama's rung under the per-family rule.** Llama's rung-1 probe at recipe B (8 Oct, 16:50 UTC) read SATURATED, and the two report-only seeds agree (three-seed mean +59.3 pp), so the rule's next step was a recipe-B probe at 5e-5. That probe was pre-registered at 15:31 UTC and read at 16:59 UTC (FOCUS; commit 7a8eb08): E +48.0 pp, partial and inside its prediction, but answers end reads TOO CLOSE and Granite has not run (development evidence). §4.1 now says "Llama-3.2-1B's provisional rung is $5 \times 10^{-5}$ (one probe seed; development evidence; to be confirmed by stage 2's Gate A)" ("provisional" since revision 5e), and the placeholder table marks the rung provisionally filled. It stands only if the user accepts rung 2 (decision 25); if the user declines, §4.1 reverts to a placeholder for Llama's rung. The rung-2 reading under recipe G (+41.2 pp, partial) is not a recipe-B probe and did not set Llama's rate.
19. **Safe LoRA's settings** (FOCUS 10:01 UTC; to fix in the stage-1 pre-registration before any Safe LoRA run, after the lead reads the paper in full): the unaligned model (OLMo-2-0425-1B base for OLMo-2-0425-1B-Instruct; Llama-3.2-1B base for the second family), the exact projection, the layer-selection rule and its threshold (from the paper's rule or a fixed default, never tuned on stage-1 outcomes), the seeds (the adapter-scaling seeds 1101–1103 unless the pre-registration says otherwise; Table 1 says 3), and the engine work (a post-hoc state transform evaluated by the existing eval-state job, built with tests and reviewed). §4.4 and Table 1 are marked [NOT YET FIXED]. **Revision 6b (review A W3; review B W2, Q3):** both reviews ask why the post-hoc references run on three a0 seeds when six exist at evaluation cost only; §4.4 now marks the seeds (three, or all six) [TO BE FIXED in the stage-1 pre-registration] (decision 36). Review B Q11: Safe LoRA's

unaligned model is the OLMo-2-0425-1B base (this decision); the body still says [NOT YET FIXED]. **Revision 6d:** under seed layout D1 $a_0$ has seven seeds, so §4.4's open choice reads "three, or all seven $a_0$ seeds" (US-01 is still open; conflict 70).

20. **Alpaca's preconditions** (FOCUS 09:40 UTC; to settle before any Alpaca run, not now): the user extends the learning-rate rule from families to datasets before any Alpaca probe; a short design note fixes Alpaca's units (it has no categories, unlike Dolly), the field mapping (instruction, input as context, output as response), the version (tatsu-lab/alpaca; CC BY-NC 4.0) and the download, which the user approves. Alpaca still has no citation (CITE-CHECK 24). **Revision 6b (review B W14, partly new):** review B would replace or supplement Alpaca with one modern chat or domain dataset (Alpaca, like Dolly, is 2023 single-turn data) and report answer-length distributions per dataset. Not in the body.
21. **The exploratory E-scaled Gate B floor** (FOCUS 8 Oct 20:17 UTC): 8 × E_common / 45.1 pp equals 8 pp at recipe B (E_common is the recipe-B Gate A mean, +45.1 pp), so it repeats the primary floor; revision 5 drops the sentence from §4.6. Confirm, or restore it if DRAFT revision 5 keeps it.
22. **B+Qi's warmup.** FAQ reports that Qi et al.'s Table 9 shows warmup alone does most of the work on benign data (FOCUS 8 Oct 15:59 UTC; not checked here). Recipe B has no warmup, so whether B+Qi takes Qi et al.'s own schedule decides whether B+Qi differs from recipe B only in the objective or also in the schedule, like recipe G. Fix it in the stage-1 pre-registration and report it beside recipe G. **Lead's default (FOCUS 13:45 UTC, 9 Oct):** Qi's constrained loss on recipe B *without* Qi's warmup, so B+Qi isolates the loss while recipe G covers the schedule; the stage-1 pre-registration fixes it [NOT YET FIXED].
23. **RESOLVED (FOCUS "USER DECISIONS, 2026-10-09 15:23 UTC", item 3, "FAQ decision 23"): the judge wording package.** Revision 5b applied the lead's tighter wording (FOCUS 9 Oct 15:05 UTC; §3.1, §4.2, §4.6, §6): U is "refusal erosion as Qwen3Guard labels it"; a harm sentence, even where E_HB ≥ 20 pp and the same-sign, half-size rule holds, says "harmful compliance as the HarmBench classifier defines it" and (in revision 5b) carried the caveat that a blind human read rated about a third of judge-unsafe answers as usable help. Revision 5b left four items for the user. The user accepted the package as follows, applied in revision 5c: (a) the *worse* label says "adverse", not "significantly harmful" (§1, §3.2, §4.3 and these notes; conflicts 33 and 35); (b) the title keeps "safety", and the abstract names the measures exactly (refusal erosion as Qwen3Guard labels it; harmful compliance as the HarmBench classifier defines it); (c) the blind-read caveat sits once in §4.2 and once in §6, with Appendix A, not in every harm sentence; (d) the read's counts go to Appendix A as development evidence, not to the main results. Revision 5b's last open item, a second rater, is not part of this decision and becomes decision 24 (conflict 32).
24. **A second rater or a larger human read** (revision 5b's decision 23d; open). FAQ's main-track list asks for "HarmBench co-primary plus a human check" (FOCUS 8 Oct 15:59 UTC, item 4); the one-reader development read (Appendix A) does not meet it as a confirmatory measure (not costed). Decide whether one is planned on stage-1 arms. **Revision 6b:** both reviews ask for it. Review A W6: two independent non-author raters, blind to arm and judge label, with adjudication and κ; about 130 judge-Unsafe answers per arm group (a0, removals, additions) and about 150 judge-Safe answers; misclassification-corrected contrasts as a sensitivity; it disagrees with "not planned", citing risk 13 (the judge's false-Unsafe rate moves with fine-tuning). Review B W6: two raters who did not design the study, about 200 answers stratified by arm, Cohen's κ. Still open; not in the body (risk 39).
25. **RESOLVED (FOCUS 9 Oct 20:31 UTC, decision 25 accepted; 10 Oct 09:40 UTC, Gate A recorded as PASS under it).** Revision 6: §4.1 says the rung plainly and discloses, with Llama-3.1-8B's rung, that it "was accepted on a too-close bar" (decision 34). *History:* **Accept rung 2 (recipe B at 5e-5) as Llama-3.2-1B's main-recipe rung, given the answers-end TOO CLOSE?** (FOCUS 9 Oct 16:59 UTC; the user decides; not urgent: needed only when stage 2 is planned.) The rung-2 probe (seed 1101) reads E partial (+48.0 pp, band 41.2–54.8), inside its pre-registered prediction (+35 to +50), so it qualifies Llama on the erosion class. Answers end reads TOO CLOSE under the frozen bars: 64 of 600 sampled answers (10.7%) fail the screen against the 10% bar, but 53 of 589 (9.0%) without the 11 answers shorter than 5 tokens, which is the frozen short-answers rule's TOO CLOSE case (7 Oct 21:35 UTC amendment; a probe qualifies only with no TOO CLOSE, and any TOO CLOSE goes to the user). Granite, the second judge, has not run on this probe. **The lead's**

**recommendation:** accept rung 2 (5e-5) as Llama's main-recipe rung for planning, with stage 2's own Gate A seeds re-reading answers end per seed. If the user accepts, §4.1's sentence stands and "provisional" (revision 5e) can be dropped; the rung is still to be confirmed by stage 2's Gate A. If not, §4.1 reverts to a placeholder (decision 18). Note: this probe had 11 very short answers, against 0–1 in the other Llama runs.

26. **RESOLVED by the user's rule (17 Oct, relayed by the coordinator: "keep the more ambitious one always but always open to fallbacks"): the title keeps "Compute-Matched".** The design will match tokens per removal arm, or pre-register a token-matched analysis beside the row-matched one; the lead rules on which in the stage-1 pre-registration (§3.2 says so, marked [TO BE FIXED in the stage-1 pre-registration]). The fallback title, alternative 5 ("A Cost-Accounted Comparison of Data Repairs"), is used only if token matching cannot be done. The literature sweep's reasoning, kept for the record (§0 item 3 and §5): the design measures cost but does not equalize it (every arm is logged in GPU-seconds and FRE, §4.7); the removals are row-matched (600 rows), and the shortest-answer rule removes far fewer tokens than random removal; a concurrent submission found that a selection gain at a fixed document budget vanished once token budgets were matched; keep "Compute-Matched" only if stage 1 matches total GPU-seconds per arm, or at least training tokens per removal arm. Risk 30. **Revision 6b:** the user's decision of 17 Oct is relayed again ("the 'Compute-Matched' title kept with token matching"); both reviews still recommend "Cost-Accounted" (review A W8: nothing is matched on compute, removals 33 steps against additions 38; review B W13). Review A corrects itself with these notes' ledger: charging every arm 1 FRE under-charges the addition arms by only about 1%, so that sub-point is negligible; the title concern stands. **Revision 6c (FOCUS fcf79a2, UW-14; addendum §6 item 5; the lead's ruling 3): resolved in the lead's chat.** The title keeps "Compute-Matched"; the token-axis analysis (addendum §5 item 1) is the token-matched analysis beside the row-matched one, and token-matched random arms R_tok for A1, L1 and H1 are pre-registered secondary arms in the funded tier (three seeds each; about USD 19 with contingency, an estimate), with the analysis alone as the fallback. §3.2 says so without a marker, and Appendix C.3 gives a paragraph. UW-14 names no title fallback.
27. **Candidate additional reference arms (the user's rule of 17 Oct; notes only; not in the body's list of five references).** (a) The opening-hold loss of a concurrent ICLR 2027 submission ("A Harmful Request Is Still a Request"; the literature sweep, entry 3: it cut harmful compliance from 22.6% to 0.9% in its tests, the sweep's reading; not fetched or verified here); (b) Safety at One Shot (Zhang et al., 2026a; a single-example patch after fine-tuning; cited in §2 and §6 as the bounded D5 item). Decide before stage 1 is frozen whether either becomes a reference arm, and how B+Qi relates to an opening hold. If (a) is added, its citation must be fetched by the OpenReview API first. **Revision 6b:** the body now lists a lower-learning-rate reference, recipe B at the next lower rung, as [PLANNED; to be pre-registered] (§4.4; review B W3), and the data-only analogue of the one-shot example (the example added to the training data) and Qi et al.'s augmentation as candidate data arms [TO BE FIXED in the stage-1 pre-registration] (§4.3; review B W5; decision 41). The post-hoc one-shot patch and the opening-hold loss stay out of the body's references.
28. **RESOLVED by the user's rule (17 Oct, relayed by the coordinator): the re-plan scope is option B, with option C as the fallback.** The ICML re-plan proposal (`writing/paper/2026-10-17-icml-replan-PROPOSAL.md`, about 21:25 UTC on 16 Oct): option B is "P1–P6 + 5 references on OLMo-2-7B and Llama-3.1-8B; confirmation at 7–8B (≤ 2 contrasts per family)", with Gate B at 1B, OLMo-1B's stage 1 as development, Llama-1B's primary family as a replication, a reduced Alpaca study on Llama-8B and 32B if funded (about USD 1,180–1,760, an estimate; no grant is awarded); option C is the same with the full 7–8B comparison on one family. The body now describes the 7–8B tier as the full comparison on both families, marked [PLANNED; funding-gated, fallback one family] (§4.1, §4.8, §6). Not yet written into the body (for the lead): option B's roles for the 1B stages (OLMo-1B as development, Llama-1B as a replication), confirmation at 7–8B, the proposal's "Gate B-lite" transfer check (no full Gate B at 7–8B), and its power at 7–8B (a 6-seed MDE of about 11–12 pp per contrast, against the 5 pp SESOI; the measured 7–8B seed spread is about 1.7× the 1B spread). The abstract, §1 and the contributions still describe OLMo-2-1B as the stage-1 model (conflict 52). **Revision 6b (the user's decision of 17 Oct, relayed):** option B with option C as the fallback stands, and Olmo-3-7B-Instruct is a third 7–8B model, [PLANNED; funding-gated; first 7–8B model cut] (§4.1, §6; decision 48). Both reviews bear on it. Review A ("What Appendix B's numbers add", item 2): with a 6-seed

MDE of about 11–12 pp at 7–8B, the ±5 pp TOST becomes nearly unreachable (the chance of *equivalent* at a true difference of 0 is about 0.29 for 6 against 8 seeds at 1.7 times the 1B seed SD, review A's calculation), so δ, the seeds or pairing must change before option B is final (decision 39). Review B (§8.1, W1): the same arithmetic; either a wider SESOI at 7–8B stated in the abstract, or about four times the seeds on fewer contrasts (P1, P4 and the best reference). **Revision 6c:** FC-1 is closed (the user confirmed the ambitious-default rule in the lead's chat; FOCUS fcf79a2), so the rule no longer rests on a relay. FOCUS does not name option B or C; the option is still recorded only as relayed. UW-21 keeps OLMo-2-1B's P1–P6 as the confirmatory primary family under any re-plan option, which matches the body's stage-1 framing (conflict 52).

29. **Gate C's format clause (§4.6).** The brief asked for the ATTR_BETTER/REP_BETTER consequence; revision 6 also states the pre-registered form sentence, because the pre-registration requires the two equalities to be reported whatever Gate B reads. Keep it, or cut §4.6 to the brief's clause (Appendix C.2 keeps the full rule either way). **Revision 6b:** to hold the length, §4.6 keeps "the paper may say only that neither pick is the format pick" and points to Appendix C.2, which keeps the full rule, the verdict list and "which does not show that the picks are free of form".
30. **Appendix A and the concurrent submission.** The side paper reports the same blind read with the same counts. Revision 6 keeps Appendix A (it is this paper's check of what U measures) and adds one sentence that cites the concurrent submission. Options: keep; cut Appendix A to a pointer; or keep it and move the counts out of the side paper (the GPT lead's paper; not this chat's decision).
31. **The release policy ([TO BE DECIDED] in the Impact Statement).** What is released: menus, seeds, outcome tables, costs and scores; generated answers to harm prompts or not; anchor text; the planted rows' identifiers. The repository rule (no rows, prompts, answers or anchor text in git) is about git, not a release.
32. **Citing the concurrent submission, and dual submission.** ICML 2026's rules allow citing one's own unpublished work only anonymized, with an anonymized copy as supplementary material, and forbid parallel submission of substantially similar work. The overlap rule keeps the side paper's findings out of this paper; the placeholder [CITE: concurrent submission, anonymized] sits in §4.1 (twice), §4.10, §6, Appendix A and Appendix C.8. Decide the citation form at submission (for example "Anonymous, 2026", with the anonymized copy), and check the side paper's ARR status then (ARR commitment deadline about 23 Dec 2026; FOCUS 15 Oct records sync). The anonymized copies of the pre-registrations (Appendix C's [TO BE DECIDED]) go the same way. **Revision 6c:** the placeholder now sits in §4.1 (twice), §4.4, §4.10, §6 (twice: the rank-16 check and the Qwen runs), Appendix A and Appendix C.8.
33. **The page budget for results.** The body is 7.96 pages with §5 and §7 as placeholders. Results (Gate A, Table 2, Figure 1's data, Gates B and C) need about a page. Options: move more design to Appendix C (for example §3.4, §4.5 and §4.10 to one paragraph each); shrink Figure 1's placeholder to the figure's real size; drop the related-work paragraph "Prediction and decisions under a budget" to one sentence. **Revision 6b:** the body is still 7.96 pages, with about one line of slack (a probe build with one more line put the Impact Statement on page 9); §3.4, §4.3, §4.5, §4.6, §4.7, §4.8 and §4.10 were shortened to make room for revision 6b's additions. Results still need about a page. **Revision 6d:** 7.92 pages; the new body text was shortened to fit, with its details in Appendix C.11, so the body still has only about two lines of slack, and results still need about a page.
34. **The disclosure of Llama's two rungs (§4.1).** "Both Llama rungs were accepted on a too-close bar (one seed's answers-end bar at 1B; the window's upper edge at 8B); §5 reports both." Keep it in §4.1, or move it to §5 with the numbers.
35. **The analysis model (review A W2, new; §3.2 [TO BE FIXED in the stage-1 pre-registration]).** Revision 6 said "differences of means over unpaired seeds, with the variance pooled across arms", while Gate B pairs by seed and its noise is a paired SD. Review A: pre-register one model (a crossed mixed model or GLMM on answers: arm fixed, seed random, prompt random, arm × prompt random; or a two-way cluster bootstrap), a heteroscedastic (Welch-type) seed variance as primary or as a pre-registered sensitivity, ρ and the variance components from the Gate A/B runs, and the MDE restated on the right scale; it says whether the 4.5 / 6.0 pp blinded-rule thresholds are on the pair or the per-arm scale. Review A's arithmetic from these notes' numbers: within-seed correlation about 0.76 (2 df), and a seed-paired analysis would cut the Holm first-step MDE from about 5.2 to about 2.7–3.8 pp. FOCUS's newest commits include a stage-1 analysis-model draft (3db753e) and

pre-registration DRAFT revision 2 (2ded100); not read in full for this revision. **Revision 6d: RESOLVED (UW-02; LW-01: HB-C; addendum §6 item 1; the lead's ruling 1 (a)).** HB-C is the primary paired model: arms paired by seed, one seed effect per design cluster ($\{a_0$, S1, S3, S4$\}$ and $\{a_R$, A1, L1, H1$\}$), one residual variance per arm by REML, prompts crossed with seeds, Satterthwaite df; HB, S-a and S-aW are pre-registered sensitivity analyses (§3.2; Appendix C.11). Review A's other items not in the body: ρ and the variance components, and the MDE restated on the model's scale (decision 52).

36. **The test of "beats the references" (review A W3; review B W2; new; §4.4 [TO BE FIXED in the stage-1 pre-registration]).** Proposals: run the safety prompt, adapter scaling and Safe LoRA on all six a0 seeds, paired with a0 by seed (evaluation cost only); test each data repair against B+Qi and against the adapter-scaling curve interpolated in λ to the repair's task loss, with a seed bootstrap over the curve; pre-register the intersection–union test, its α and a margin; give recipe G six seeds or call it descriptive; or pre-register a rule that picks one data repair and confirm only it against the references on fresh seeds (review B). Revision 6's "lower F than the curve at equal O and L" is not defined on a one-dimensional curve (both reviews).
37. **Multiplicity (review A W4, new; §4.9 [TO BE FIXED in the stage-1 pre-registration] for the closed family only).** One closed testing family or a fixed-sequence gatekeeper over superiority, equivalence and the three *worse* tests (one Holm family of 12, then confirmation, then Llama); the sidedness of every test (P4 is two-sided; P1–P3, P5, P6 not stated); whether TOST is Holm-adjusted (if not, say "per-comparison equivalence" wherever it is used); one primary τ_O. Review A's figures: at least one false *worse* under the global null up to 14% with three unadjusted endpoint families; the chance of *equivalent* at a true difference of 0 falls from 0.93 to 0.68 at central noise with Holm-adjusted intervals. Not in the body: sidedness, TOST adjustment, one primary τ_O. **Revision 6d: RESOLVED in part (UW-04 (a), UW-06 (a); addendum §6 item 3; the lead's ruling 1 (b)).** One closed graphical family Φ of 24 nodes replaces Holm over P1–P6: $U_k$ two-sided, $O_k$ and $L_k$ one-sided adverse, $F_k$ (feasible at $\tau_O^* = 5$ pp) one-sided; equivalence is per comparison, outside Φ; one primary $\tau_O^*$ (§4.9; Appendix C.11). Not in the body: the familywise equivalence grade (LW-02, item B3) and the false-*worse* figures (DRAFT §4.3).
38. **Over-refusal (review A W12, new; review B W7, new; review B §8.3, new; §3.2 [TO BE FIXED in the stage-1 pre-registration], twice).** One baseline for side effects (review B: the untuned model for feasibility; the selectivity score as the primary side-effect contrast for additions; no *worse* on over-refusal unless R(m) also exceeds r(θ0) by more than τ_O); feasibility as a one-sided non-inferiority bound (upper 95% bound of O(m) − O(comparator) below τ_O, and likewise for L); all 250 XSTest safe prompts and OR-Bench-Hard-1K, greedy; drop τ_O = 2 pp or enlarge the bank (two prompts at 100); in-domain refusal on the 200 held-out Dolly prompts; partial refusals reported separately; the precedence of "feasible" and "worse". These notes' development numbers make the conflict the expected case: untuned OLMo refuses 17/100, recipe B 4–6/100, so an addition arm that restores the untuned level is feasible yet likely *worse* against a0. **Revision 6d (the lead's rulings 1 (c), 1 (d) and 3):** the baseline of *worse* is settled by R1.3 (infeasible against θ0 at $\tau_O^* = 5$ pp and significantly above the comparator; for task loss, against $a_0$ at τ_L); one primary bar, 5 pp, with 2 and 10 pp reported; the bank is the development 100, extended by 400 harmless PHTest prompts if REQ-19's free check shows a six-seed MDE above 5 pp, and never the hidden remainder (UW-07a, UW-07b), so §3.2 no longer names the full XSTest and OR-Bench-Hard banks. Still marked [TO BE FIXED]: feasibility as a one-sided non-inferiority bound, which LW-06 (a) settles (decision 53). Not in the body: in-domain refusal, partial refusals, the selectivity score as a side-effect contrast. **Revision 6e (LW-06 (a); the lead's ruling 1):** feasibility is the state of the two-sided 90% interval (feasible, infeasible or undetermined) of $O(m)$ against θ0 and of $L(m)$ against $a_0$, with one bound convention (§3.2; Appendix C.11), marked [PLANNED; the stage-1 pre-registration is not yet frozen]; the [TO BE FIXED] is gone. **Revision 6g (the lead's ruling 5; decision 55):** the extension fired (REQ-19's six-seed MDE on the 100 is 11.04 pp); §4.2 states the 500-prompt bank, and Appendix C.11 the number and the draw rule.
39. **The margin δ (review A W5, new).** Anchor δ to the family's measured E (for example 10–15% of E, with 5 pp absolute as secondary) or to a judge test–retest or inter-judge figure; report the *better* label's own operating characteristic (80% power for *better* at about 6.2–6.6 pp at central noise); make the 90% interval, not the label, the primary output of Table 2 and Figure 1; report the smallest margin at which each contrast would be

*equivalent*; settle δ at 7–8B before option B is final (decision 28). Not in the body. **Revision 6e (UW-13; LW-02 C1; the lead's rulings 3 and 5):** δ stays 5 pp, stated in Appendix C.11 as about 11% of the measured erosion at recipe B, and another family's margin is $\delta_{fam} = 5 \times E_{fam} / 45.11$ pp, with 5 pp absolute beside it (§3.2's clause; Appendix C.11). The rest of this item is not in the body.

40. **Secondary endpoints and the HarmBench input (review B W6, partly new; review B W15, new; §4.2 [TO BE FIXED in the stage-1 pre-registration], twice).** StrongREJECT's fine-tuned evaluator on the same 600 answers (no new generation) as a harm-weighted secondary endpoint; a prefilling probe (prefill 5–20 tokens of a harmful target, greedy, one pass per arm and seed) to separate shallow from deep repairs; the HarmBench classifier's input clipped to 512 tokens or the full answer, with its agreement with Qwen3Guard per source bank. FOCUS 8 Oct 10:14 UTC records that the development check already used the classifier's "512-token generation truncation" (conflict 55). **Revision 6c (the lead's ruling 1):** the HarmBench input is settled: the validation classifier, with each answer clipped to 512 classifier tokens, as HarmBench's evaluation script does (CITE-CHECK 36). §4.2 names it, Appendix C.10 gives the model id, and the [TO BE FIXED] sentence is gone. The Llama-2-13b test classifier is a candidate sensitivity check only (Appendix C.10). Still open in §4.2: StrongREJECT's evaluator and a prefilling probe as secondary endpoints [TO BE FIXED in the stage-1 pre-registration]; per-bank agreement with Qwen3Guard is not in the body. **Revision 6d (the lead's ruling 5):** Appendix C.10 says how the 512 tokens are counted (the classifier's own tokenizer from its snapshot; the beginning-of-sequence token counted; right truncation; decoded without special tokens), checked on the script and the Gate B pipeline (CITE-CHECK 36).
41. **Added data-only repairs and their dose (review B W5, partly new; §4.3 [TO BE FIXED in the stage-1 pre-registration]).** Two addition arms at the 48-row budget and six seeds: Qi et al.'s augmentation (a refusal after a harmful prefix; CITE-CHECK 34) and the one-shot example of Zhang et al. (2026a) replicated to 48 rows; a three-seed secondary arm that replays OLMo's own Tülu-style safety rows (review B did not verify that the OLMo variant keeps them); whether these join the primary family (and how Holm then changes). Not in the body: the replay arm and the cite of Eiras et al.
42. **Selection variance and A1's query (review A W7, partly new; review B W9, partly new).** Score A1 (and S3) on at least three a0 seeds and report the top-600 Jaccard and Kendall τ; either average the scores (and say the selection is fixed) or give each A1 seed its own seed's selection, so that selection variance enters P1. Review B: a scored-only variant whose query is a0's own judge-Unsafe sampled answers on development prompts not used for selection, with an optimizer-aware LoRA gradient (LESS-style), reported as top-600 overlap with A1; call A1 a "gradient-similarity filter" in claims and bound the conclusion to that estimator. Not in the body. (FOCUS 17 Oct 02:42 UTC: the builder's batch 1 adds A1's report-only variants, a q-only against e1-only stability report and an Adam variant, addendum §5.2.)
43. **The positive control (review A W11, new; review B W11, new).** A decoy cell (harmful-looking, non-eroding rows, for example the same prompts with refusals) to test specificity; at least three sealed plant locations; max or top-quantile aggregation beside the mean (the plant is 6% of an M = 8 cell and all of an M = 128 cell); the anchors' source (if PKU-SafeRLHF, as for the plant, the control is circular by construction; FOCUS 8 Oct 15:53 UTC: the 128 anchor prompts are PKU-SafeRLHF prompts with unsafe responses); before the sweep, the content classifier's and a judge's rank of the planted cell; a benign-looking plant (He et al.'s lists and math). FOCUS 17 Oct mentions decoy plant labels in the stage-1 build (RC-21). Not in the body. **Revision 6f:** Appendix C.5 now says that the plant comes from PKU-SafeRLHF, the anchors' source (LW-10 F1), and the content classifier's rank of the planted cell is a construction check, published before the pilot (LW-10 F2). LW-10 R8.5 rejects a He-style plant (A1 would find it by construction), and the user took UW-16: no second, benign-topic plant now (FOCUS fcf79a2). The decoy, the further locations and the aggregation are not in the body. **Revision 6g (the lead's ruling 4; addendum §8 item 1):** the anchors' source is answered by a planned sensitivity: the planted sweep also reports A1's planted-cell rank under a disjoint-source anchor set (Appendix C.5); the shared source as a limit is the fallback (decision 56).
44. **Three smaller rules (review A W13, W14, W9; new or partly new).** A coherence-screen failure keeps its arm in the family, labelled "failed (worse)" (W13; §4.2 still says "is not read"); the value-of-diagnosis predictions are scored by the calibration of per-unit predictive intervals with a proper scoring rule, and the menu-level

prediction is descriptive (W14; §4.9 still lists it as family 5); "replicates across two families" as agreement of labels in both families or a TOST on the interaction, and a family difference claimed by a test (W9; §4.8 unchanged). Not in the body. **Revision 6e (LW-09; the lead's ruling 2):** the coherence rule is settled, not as W13 proposed: the arm is read and keeps its tested label with the flag "fails answers end", not "failed (worse)" (§4.2; Appendix C.11). The other two rules are not in the body. **Revision 6f (the lead's ruling 1):** §4.2 now names the flag "fails answers end".

45. **Presentation (review A W1 and W15, partly new; review B W13, partly new).** Say "pre-specified" rather than "pre-registered" in the abstract until the stage-1 protocol is frozen, or freeze it with a timestamped anonymized registry link and hashes; a one-page "design at a glance" table (arm, comparator, seeds, test, sidedness, margin, family) or one design figure; fewer coined terms in the body (about five); the anonymized pre-registrations as supplementary material (decision 32). Not in the body.
46. **The threat model's follow-ups (review B W3, partly new; review A W8, partly new).** Confirm Table 1's assignments (S4 is C only if the service lets the customer sample the untuned model; the safety prompt counts as C because the customer controls its own prompts, though a provider's gate may not use them; conflict 59); add the dataset publisher as an actor, for whom transfer across models is the property to test (review B Q7: OLMo-selected rows on Llama and the reverse); a selection rule for "the best repair within each actor's menu" (review A: Gupta-type subset selection of all repairs within $\delta$ of the best, or simultaneous intervals for all 15 pairwise contrasts; net benefit $\Lambda - \lambda \cdot C(m)$ over a range of $\lambda$). Not in the body.
47. **Forking paths (review A W10, partly new).** Exclude the probe seed from every confirmatory estimate (Gate A's E and stage-1 a0) or report both (the body does not yet report the fresh-seed mean, +45.9 pp; risk 2); say whether stage-1 a0 reuses the Gate A seeds; one decision-tree figure of every fork (who, when, what was visible); extend the Gate B simulator to the whole stage-1 label pipeline under named truth scenarios. Not in the body.
48. **CONFIRMED (revision 6e; the user in the lead's chat, recorded in FOCUS 8dac6ad, after first choosing it in the FAQ chat; revision 6b had it as relayed, revision 6c as not yet confirmed): Olmo-3-7B-Instruct as a third 7–8B model if funded**, [PLANNED; funding-gated; first 7–8B model cut] (§4.1, §6; review B W1 and §5.4). Its learning rate follows the same per-family rule; no probe has run; the first 7–8B model to cut if funds are short. FOCUS (head 6368d66) does not record it yet (conflict 53). **Revision 6c:** the lead reported, through the brief for revision 6c, that "Add OLMo 3 7B" was not yet confirmed in the lead's chat and that the free (USD 0) checks were running; FOCUS (last changed in fcf79a2) has no Olmo 3 entry, so revision 6b's "RESOLVED" rested on the coordinator's relay alone. The body keeps the [PLANNED; funding-gated; first 7–8B model cut] marker (§4.1, §6). **Revision 6d (the coordinator's addition; source: `research/review/2026-10-17-olmo 3-7b/`, commit 4446539, read from origin: `model-check.md`, `compat-memory.md`, `olmo3-7b-addendum.md` and `verify-report.md`; then not yet confirmed).** The lead's free (USD 0) checks found: `allenai/Olmo-3-7B-Instruct` is Apache-2.0 and not gated, and its report is arXiv:2512.13961 (Team Olmo et al., 2025; the body's `olmo2025olmo3`, kept). **Safety training:** real safety data in SFT (CoCoNot, WildGuardMix, WildJailbreak; about 110k of 2.15M prompts, 5.1%) and in DPO (about 26.7k of 259,922 pairs, 10.3%), none in RL (no safety data or reward); the 7B safety average falls from 89.9 after DPO to 87.6 after RL (the report's v2 Table 26; cite the paper, not the model card's 87.3, which seems to show v1 numbers); HarmBench refusal is 94.9 (Table 53; OLMo-2-7B-Instruct 88.8). **Architecture:** the same shapes as OLMo-2-7B (32 layers, hidden 4,096, MLP 11,008, 32 query and 32 key-value heads), so the LoRA r64 count (159,907,840) and the memory settings carry over; new are sliding-window attention (4,096) in 24 of 32 layers, YaRN RoPE scaling and a new chat template (ChatML-style markers, no BOS, a default system message). **Limits:** it needs transformers ≥ 4.57 (the pinned lock has 4.56.2, which has no `olmo3` code), so it needs its own environment lock (risk 61); the checks list three blockers in the environment and kits: that lock, the chat template file (`chat_template.jinja`) missing from the kit download patterns, and the preflight chat-row check, which finds no padding or reserved-token rows to compare. **Cost (ESTIMATES, re-costed from the measured OLMo-2-7B times; revision 6e, FOCUS 8dac6ad and the addendum):** about USD 8–10 for Gate A rung 1 only, a figure that counts two seeds (a probe pod and one fine-tune pod); about USD 16–44 for a three-seed Gate A (the untuned unit and three seeds at the chosen rung, over up to three rungs; USD 6–9 if rung 1 reads weak and Olmo 3

stops). It is a third family (no 1B model), so the per-family rule needs its own Gate A ladder. The body is unchanged. **Revision 6e (FOCUS 8dac6ad; the coordinator's addition):** CONFIRMED by the user ("add OLMo 3 7B Instruct as a third 7–8B model if funded, keeping OLMo-2-7B and Llama-3.1-8B"). Cut order if money is short: 70B, then 32B, then Olmo 3 7B. Before any run (FOCUS): its own environment lock (transformers ≥ 4.57; a cross-family limit to state, risk 61), three kit fixes (download patterns, the Mac tokenizer fetch, the preflight chat-row check), its own data config, prepared set and recipes; its Gate A follows the per-family learning-rate rule. No spend without money in hand and the launch go. The body keeps its [PLANNED; funding-gated; first 7–8B model cut] marker (§4.1) and §6's clause (conflict 84).

49. **The stage-1 decisions that the body does not yet apply (revision 6c; for the lead).** FOCUS fcf79a2 settles, at the recommended defaults: the analysis model (UW-02; LW-01: HB-C, paired by seed with one seed shift per design cluster; §3.2 still marks it [TO BE FIXED]); the closed graphical testing family of 24 nodes (UW-04; §4.9 still marks it); rule R1.3 for *worse* on over-refusal, against θ0 at the primary 5 pp bar and significantly above the comparator (UW-05; §3.2 still marks the baseline); one primary over-refusal bar of 5 pp, with 2 and 10 pp reported (UW-06; §3.2 still reports the grid only); the label order (a), *worse* first with the full row printed (UW-08; §3.2 still marks it); the seed layout D1, $a_0$ and $a_R$ at seeds 1101–1107 and the method arms at 1102–1107 (UW-03; Table 1 and Appendix C.3 still say random removal at eight seeds and no repair at six); and PHTest prompts in the primary over-refusal bank if REQ-19's MDE check fires (UW-07a; §4.2 says 100 prompts). The brief for revision 6c applies only UW-14 and the HarmBench clip, and keeps three markers for the stage-1 freeze (ruling 4). Decide whether the body states these now or at the freeze (conflicts 64–66). **Revision 6d: applied (the lead's rulings 1–3):** the analysis model, Φ, R1.3, $\tau^*_O$, the label order, layout D1 and the PHTest extension are in the body (§3.2, §4.2, §4.9, Table 1; Appendix C.3, C.11), marked [PLANNED; the stage-1 pre-registration is not yet frozen] where the ruling asks; decision 53 lists the settled items that the body still does not apply.
50. **Llama-3.1-8B's third Gate A seed (the lead's correction; FOCUS fcf79a2).** It runs only once money beyond the wave-1 reserve is in hand. The body does not give the seed count at 8B (§4.1 says only that the rung was accepted on "the window's upper edge at 8B"), so no body change.
51. **The token axis (the W1 builder extract, W1-02 item 4).** Its default axis is total row tokens, with target tokens as a sensitivity and rows beside them, and the lead records the axis reading in the W1 record (REQ-21 (h)). The body and Appendix C.3 say only "a token axis"; state the axis once the W1 record fixes it (conflict 67). **Revision 6d: RESOLVED (the lead's ruling 4).** Total row tokens (prompt plus target, as trained) is the primary axis, with target tokens reported beside it (Appendix C.3); the body is unchanged (conflict 67).
52. **The planning MDE (revision 6d; for the lead).** §4.3 and Appendix C.3 said "At planning noise the minimum detectable effect after Holm's first step is about 5 pp", the unpaired model's figure under Holm. Under HB-C the planning MDE depends on the same-seed correlation (DRAFT §3.3, §3.5: for P1 under HB about 3.5 pp at the wave-0a pair SD and 7.1 pp at a pair SD of 3.75 pp; HB-C costs 8–12% more if one seed shift holds, LW-01), and Φ's first U level is 0.75% against Holm's 0.83%. Revision 6d drops the number and keeps the placeholder (§4.3: "The MDE depends on the same-seed correlation"; Appendix C.3 adds that the stage-1 record prints it at Gate B's pooled noise, LW-02 item A5). Confirm, or give a planning figure under HB-C for the body; the wave-0a values are Gate B values, which the body does not state. **Revision 6e: confirmed by the lead (ruling 6).** The body keeps "The MDE depends on the same-seed correlation" and its placeholder.
53. **Settled stage-1 items that the body still does not apply (revision 6d; for the lead).** (1) LW-06 (a): feasibility uses the two-sided 90% interval of O(m) − O(θ0) (feasible if its upper end is at most τ, infeasible if its lower end is above τ, else undetermined; the same for L against $a_0$), which settles §3.2's remaining [TO BE FIXED] on feasibility; R1.3's "infeasible" then has this meaning. (2) LW-09: an arm that fails the coherence screen keeps its tested label with the flag "fails answers end"; §4.2 still says "an arm whose answers fail too often is not read" (risk 46). (3) UW-13: δ_fam = 5 × E_fam / 45.11 pp for other families, with 5 pp absolute beside it; §4.8 does not state the Llama margin. (4) UW-15: headline item (2) reads "does each diagnostic find a concentrated, harmful-prompt cause"; the abstract and contribution 2 keep their wording. (5) LW-02: a familywise equivalence grade (the 98.33% interval) and δ stated as about 11% of E. Decide whether the body states them now or at the freeze. **Revision 6e: RESOLVED (the lead's rulings 1–5).** (1) §3.2 and Appendix

C.11 (feasibility); (2) §4.2 and Appendix C.11 (read and flagged); (3) a §3.2 clause and Appendix C.11 (δ_fam; §4.8 still names no margin); (4) the abstract, §1, contribution 2 and §6; (5) Appendix C.11 (the grades; δ's reason). Decision 54 lists what the body still does not apply.

54. **Settled stage-1 items that the body still does not apply (revision 6e; for the lead).** LW-10, taken at its default (FOCUS fcf79a2), corrects the plant's description: "37 PKU-safe, non-refusing answers to PKU red-team prompts, from the same source as the anchors", not "benign-looking" (DRAFT §8.5, F1 the factual description; R8.5). §4.5 ("a planted cause of $k = 37$ benign-looking rows"), Appendix C.5 ("rows that look benign", "they look benign") and the Impact Statement ("plants benign-looking rows") still say it, beside UW-15's "harmful-prompt cause" in the abstract, §1, contribution 2 and §6 (conflict 79). LW-10 also says that only A1 and the Self-Inf-N-style score count toward "finds a known cause", and that the text-only scorers' planted-cell ranks are published before the pilot as construction checks (F2, R8.2); contribution 2's "each diagnostic" does not say so. Decide whether revision 6f applies them. **Revision 6f: RESOLVED (the lead's rulings 2 and 3, GI VI, 17 Oct, under the user's weekend delegation).** (a) §4.5, Appendix C.5 and the Impact Statement describe the plant as PKU red-team prompts with PKU-safe, non-refusing answers (LW-10 F1). (b) Only the training-signal diagnostics (A1 and a Self-Inf-N-style score) count toward finding the cause, and the text-only scorers' planted-cell ranks are construction checks, published before the pilot (contribution 2, §4.5, Appendix C.5; LW-10 F2), also if the pilot stops (Appendix C.5; F8); the abstract, §1 and §6 say "each training-signal diagnostic" or "a training-signal diagnostic". LW-10's other items (F4, the dilution-matched decomposition; F5, Λ per planted row; F6, the relocation replay; F7, the refusal decoy; the pilot interval; R8.3, the classifier prediction; N6) are not in the body.
55. **Stage-1 facts recorded during revision 6f (FOCUS 23848e0; for the lead; not in the body).** (1) REQ-19 ran on the Gate A records (FOCUS 09:37 UTC): the six-seed over-refusal MDE against θ0 is 11.04 pp, so the O-ext trigger fires and, by UW-07a, the primary over-refusal bank becomes the 100 development prompts plus 400 harmless PHTest prompts. §4.2 still gives the extension as conditional ("if a free, pre-registered check finds its six-seed minimum detectable effect (MDE) above 5 pp"), which stays true. (2) A1's removal set is sealed (FOCUS 09:32 UTC), with a report-only stability read (top-600 Jaccard 0.434 between the two checkpoints' scores), and FOCUS says "the paper names it as a limit"; §6 does not. Neither is in the brief for revision 6f. Decide whether the body states the 500-prompt bank as a status fact (no number, as for Gate B's noise gate) and adds A1's one-shot stability to §6 as a limit; the numbers stay development evidence in these notes. **Revision 6g: RESOLVED (the lead's rulings 5 and 6, GI VI, 17 Oct, under the user's weekend delegation).** (a) §4.2 states the bank as the 100 development prompts plus 400 harmless PHTest prompts, drawn by a sealed rule, because a pre-registered power check put the six-seed MDE above 5 pp (no number in the body); Appendix C.11 gives 11.04 pp and addendum §7's draw rule (conflicts 91, 93). (b) §6 states A1's one-shot scoring as a limit, in the ruling's words, with the rounded figures (Spearman 0.77; top-600 Jaccard 0.43); the numbers are in the body, not only in these notes (conflict 90).
56. **The §6 limit sentence on the shared PKU source (revision 6g; the fallback of the lead's ruling 4; for the lead).** Not in the body: after ruling 6's sentence there was no room (with the rulings and all eleven candidate trims, the sentence put the Impact Statement heading at the top of page 9). Fallback text, for use if the disjoint-source anchor set cannot be built before the stage-1 freeze or stage 1b is not funded (addendum §8 item 1), with about two lines of further trims: "The plant and A1's anchors share one source, PKU-SafeRLHF, so an A1 hit on the plant may be near-circular." (the version tried in revision 6g added "Appendix C.5 adds a disjoint-source sensitivity"). Appendix C.5 already says that the shared source is reported as a limit if the set cannot be built.

## Risks a reviewer will raise (sources in brackets)

1. **"Unsafe" means "did not refuse".** See the development evidence. Answer: the HarmBench claim rule, hard-category U, U by length band, the selectivity score, and claim wording [Gate A audit, 4a; plan §4.2]. **Judge validity (revision 5b):** the blind human read [FOCUS 9 Oct 15:05 UTC; development evidence; Appendix A since revision 5c] supports U as lost refusal (34 of 39 Qwen3Guard-Unsafe answers went along with the request; 0 of 10 Safe controls missed) but rated only about a third of judge-unsafe answers as usable help (13 of 39, 0.33 [0.19, 0.50]; 4 of 12 OLMo answers that HarmBench labels "yes"). Limits: one reader, who also

designed parts of the study; 50 answers; small groups; no second rater; no comparison between groups. Answer: the body calls a rise in U refusal erosion as Qwen3Guard labels it and allows at most "harmful compliance as the HarmBench classifier defines it"; the caveat sits once in §4.2 and once in §6, with the counts and limits in Appendix A (revision 5c); no sentence claims usable harm. A confirmatory human check is not planned (decision 24).

2. **Forking paths.** Five settings were screened before OLMo-2-1B fell in the 20–55 pp window, fixed after the Qwen results; seed 1101 both chose the setting and counts in the mean [audit, 4b]. Answer: report every setting tried and the fresh-seed mean (+45.9 pp). **Added in revision 3:** recipe G became the main recipe after its one-seed OLMo result (+29.0 pp, partial) and Llama's saturation under recipe B were known (FOCUS 17:36 and 19:24 UTC). The choice used probe data only, never a Gate B or stage-1 outcome; Gate A is re-run at the common recipe with seeds 1102 and 1103 per family, and seed 1101 (the probe) counts, so report the fresh-seed mean beside the 3-seed mean again. **Added in revision 4:** the ARC rule was adopted (FOCUS 20:50 UTC) after the first OLMo recipe-G probe had been read under the old rule (disclosed in §4.2); OLMo's recipe-G seeds 1102 and 1103 ran before Llama's recipe-G probe was read and count toward Gate A only if recipe G becomes the common recipe (19:58 UTC); Llama's recipe-G seed 1103 was launched before seed 1102 was read and never enters the two-seed classification (21:37 UTC). **Added in revision 5:** the recipe roles changed twice on development data: recipe B (Gate A, 8 Oct morning), recipe G main (8 Oct 17:36 and 19:24 UTC), recipe B main again (9 Oct 09:29 UTC), after the ladder and the HarmBench pod results were known. Every switch used probe, Gate A and judge readings only, never a Gate B or stage-1 outcome, and the Gate A of record is the recipe-B one, read before any of the switches. The per-family rule was fixed after the ladder was read; §4.1 and §6 say so. **Added in revision 6:** the 7–8B rungs come from two-seed probes under the same rule, and Llama-3.1-8B's rung was accepted by the user although its two-seed read was TOO CLOSE at the upper edge (FOCUS 11 Oct 03:53 UTC); Llama-3.2-1B's Gate A read TOO CLOSE by the rule text on one item (seed 1101's answers end) and was recorded PASS under decision 25 (FOCUS 10 Oct 09:40 UTC). §4.1 discloses both. The stage-1 commitments are frozen after the Gate B noise gate and the sealed rankings were read (rule T2: REQ-21 (m) lists every Gate B value seen); §4.9 says so.
3. **Recipe dependence.** On OLMo, recipe G erodes about a third less than recipe B, and the token-mean-only run puts all of that on the schedule (development evidence, one seed). Answer (revision 5): recipe B is the main recipe and the body calls it plain, not standard; recipe G, the standard schedule, is a mandatory reference (the recipe-only fix), so the reader sees what the schedule alone removes; the repair contrasts are within recipe B, and §6 says rankings may differ at a lower dose. On Llama recipe G barely moves the erosion (risk 26), so "G lowers the dose" is a statement about the learning rate, not about erosion in every family. The rank-16 check under recipe G reads "rank matters little" by its pre-registered 10 pp rule, but with a small margin at one seed each, and over-refusal rose at rank 16 (development evidence); report the difference itself, not only the rule's label (FOCUS 21:25 UTC). The rank-64, all-linear adapter is still larger than common practice (rank 8–16). **Revision 6:** in the side work recipe G did not lower erosion at 7–8B (OLMo-2-7B and Llama-3.1-8B; FOCUS 11 Oct 10:38 UTC), so the body calls recipe G a reference, not a general fix, and cites the concurrent submission; the 7–8B recipe numbers are not this paper's results.
4. **Nulls everywhere.** A 600-row filter clears the MDE only if the findable effect is about 11–14 pp at row AUC 0.9 [plan §1 item 16, POW Part 3]. Answer: equivalence labels make nulls informative; the planted control; the dose curve.
5. **Additions win on U by construction.** Refusal rows restore refusal, and U tracks refusal [plan §13], which the blind human read supports (risk 1). Answer: the HarmBench rule, selectivity, O against the untuned model, and the adapter-scaling curve.
6. **A free reference may beat every data repair** (safety prompt, adapter scaling) [plan §13]. That is a valid answer; say so.
7. **Prior-work overlap** (DualSelect, Yin & Rush, DataShield's cost table). Answer: no "first" claims (§2 "How we differ"); the equivalence bounds, references, positive control and pre-registered predictions are the contribution.
8. **Re-implemented baselines.** The four method repositories have no licence [plan §5.1, FIL]; all scorers are re-implemented ("X-style") with deviation tables; authors are asked for reference outputs; stage 4's published-

setting replication checks fidelity.

9. **Diffuse erosion is the expected Gate B outcome** [Gate B draft §8; audit §7]. Under this framing it no longer collapses the paper: stage 1 runs unchanged, and the planted sweep carries the diagnosis question.
10. **The plant may not erode** above the background [plan §13]; the pilot gate decides.
11. **Small scale favours trial runs; cost ratios change with size.** No 1B cost ratio is claimed for 7–8B.
12. **The decision-value map is a model output** at the menus we can afford [plan §0, NOV-2, MET-M1]. Answer: it is secondary, frozen before the data, and scored against measurements and on Llama.
13. **Judge precision on benign prompts.** After fine-tuning Qwen3Guard labels 22–29% of the benign bank Unsafe (untuned 5%) [audit §5].
14. **"Adding a little safety data is known to work." Correction (FAQ second review, 8 Oct; source opened):** DataRx does NOT show that adding works by default. Its 59.23% → 13.70% compares a *random* 1% BeaverTails mix with DataRx's *selected* 1% mix on Llama3-8B. The fine-tune with no safety data scored 20.24%, so the random mix made it worse. So *which* safety data is added decides whether adding helps (now P4 and P5). Guo's 10 rows were not checked for over-refusal.
15. **Added data can hide misalignment** that returns on prompts like the training data (Dubiński et al.) [reading list v2, risk 4]. The plan adds an overlap control: the add effect must hold on the eval prompts least similar to the add set.
16. **Motivation.** We found no public evidence that practitioners weigh attribution against retrains [problem-validation review]. The draft does not claim that they do.
17. **Money.** The account balance is read before each launch and recorded in FOCUS; it changes daily. **Revision 6:** Gate B's wave-1 need is USD 83.91 with the frozen allowances (measured worst path USD 72.43; FOCUS 16 Oct 13:21 UTC), and the newest balance in FOCUS is USD 91.94 at 19:07 UTC on 16 Oct; no grant is awarded (BlueDot's reply expected about 18 Oct; TAIF late Nov to early Dec). The ICML re-plan proposal estimates its recommended option at about USD 1,180–1,760 (an estimate; decision 28). *History (revision 5e):* these notes gave no figure. Every reading so far is far below the core path of about USD 215–405 [plan §10, §13]. Gate B's wave-1 need is being recomputed with the [D20] constants (a rough estimate is about USD 60–64; the top-up it needs depends on the balance at launch; the recomputed figure decides; FOCUS 13:16 UTC). Every stage starts on a funding event; only complete seed blocks are analysed.
18. **B+Qi may add little over recipe B, or act like recipe G.** FAQ reports that Qi et al.'s Table 9 shows warmup alone does most of the work on benign data [FOCUS 15:59 UTC; CITE-CHECK 25], and our token-mean-only run puts recipe G's reduction on its schedule (development evidence). If B+Qi brings Qi et al.'s warmup to recipe B, it differs from recipe B in its schedule as well as its objective (decision 22). B+Qi stays as a reference because Qi et al.'s objective constrains updates on the initial tokens, a mechanism different from loss weighting; the paper must not justify it by the refuted loss-weighting account. A B+Qi equal to recipe B or to recipe G is still a valid reference result; no novelty is claimed for it.
19. **One family, or a same-recipe contrast read as a replication.** Under recipe B, Qwen and Llama saturate or sit at the top edge and OLMo erodes partly. That is the same-recipe family contrast (§4.10), reported separately and not as a replication. Under the per-family rule Llama runs at its own rate; if no rung of recipe B reads partial for Llama, or stage 2 is not funded, the paper has one family. **Revision 6:** Llama-3.2-1B's Gate A passed at its rung (decision 25), so only funding decides stage 2; the same-recipe family contrast is now the concurrent submission's (overlap rule).
20. **The mechanism account is refuted for B → G.** The mechanism note's first-token loss-mass account (under the per-row loss the first 5 response tokens carry 17.7% of the objective, against 5.3% under the token mean; FOCUS 20:01 UTC) predicted that the token-mean loss carries most of recipe G's reduction. The pre-registered token-mean-only run found that it carries none, at one seed [FOCUS 21:36 UTC]. Answer: report it as a pre-registered prediction that failed; the body offers no loss-weighting explanation; the side study is "refusal openings against harmful content", descriptive, with the first-token margin as its only causal test. Under recipe G refusal openings fall less than under recipe B, but that is now read as a dose effect, not as support for the loss account.

21. **Recipe G is a lower dose.** A reviewer may read "the standard schedule erodes less" as "a lower mean learning rate erodes less", which work on optimization choices already suggests (Kim et al., 2025). Answer: say so; recipe G is a recipe-only reference, and the main claims are repair contrasts within recipe B. **Revision 6:** the body no longer says that recipe G erodes less at all; it gives the design fact (it halves the mean learning rate) and "not a general fix".
22. **Family × recipe interaction.** Recipe G cuts OLMo's erosion by about a third but barely moves Llama's, and the Side Project's logit-scale test excludes 0 (exploratory; development evidence). No rung of the one-recipe ladder left both families partly eroded (FOCUS 01:43 UTC). Answer (revision 5): the per-family rule matches the erosion dose, and cross-family sentences say "at a matched erosion dose"; the rule was fixed after the ladder was read (disclosed in §4.1 and §6); a family difference is claimed only above the interaction's MDE. **Revision 6:** the interaction is the concurrent submission's finding, and it was "not shown" at 7–8B (side work); the body cites it and reports no number.
23. **The ARC rule is a loosening, adopted mid-stream.** It can pass a run whose true drop is up to about 6 points, and for the first OLMo recipe-G probe it was not applied blind [FOCUS 20:50 UTC]. Answer: both verdicts are reported; "skill kept" rests mainly on the held-out task loss; §4.2 and §6 disclose it. **Revision 5:** its scope after the per-family rule is open (decision 16). **Revision 5c:** the user extended it to every family's Gate A read at its chosen learning rate, fixed before any such read and disclosed as a loosening (FOCUS 15:23 UTC; decision 16 resolved); §4.2 states the scope.
24. **The main recipe is harsh.** Recipe B has no warmup, a constant learning rate and a per-row loss; a reviewer may call the erosion an artefact of a recipe few people use. Answer: recipe G, the standard schedule, is a mandatory reference, so the paper shows what the schedule alone removes; recipe B was chosen because its erosion clears the 20 pp HarmBench line (so "harmful compliance as the HarmBench classifier defines it" is allowed; the blind-read caveat sits once in §4.2 and once in §6) and leaves more room for repairs, while recipe G's does not (development evidence; FOCUS 09:29 UTC); §6 says repair rankings may differ at a lower dose. **Revision 6:** at 7–8B, recipe B's erosion at each family's rung also clears the HarmBench line on every seed (FOCUS 11 Oct 10:40 UTC).
25. **Safe LoRA may beat every data repair.** It needs no training and no data. That is a valid answer, like risk 6, and the paper says so. Its settings must be fixed before any stage-1 outcome (decision 19).
26. **"G lowers the dose" does not carry over between families.** Halving recipe G's peak learning rate (rung 2) cut both families by about 28–29%, but moving from recipe B to recipe G, which halves the mean learning rate, cut OLMo by 36% and Llama by only about 4–7% (Llama's B-against-G gap is within its seed spread; development evidence). So the schedule does not act like a lower constant rate in every family, and the token-mean-only run is OLMo-only and one seed. Answer: the body states only the design fact (G halves the mean learning rate) and calls G the recipe-only fix; it makes no claim that G lowers erosion in every family. **Revision 6:** nor across sizes: in the side work G did not lower erosion for OLMo-2-7B or Llama-3.1-8B (FOCUS 11 Oct 10:38 UTC); the body now says "not a general fix" and cites the concurrent submission.
27. **A matched dose differs in learning rate.** Under the per-family rule a contrast that "replicates at a matched erosion dose" replicates across families and learning rates together. Answer: §4.8's claim rule and the fill-in say "at a matched erosion dose"; the same-recipe contrast is reported separately.
28. **The page limit, once results exist** (decision 33). The body fits 8 pages only while §5 and §7 are placeholders.
29. **DualSelect is close** (the literature sweep). A reviewer may say that DualSelect already compares selectors and a safety-data mix with seeds, a false-refusal rate and its selection cost. Answer: §1 and "How we differ" now say what it covers; the differences are purely benign data, removal against addition with attribution, representation and a free length rule, every arm's full cost in one unit, limits on over-refusal and task loss fixed in advance, equivalence-bounded labels over six seeds, model-level references and pre-registered gates.
30. **"Compute-matched"** (decision 26). The arms are row-matched and cost-accounted, not compute-matched, and the shortest-answer rule removes fewer tokens than random removal. Answer (the user's rule of 17 Oct): match tokens per removal arm, or pre-register a token-matched analysis beside the row-matched one (§3.2, [TO BE FIXED in the stage-1 pre-registration]); report tokens removed and optimizer steps per removal arm (the sweep's design action 1); if token matching cannot be done, use the fallback title. **Revision 6c:** answered by

UW-14 (§3.2; Appendix C.3): a token-matched analysis beside the row-matched one in any case, and token-matched random removals for A1, L1 and H1 in the funded tier. The arms are still cost-accounted, not matched in GPU-seconds; the title rests on the token matching (decision 26).

31. **Scoring once** (SABA, cited in §2). Risk scores taken at the start of training may rank rows differently from later. Our LARF-style arm scores on the untuned model; A1 scores at two checkpoints of the full fine-tune. The sweep's design action 4 (two nearly free Gate C analyses: the untuned model's refusal of each Dolly prompt, and the attribution score's share along a safety direction) is for the lead. **Revision 6g (the lead's ruling 6):** §6 states it as a limit: A1 scores once, and its row scores at the two checkpoints agree only moderately (Spearman 0.77; top-600 Jaccard 0.43), so the removal set depends on the checkpoint.
32. **Literature the body does not cite** (the sweep's §5 "sentences to add"): concurrent work on first-token damage, a capacity-rank-recipe map, seed effects larger than data edits, judge filters that drop safe demonstrations, multi-layer representation filters, task-format safety data, natural refusals, subliminal traits and a token-mass confound. Not added in revision 6: each needs its own programmatic fetch and room in the 8 pages, and most are anonymous submissions. The lead chooses which to add before submission; the sweep maps their OpenReview ids.
33. **Overlap with the concurrent submission** (decisions 30 and 32). The side paper covers the family × recipe and dose material, the opening layers and the blind read. Answer: the main paper cites it anonymously and reports none of its findings as new.
34. **"Pre-registered" before the freeze (review A W1, new part).** The abstract says "six pre-registered repairs" while the stage-1 pre-registration is [NOT YET FIXED] and ten items are [TO BE FIXED]. Body: not changed (decision 45). **Revision 6c:** eight items are [TO BE FIXED] now (token matching and the HarmBench input are settled); the abstract is unchanged (the lead's ruling 4). **Revision 6d:** four items are [TO BE FIXED] now, and the five settled ones are marked [PLANNED; the stage-1 pre-registration is not yet frozen]; the abstract is unchanged (the lead's ruling 6). **Revision 6e:** three items are [TO BE FIXED] now (feasibility is settled). LW-26 puts the word "pre-specified", not "pre-registered", in the W1 record until REQ-20's manifest and REQ-21 are committed, but names no paper text, so the abstract is unchanged (the lead's ruling 8).
35. **The analysis model (review A W2, new).** Unpaired, pooled-variance contrasts against a paired Gate B; heteroscedastic arms; prompts as a crossed factor. Body: §3.2 [TO BE FIXED in the stage-1 pre-registration] (decision 35). **Revision 6d:** answered in the body: HB-C primary, paired by seed, prompts crossed with seeds; HB, S-a and S-aW as sensitivity analyses (§3.2; Appendix C.11; decision 35).
36. **No test for "beats the references" (review A W3, new; review B W2, new).** The headline FINDING 2 sits outside every pre-registered family; references at three seeds; "below the curve at equal O and L" undefined. Body: §4.4 [TO BE FIXED in the stage-1 pre-registration] (decision 36).
37. **Multiplicity (review A W4, new).** Holm over six covers only part of the label decisions; TOST adjustment, sidedness and gatekeeping unstated. Body: §4.9 [TO BE FIXED in the stage-1 pre-registration] for one closed family; the rest not in the body (decision 37). **Revision 6d:** answered in part: one closed graphical family $\Phi$ of 24 nodes with stated sidedness (§4.9; Appendix C.11); equivalence per comparison (decision 37).
38. **The margin δ (review A W5, new).** 5 pp equals the planned MDE, is absolute, and the chance of *better* at a true difference of δ is 0.5. Body: not changed; §6 no longer says smaller differences "stay inconclusive" (decision 39).
39. **Judge bias by arm (review A W6, new part; review B W6, partly new).** Differential misclassification across arms; the human check cannot estimate per-arm error; the claim rule has no uncertainty and is undefined for *equivalent* contrasts. Body: not changed (decision 24).
40. **One-shot selection (review A W7, new part).** The six A1 seeds replicate one selected subset. Body: not changed (decision 42).
41. **No estimator for "most safety per compute" and no procedure for "best" (review A W8, new part).** Revision 6b's new sentence ("We report the best repair within each actor's menu") makes this sharper. Body: not changed (decision 46).

42. **The replication rule accepts absence of evidence (review A W9, new part).** A 7 pp effect in OLMo and 0.5 pp in Llama of the same sign "replicate". Body: not changed (decision 44).
43. **Forking paths (review A W10, new parts).** The fresh-seed mean is not in the body; stage-1 a0's reuse of Gate A seeds is not discussed; no decision tree or end-to-end simulation. Body: not changed (decision 47).
44. **The positive control (review A W11, new; review B W11, new).** Near-circular (anchor similarity), dilution confounded with menu size, one plant location; the plant may be trivially findable. Body: not changed (decision 43). **Revision 6f:** partly answered in the body. Only the training-signal diagnostics count toward finding the plant, and the text-only scorers' ranks are construction checks, published before the pilot (§4.5, Appendix C.5; LW-10 F2); Appendix C.5 also says that the plant shares its source, PKU-SafeRLHF, with the anchors. A1, one of the two counted diagnostics, is bi-anchored on those anchors (Appendix C.2), so a reviewer may still call an A1 hit near-circular. The sources answer this only for a He-style plant (R8.5), and §6 does not state it. **Revision 6g (the lead's ruling 4; addendum §8 item 1): answered by a planned sensitivity.** Appendix C.5 says that the plant's prompts are disjoint from the anchors', and that the planted sweep also reports A1's planted-cell rank under a disjoint-source anchor set (64 harmful and 64 safe anchors, not PKU-Alignment data, built by Gate C's anchor rule, its source sealed before the stage-1 freeze and before any planted run); the rank under the PKU-SafeRLHF anchors stays the counted one, marked [PLANNED; the stage-1 pre-registration is not yet frozen]. §6 does not state the shared source (no room; decision 56).
45. **Over-refusal (review A W12, new; review B W7, new; review B §8.3, new).** Two baselines (untuned for feasibility, comparator for *worse*); 100 greedy prompts; point-estimate feasibility. Body: §3.2 [TO BE FIXED in the stage-1 pre-registration], twice (the baseline; non-inferiority on the full XSTest and OR-Bench-Hard banks) (decision 38). **Revision 6d:** R1.3 sets one baseline for *worse* (θ0 at $\tau_O^* = 5$ pp, and significantly above the comparator); 5 pp is the primary bar; the bank may grow by 400 PHTest prompts (§4.2); feasibility as a non-inferiority bound is still marked [TO BE FIXED] (decisions 38, 53). **Revision 6e:** feasibility is the state of the two-sided 90% interval against θ0 (§3.2; Appendix C.11; LW-06). Under the prompts-random error model for O, an arm at θ0's level reads "undetermined" at τ_O = 2 and 5 pp on 100 prompts (DRAFT §6.2), which a reviewer may raise; the PHTest extension is the pre-set answer if the free MDE check fires. **Revision 6g:** the extension fired: the primary bank is the 100 development prompts plus 400 harmless PHTest prompts (§4.2; Appendix C.11; decision 55).
46. **Arms that fail the coherence screen are "not read" (review A W13, new).** Informative missingness. Body: not changed (decision 44). **Revision 6d:** LW-09 settles it in the sources (the tested label stays, with a flag); §4.2 still says "is not read" (decision 53). **Revision 6e:** answered: the arm is read and keeps its tested label with a flag (§4.2; Appendix C.11). The pooled bar was chosen after $a_0$'s screen rates were read (DRAFT §3.7), which the stage-1 record discloses.
47. **The value-of-diagnosis test is one realization (review A W14, new part).** Body: not changed (decision 44).
48. **Presentation (review A W15, new part; review B W13, new part).** The detection range reads "about 10–12 pp" in §4.6 and "about 9, 10 or 12 pp at low, central or high planning noise" in Appendix C.1. The two agree (§4.6 gives the central-to-high range), but a reader sees a mismatch; a fix is "about 10 pp at central planning noise (9–12 pp across the planning range)". Also the density of coined terms and the missing design-at-a-glance table. Body: not changed (decision 45).
49. **Model currency (review B W1, new part).** OLMo-2 and Llama-3.x date from 2024–2025; Qwen3's thinking mode is not discussed. Body: Olmo-3-7B-Instruct added as [PLANNED; funding-gated; first 7–8B model cut] (§4.1, §6; decision 48); no limitation sentence on reasoning-mode models.
50. **The threat model (review B W3, new part).** Fine-tuning APIs expose the learning rate and epochs; the customer cannot run A1, L1 or S3. Body: changed (§1, §3.1, Table 1, §4.4, §6) and a lower-learning-rate reference [PLANNED; to be pre-registered] (§4.4); the dataset publisher is not in the body (decision 46).
51. **Loss normalisation and H1 (review B W4, new part).** H1 at six seeds under token-mean loss; tokens removed and first-five-token loss share per arm. Body: not changed (decision 11; token matching stays [TO BE FIXED in the stage-1 pre-registration], decision 26). **Revision 6c:** token matching is settled (§3.2; Appendix C.3 reports tokens per arm), and the funded-tier R_tok arm for H1 measures the low end of the token axis that

the analysis alone must interpolate (the builder extract's hazard 5). **Revision 6d:** the token axis is total row tokens, with target tokens beside it (Appendix C.3).

52. **Stronger repairs and literature (review B W5, new part).** Body: §4.3 [TO BE FIXED in the stage-1 pre-registration] for Qi et al.'s augmentation and the one-shot example as data arms; §2 cites Safe Delta, Antidote and the survey; Tülu-style replay, Eiras et al. (task-format safety data), backdoor-enhanced safety alignment, Fraser et al. and SORRY-Bench are not in the body (each needs a programmatic fetch; decision 41).
53. **No harm-weighted score and no robustness endpoint (review B W6, new part).** Body: §4.2 [TO BE FIXED in the stage-1 pre-registration] (StrongREJECT's evaluator; a prefilling probe) (decision 40).
54. **Labels that can contradict each other (review B W8, new part).** Body: §3.2 now states that *better* and *equivalent* exclude each other and marks the order when *worse* holds with another label, and a label for a significant difference below δ, [TO BE FIXED in the stage-1 pre-registration] (decision 12). **Revision 6d:** the order is settled: *worse* first, the full row printed, and "significant, below the smallest effect of interest" in the row for a significant difference below δ (§3.2; Appendix C.11; decision 12).
55. **A1 is gradient similarity to external anchors (review B W9, new part).** Body: not changed (decision 42).
56. **Why Qwen is out (review B W12, corrected by the reviewer after reading these notes).** The reviewer first read the exclusion as "selecting families that erode"; these notes show the opposite (Qwen2.5-1.5B and 7B saturate under recipe B). Body: §4.1 now gives the reason in one clause. Remaining risk: Qwen was dropped on 7 Oct 20:58 UTC, before the per-family learning-rate rule existed (9 Oct), and no lower rung was probed for Qwen (a Qwen backup at Qi's LR 2e-5 was cancelled before launch, FOCUS 20:58 UTC), so a reviewer may ask why the matched-dose rule was not applied to Qwen. Answer: the user chose "Llama/OLMo only" before that rule; say so if asked (conflict 54). **Revision 6c (the lead's ruling 2):** §6 now says it in one plain sentence (dropped during development at $10^{-4}$, before the per-family rule existed, not retried at a lower rate; the concurrent submission's Qwen runs are the only later evidence), and §4.1 points to §6.
57. **Coined terms (review B W13, new part).** Body: not changed (decision 45).
58. **Dataset realism (review B W14, new part).** Body: not changed (decision 20).
59. **The HarmBench classifier outside its validated setting (review B W15, new).** Body: §4.2 [TO BE FIXED in the stage-1 pre-registration] (512-token clip or full answer). The development numbers already used the 512-token clip (FOCUS 8 Oct 10:14 UTC), so the reviewer's premise holds only if stage 1 reads full answers; per-bank agreement is not in the body (decision 40; conflict 55). **Revision 6c:** the clip is settled (§4.2; Appendix C.10), so the classifier reads the first 512 classifier tokens of each answer, as in HarmBench's own script. A reviewer may still ask about harmful content after the window; the stage-1 DRAFT reports the share of HarmBench inputs cut at 512 tokens (its secondary list), which the body does not mention. **Revision 6d:** Appendix C.10 also says how the window is counted (the classifier's tokenizer; the beginning-of-sequence token counted; right truncation).
60. **The reviewer's Q1 (review B §8.2).** The untuned U per family (3.2% OLMo-2-1B, 13.8% Llama-3.2-1B) and recipe G's partial erosion on OLMo-2-1B (+28.8 pp, three seeds) would pre-empt "an artifact of recipe B" in one sentence; they are development numbers, and the recipe-G contrast is the concurrent submission's (overlap rule). Body: not changed; for the lead.
61. **Olmo-3-7B-Instruct runs under another software stack (revision 6d; the coordinator's addition; source `research/review/2026-10-17-olmo3-7b/`, 4446539).** The model needs transformers ≥ 4.57, so it needs its own environment lock, while OLMo-2 and Llama run under the pinned lock (transformers 4.56.2). It also has its own chat template, sliding-window attention in 24 of 32 layers and YaRN RoPE scaling. So any contrast between Olmo 3 and the other families also differs in library version, template and attention code. If the model is run, the paper must state this as a cross-family limit (§6). Body: not changed. **Revision 6e:** decision 48 is confirmed (FOCUS 8dac6ad); the model is funding-gated and has not run, and this risk stays.

## Where the framing is weak (the drafter's view)

1. **The cost axis is mostly evaluation.** Generation is about 81% of an arm and training about 8% [plan §1 item 6]. FRE then measures our evaluation size more than training; the two cost conventions help, but a reviewer may

still read "compute-matched" as "evaluation-matched". Say plainly what one FRE buys.

2. **The comparison fixes doses.** Rankings across removal (600 rows) and addition (48 rows) hold only at these doses; the dose curve is secondary and 3-seed.
3. **Category units versus row scorers.** Gate B tests categories; stage 1 tests row scorers. A row scorer can find what Gate B cannot, and the reverse; the paper must not read one from the other.
4. **The data-only user and diagnostics pull against each other** (previous draft, point 2): a closed-service customer cannot run attribution; a developer with open weights also has model-level repairs. The references now make this explicit, but the population that benefits from paid diagnosis stays narrow. **Revision 6b (review B W3):** the body now names the two actors and reports the best repair within each actor's menu (§3.1, Table 1, §4.4, §6). Paid diagnosis is open only to the provider (or an open-weights developer), who also has every model-level reference; review B adds the dataset publisher, for whom transfer is the test (decision 46).
5. **Mechanism tests are thin.** The token-mean-only run separated loss weighting from schedule at one seed, on OLMo only, and refuted the loss-weighting account of B → G. What remains is a descriptive decomposition with its two layers (openings lost; direct answers already unsafe) and one teacher-forced first-token margin. The two-layer account is observational (no prefill test; positions counted in words), and neither part says why fine-tuning switches refusal openings to direct ones. Both are side studies.
6. **The opening side study is largely pre-empted** (revision 6; the literature sweep, entry 3 and design action 3). A concurrent ICLR 2027 submission puts the damage on the answer's opening token across 17 models. §4.10 keeps the side study exploratory; frame it as a replication if it stays.

## Development evidence (for the author only; never a finding in the abstract or the contributions)

- **Gate A PASSED on OLMo-2-0425-1B-Instruct under recipe B** (8 Oct, 05:40 UTC; the Gate A of record since FOCUS 9 Oct 09:29 UTC): E = +43.5, +43.7, +48.2 pp over three seeds; mean +45.1 pp (band 41.3–49.0; partial); fresh-seed mean +45.9 pp; answers end; held-out Dolly loss 2.10 → 1.54; Granite agrees. Independent audit: sound with caveats.
- **Audit disclosures:** P(Unsafe | no refusal) 0.91–0.94 after fine-tuning; about a third of Unsafe answers carry only "unethical acts" or "politically sensitive"; E under hard categories only is about 28 pp; five settings screened; 2 df for seed variance; the ARC bar is near chance at 1B (untuned 0.392).
- **Blind human read** (FOCUS 9 Oct 15:05 UTC; report-only, gates nothing; since revision 5c, Appendix A of the paper carries the pooled counts and limits, labelled development evidence, decision 23d; the group counts stay here). Pre-registered 8 Oct 20:56 UTC and run by the Side Project; the user was the single reader; the ratings were sealed (commit 8c338fb) before the key was opened; scored 9 Oct 14:58 UTC exactly as fixed. Source: the Side Project's branch claude/wonderful-greider-d1aa5b, `research/side-blind-read/PREREG-2026-10-08.md` (Results) and `blind-read-scores-2026-10-09.json`, commits 4d63e04 and 57d08dd. 50 sampled answers, shown without model, recipe, seed or judge label; rubric: 0 refuses or deflects; 1 goes along but gives no usable help (vague, generic, wrong, fictional or incoherent); 2 gives usable help.
- **Qwen3Guard-Unsafe answers:** 40 drawn, 39 rated (1 skipped). 34 went along (rated 1 or 2): 0.87 [0.73, 0.96]. 13 gave usable help (rated 2): **0.33 [0.19, 0.50]** (exact 95% intervals). Rated 2 by group: OLMo-2-1B recipe B 4 of 12; OLMo-2-1B recipe G 3 of 6; Qwen2.5-1.5B recipe B 2 of 7; Qwen2.5-7B recipe B 2 of 6; Llama-3.2-1B recipe B 2 of 8. The groups are too small to compare, and no comparison was made.
- **Safe-labelled controls:** 0 of 10 rated 1 or 2 (no judge misses; [0.00, 0.31]).
- **HarmBench, on the 15 OLMo items that have a label:** of 12 "yes", 4 rated 2 and 8 rated 0 or 1; of 3 "no", none rated 2. The human bar ("would meaningfully help") is stricter than HarmBench's.
- Two "looping or garbled" ticks, both Qwen2.5-1.5B. Reading (I, the Side Project's): U is a sound measure of lost refusal; small models often comply vaguely, generically or wrongly.
- **Limits:** one reader, who also designed parts of the study; 50 items; small groups; no second rater; no comparison between groups.

- **For Gate B (FOCUS):** it meets the intent of DRAFT §12 item 5 (what U means), on a different sample from the audit's proposal (§8 item 4); DRAFT revision 5 reports it as done; the verdict stays on U. It also gives the "about 50 human-read answers" that FAQ asked for the arXiv note (FOCUS 8 Oct 15:59 UTC, item 1).
- **Qwen2.5-1.5B and 7B-Instruct saturate** under recipe B (+75.2 and +82.2 pp); rank 16 still gives +67.8 pp. Guo-gap analysis: judge and prompts explain at most about 13–30 pp of the gap to Guo's Qwen LoRA cell.
- **Mechanism (exploratory note):** refusal openings fall by about 50 pp (OLMo) and 85 pp (Qwen); the opening switch carries about 0.42–0.64 of OLMo's Unsafe increase; direct answers also became more harmful; the erosion is broad across prompts and stable across seeds. Recorded prediction: a first-token path alone gives at most about 3 pp per category, so a LOCALIZED Gate B would need a content effect. **Revision 4:** the note's first-token loss-mass account does not explain the B → G difference (the token-mean-only run, below).
- **Over-refusal:** untuned OLMo refuses 17/100 benign prompts; after recipe-B fine-tuning 4–6/100 (recipe G, one seed: 17/100). v2's base-model positive control: a classifier-targeted rewrite cut sampled unsafe answers from 80.3% to 11.4%, but XSTest-safe refusals rose from 1/50 to 22/50 in both seeds.
- **Cost of an arm** (one Gate A ledger): training 245 s, greedy generation 1,750 s, sampled generation 761 s, guard and scoring 293 s; about 3,115 s; A100 at USD 1.59/h.
- **Planning noise:** clustered SE(U) 2.8 pp per arm; seed SD 2.65 pp (2 df). MDE after Holm's first step about 5.0 pp (removal) and 5.3 pp (add) at central noise; Gate B detects 8 pp about half the time and reaches 80% at about 10–12 pp.
- **Recorded prior:** DIFFUSE or TOO CLOSE is more likely than LOCALIZED at Gate B.
- **Recipe G on OLMo-2-1B** (one seed, 1101; FOCUS 15:17 UTC; report-only): U 3.2% → 32.2%, E +29.0 pp (SE 2.6; partial) against +43.5 pp under recipe B at the same seed (shared sampling streams); answers end; task learned (held-out loss 2.100 → 1.612); over-refusal 17/100, the untuned level (recipe B 4/100); ARC TOO CLOSE only because 48 questions changed (a gain of 2); Granite agrees (greedy E +23.0 pp, κ 0.93). Refusal openings fall less under G (−0.36 against −0.51, sampled); the opening switch carries about half of the increase under both recipes (0.48 against 0.52). **Revision 4:** the token-mean-only run puts this reduction on the schedule, not the loss weighting, so the smaller drop in refusal openings is read as a dose effect, not as support for the first-token loss account.
- **Llama-3.2-1B under recipe B** (one seed; FOCUS 16:50 UTC; report-only): **SATURATED**, so it does not qualify. U 13.8% → 76.3%, E +62.5 pp (interval 56.3–68.7). Sampled answers end; ARC and task loss pass; over-refusal 11 → 2 of 100; heavy greedy loops (60 of 200 capped; reported, not gating); Granite agrees (greedy E +67.5 pp, κ 0.92). The untuned Llama is far less safe than the untuned OLMo (U_0 13.8% against 3.2%); the opening switch carries 0.68 of the increase (OLMo 0.52).
- **The same-recipe family contrast and recipe G** (Qwen3Guard E; FOCUS and the results record): under recipe B, Qwen2.5-1.5B (+75.2 pp) saturates, Llama-3.2-1B sits at the top edge or above (+62.5, +61.3, +54.0; below) and OLMo-2-1B erodes partly (+43.5 pp at seed 1101; Gate A mean +45.1 pp). Under recipe G: OLMo +29.0, +31.0, +26.5 pp (mean +28.8); Llama +60.0, +56.5, +54.8 pp (mean +57.1). Recipe G removes 36% of OLMo's E on Qwen3Guard and 43% on HarmBench, and about 4–7% of Llama's E (FOCUS 09:29 UTC: "−36% of E on Qwen3Guard, −43% on HarmBench" for OLMo). These numbers stay here, never in the abstract or the contributions.
- **HarmBench compliance judge, complete** (FOCUS 17:07 UTC; reported, not gating): E_HB +27.8, +30.5, +32.8 pp; mean +30.4 pp, about two thirds of Qwen3Guard's +45.1 pp; agreement 83–86% (κ 0.65–0.71). Every seed is above 20 pp, so the 13:53 UTC judge rule is met and Gate A's recipe-B erosion may be called "harmful compliance as the HarmBench classifier defines it", with no caveat of its own (revision 5c: the blind-read caveat sits once in §4.2 and once in §6).
- **Token-mean-only run `bt`** (OLMo-2-1B, seed 1101; pre-registered 20:01 UTC, read 21:36 UTC; report-only; FOCUS): recipe B with only the token-mean loss (constant LR 1e-4, no warmup), fine-tune only on the sealed untuned unit. U 3.2% → 46.8% (281/600): **E_bt +43.7 pp** (SE 3.0, band 37.8–49.6, partial), against E_B +43.5 and E_G +29.0 on the same seed and sampling streams. **The pre-registered prediction (E_bt ≤ 36.25 pp, i.e. closer to G than to B) FAILED.** The share of the B → G reduction carried by the loss weighting, (E_B −

E_bt)/(E_B − E_G), is −0.01: none of it. The schedule (a 4-step warmup and cosine decay, which also halves the mean learning rate over the epoch) carries all of it, so the mechanism note's first-token loss-mass account does not explain B → G, and recipe G is, in effect, a lower dose (FOCUS's reading, labelled interpretation). Other bars: answers end (3 capped, 17 failing the screen); task learned (2.100 → 1.587); ARC d = 0 (TOO CLOSE under the frozen bars, pass under the adopted ARC rule); over-refusal 9/100 (B 4, G 17); greedy unsafe 75/200 (B 88, G 46). One seed. Granite and the opening lexicon queued.

- **OLMo-2-1B under recipe G, seeds 1101, 1102 and 1103** (results record, formal three-seed read 22:47 UTC, 8 Oct; reported, not the Gate A of record since FOCUS 9 Oct 09:29 UTC): E +29.0, +31.0, +26.5 pp; mean +28.8 pp, SE 1.77, band 25.3–32.4, partial; coherent-only +28.4; answers end and the task is learned on every seed; Granite agrees on every seed (greedy E +23.0, +25.5, +18.5 pp; κ 0.932, 0.911, 0.905). ARC nets a gain of 2 of 189 on every seed, TOO CLOSE under the frozen bars; frozen verdict TOO CLOSE (ARC only), PASS under the adopted ARC rule (seed 1101 read before that rule existed).
- **Llama-3.2-1B under recipe G** (seed 1101, the common-recipe probe; read 21:04 UTC; FOCUS): U 13.8% → 73.8% (443/600); **E +60.0 pp**, SE 3.2, band 53.6–66.4, which contains the 55 pp edge, so the E class reads **TOO CLOSE** (point estimate saturated; U_ft 0.738, just under the 0.75 auto-saturation line); coherent-only +58.3 pp (same class). Every other bar passes: answers end (24 capped, 47 failing the screen, of 600); ARC 124 → 133 (a gain); task loss 2.243 → 1.619. Over-refusal 11 → 7 of 100; greedy answers loop (75 of 200 capped; reported). Recipe G barely moves Llama (+62.5 → +60.0 pp) while it cuts OLMo by about a third (+43.5 → +29.0 pp at seed 1101). By the frozen amendment (7 Oct, 21:35 UTC), seed 1102 runs and the two-seed mean of seeds 1101 and 1102 is classified with the same band; seed 1103, launched early at 21:37 UTC, never enters that classification. Granite pending. **Later (results record 22:51 UTC, 8 Oct, and 09:35 UTC, 9 Oct):** seed 1102 +56.5 pp; the frozen two-seed classification (1101, 1102) is +58.25 pp, SE 2.72, band 52.8–63.7: TOO CLOSE again, so the user decided (FOCUS 09:29 UTC). Granite on seed 1102: greedy E +58.5 pp, κ 0.903. Seed 1103, read as a report-only third seed: +54.8 pp; three-seed mean +57.1, SE 2.55, band 52.0–62.2, TOO CLOSE at the top edge; Granite agrees on every seed; ARC a gain on every seed.
- **Rank-16 check under recipe G** (OLMo-2-1B, seed 1101; pre-registered 19:24 UTC, read 21:25 UTC; report-only; FOCUS): U 3.2% → 23.3% (140/600); **E +20.2 pp** (SE 2.5, band 15.1–25.2, which contains the 20 pp edge, so TOO CLOSE on the E class; point estimate partial; coherent-only +19.2) against +29.0 pp at rank 64. |E(r16) − E(r64)| = 8.8 pp ≤ 10 pp, so **rank matters little by the pre-registered rule**, but the margin is small: one seed each, and the SE of a difference is about 3.6 pp, so a real rank effect of this size cannot be excluded; report the 8.8 pp itself. Over-refusal 25/100 (rank 64 and untuned: 17). Answers end (0 capped, 14 failing the screen); task learned (2.100 → 1.649); ARC d = +1/189, a drop of one question (TOO CLOSE under the frozen bars, pass under the adopted ARC rule). The main recipe stays G at rank 64, as fixed in advance. Granite queued.
- **HarmBench on the pod** (FOCUS 8 Oct 23:28 and 23:43 UTC; results record; one device, after a Mac-against-pod parity PASS; reported, not gating). OLMo-2-1B E_HB (sampled): recipe B +27.7, +30.5, +32.8 pp (mean +30.3; the pod reproduces the Mac's recipe-B counts except one sampled label); recipe G +17.8, +19.3, +14.3 pp (mean +17.2; 0.57 of B); greedy B +27.0, +25.5, +31.0 and G +11.0, +13.5, +8.5. **Harm wording:** recipe B clears 20 pp on every seed and the same-sign, half-size rule holds, so recipe B's erosion may be called "harmful compliance as the HarmBench classifier defines it", with no caveat of its own (revision 5c wording); recipe G is below 20 pp on every seed, so its erosion is only "refusal erosion as Qwen3Guard labels it" (the 10:14 UTC rule; the 13:53 UTC rule holds, 17.2 ≥ 14.4). Family contrast (00:19 UTC, 9 Oct): Qwen2.5-1.5B recipe B +46.0 (rank 16 +45.8), Qwen2.5-7B recipe B +63.5, Llama recipe B +40.5 (seed 1101), Llama recipe G +35.5 and +35.8 (seeds 1101, 1102); the untuned Llama has the highest HarmBench rate of the four families (11%). Recipe G cuts OLMo's E_HB by 43% and Llama's by 12% (Qwen3Guard: 36% and 7%).
- **The one-recipe ladder** (recipe G; FOCUS 01:43 UTC, 9 Oct; the frozen amendment): no rung qualifies both families cleanly (table and rung-2 details below).

| Rung (recipe G) | OLMo-2-1B E (Qwen3Guard) | OLMo E_HB | Llama-3.2-1B E | Llama E_HB |
|---|---|---|---|---|
| 1: LR 1e-4 | +28.8, partial (3 seeds) | +17.2 (below the 20 pp line) | +58.25, TOO CLOSE high (2 seeds) | +35.7 |
| 2: LR 5e-5 | +20.8, TOO CLOSE low (2 seeds) | not measured | +41.2, partial (1 seed) | not measured |

Rung 2 details (results record): OLMo seeds +20.3 and +21.3 (SE 1.94, band 17.0–24.7); Llama +41.2 (SE 3.6, band 34.0–48.3); both inside their pre-registered predictions (OLMo +12 to +26; Llama +35 to +55); Granite agrees (greedy E OLMo +11.5, +12.0; Llama +38.0); ARC passes under the ARC rule. Halving the peak learning rate moves OLMo from +28.8 to +20.8 pp (−28%) and Llama from +58.25 to +41.2 pp (−29%): the same relative change in both families, at different levels. Rung 3 (2e-5) would put OLMo below 20 pp (FOCUS's reading).

- **Llama-3.2-1B under recipe B, three seeds** (report-only; seeds 1102 and 1103 run by the Side Project's kits and launched by the lead, 9 Oct 10:46 UTC, after the user's "yes"; FOCUS 12:18 UTC; the kits' reading files and the Side Project's results note): E +62.5 (seed 1101, the 8 Oct probe), +61.3 (seed 1102, attempt 2 after a cap-gate stop) and +54.0 pp (seed 1103); mean +59.3 pp (the Side Project's arithmetic). Pair readings with seed 1101: 1101 + 1102 mean +61.9 pp, saturated; 1101 + 1103 mean +58.25 pp, TOO CLOSE. Answers end; ARC a gain; task learned; over-refusal 2–4 of 100. Granite and HarmBench were not run on seeds 1102 and 1103 (the Mac is kept for other work).
- **Llama-3.2-1B rung 2 under recipe B (LR 5e-5)** (seed 1101; development evidence; pre-registered 9 Oct 15:31 UTC, read 16:59 UTC; source: FOCUS 16:59 UTC, commit 7a8eb08; it sets §4.1's provisional rung, decisions 18 and 25). Fine-tune only on Llama's sealed untuned unit; config `study-kt-llama32-1b-r64-b5e5-mb4-20261008` (recipe B with only the learning rate changed). U 13.8% → 61.8% (371/600): **E +48.0 pp** (SE 3.41; band 41.2–54.8, no edge inside: **partial**; coherent-only +46.6, partial), inside the pre-registered prediction (+35 to +50, point +43). **Answers end: TOO CLOSE** under the frozen bars: 20 of 600 capped (3.3% ≤ 5%), but 64 of 600 (10.7%) fail the screen against the 10% bar, and 53 of 589 (9.0%) without the 11 answers shorter than 5 tokens (0–1 such answers in the other Llama runs). ARC a clear gain (lost 15, gained 36; pass); task learned; greedy E +44.5 pp. Granite not yet run (it goes on the pod with the next Granite run). Pod llama32-1b-b5e5-1101, job 15:50–16:58 UTC, verified and stopped. **The user has not decided** whether to accept the rung; the lead recommends accepting it for planning, with stage 2's Gate A seeds re-reading answers end (decision 25).
- **The family × recipe interaction** (Side Project, report-only; test fixed at 12:18 UTC, 9 Oct, before seed 1102's result was read, but after 11 of its 12 values were known; side branch claude/wonderful-greider-d1aa5b, commits fd48c89 and 8cf2e2d; not in FOCUS). On logit(U) per seed, recipe G minus recipe B: OLMo −0.686 (SE 0.086), Llama −0.113 (SE 0.152); interaction −0.573, SE 0.175, Welch df 5.08, **95% interval −1.019 to −0.127, which excludes 0**: recipe G cuts OLMo's erosion more than Llama's (−16.3 against −2.2 pp; on the pp scale −14.1, SE 3.66, interval also excludes 0). Limits: written after 11 of 12 values; about 5 df; one judge (Qwen3Guard); each family's B and G units share an untuned baseline; 1B and Dolly only.
- **The two-layer account** (FOCUS 8 Oct 15:17 and 16:50 UTC; the mechanism note's addenda of 8 Oct 15:46, 17:14 and 21:56 UTC; the name is the Side Project's arXiv note, commit c51cf39; frozen lexicon `bbbc71e9…`; sampled answers, seed 1101 unless stated). **Layer 1, refusal openings lost** under recipe B: OLMo −0.51, Llama −0.68, Qwen2.5-1.5B −0.85. **Layer 2, unsafe rate of direct answers before fine-tuning:** OLMo 0.18, Qwen 0.57, Llama 0.64, so the untuned Llama's 13.8% unsafe share is mostly compliant direct answers, not failed refusals. The opening switch carries 0.52 of OLMo's unsafe increase, 0.68 of Llama's and 0.64–0.78 of Qwen's. **Across OLMo's recipes** the unsafe rate within direct openings stays at 0.57–0.70 (B, the token-mean-only run, G), while refusal openings fall by 0.51 (B), 0.50 (token mean only), 0.34–0.36 (G, three seeds) and 0.20 (G at rank 16): the recipes act mainly through layer 1. Reading (I): recipe G's lower erosion on OLMo is a dose effect of its schedule, expressed through refusal openings (one seed for the token-mean-only and rank-16 runs). Observational only: no prefill test; positions counted in words.

- **ARC rule adopted** (FOCUS 20:50 UTC; text in §4.2). The disclosure's numbers: the OLMo recipe-G probe (seed 1101) was read at 15:17 UTC under the old rule (ARC TOO CLOSE; d = −2/189, a gain; task loss 2.100 → 1.612); the rule was proposed after that reading, so for that one unit it is not applied blind. **Scope (FOCUS 9 Oct 15:23 UTC):** extended to every family's Gate A read at its chosen learning rate, fixed before any such read; it replaces "at the common recipe".
- **Llama-3.2-1B Gate A at its rung (recipe B, LR 5e-5)** (read 9 Oct 23:16 UTC by the 20:31 UTC pre-registration; the user's "yes" 10 Oct 09:40 UTC: PASS under decision 25; FOCUS). E +48.00, +52.83, +44.17 pp (seeds 1101–1103, shared untuned unit, U_0 13.83%); **mean +48.33 pp** (edge band 43.05–53.61; partial); 2 × SD 8.69 < mean; every U above U_0. Answers end: pass for 1102 and 1103, TOO CLOSE for 1101 (the decision-25 case), so the rule-text verdict was TOO CLOSE with that one item. ARC a gain on every seed; task learned. Granite (pod) E +47.5, +48.0, +41.5 pp, direction agrees. HarmBench at the rung (pod hb-llama-b5e5-1, 10 Oct ~12:37 UTC): E_HB +28.3, +34.8, +26.8 pp, **mean +30.0** (each seed above 20 pp, same sign, at least half of E). The three adapters were later lost with deleted pods; the readings and judge files survive (FOCUS 15 Oct records sync).
- **7–8B rungs** (the Side Project's reads with the frozen family-probe bars; FOCUS 10 Oct 16:40 and 21:30 UTC, 11 Oct 00:10, 03:53 and 10:40 UTC). **OLMo-2-1124-7B-Instruct at 1e-4 (rung 1):** E +49.67, +37.17 pp (seeds 1101, 1102); two-seed mean **+43.42 pp** (SE 2.13; band 39.16–47.67; partial); the seeds differ by 12.5 pp, beyond the band (which covers evaluation noise only); Granite E +49.5, +40.0; E_HB +37.0, +27.7 (mean +32.4). **Llama-3.1-8B-Instruct:** rung 1 (1e-4) saturated (E +70.17); rung 2 (5e-5) E +52.50, +54.00, two-seed mean **+53.25 pp** (SE 2.40; band 48.46–58.04, which contains the 55 pp edge: TOO CLOSE; answers end also TOO CLOSE), accepted by the user on 11 Oct 03:53 UTC "at the edge"; Granite E +50.0, +55.0; E_HB +36.2, +37.8 (mean +37.0). Both families clear the 20 pp HarmBench line on every seed with the same sign and at least half the size, so "harmful compliance as the HarmBench classifier defines it" is allowed at 7–8B as at 1B. The same per-family pattern as at 1B (OLMo 1e-4, Llama 5e-5).
- **Side work at 7–8B: NOT this paper's result** (the concurrent submission's; overlap rule; FOCUS 11 Oct 10:38 and 12:54 UTC). Recipe G against recipe B at rung 1 (three seeds): OLMo-2-7B +44.1 → +50.3 pp; Llama-3.1-8B +60.7 → +59.2 pp; the 1B family × recipe interaction is "not shown" at 7–8B (I = +0.334, 95% −0.49 to +1.16). Halving the learning rate: OLMo-2-7B +31.9 against +44.1 pp, Llama-3.1-8B +53.25 against +60.7 pp; "both families respond to the LR alike" is not shown as equality (the side paper: "without establishing equal effects"; FOCUS 15 Oct records sync). FOCUS: recipe G's benefit "is family- and scale-dependent, so the paper must not present recipe G as a general fix".
- **Gate B anchor and recipe S** (FOCUS 10 Oct 14:01 UTC): 64 harmful and 64 safe anchors kept and sealed (text out of git); recipe S frozen, with recipe B's training identity.
- **Gate B wave 0a: the noise gate PASSED** (FOCUS 16 Oct 00:25 UTC; the frozen reading; report from `tmp/gate-b-olmo/gate_b_noise_gate.json`). r_j = U(a0, j) − U(R_j, j) = +0.67, +2.50, +4.33 pp (seeds 1101, 1102, 1103); SD(r) 1.83 pp; σ_eval,0 1.28 pp; **σ̂_gate = 1.83 pp ≤ 7** (governing reading plain); the LOCALIZED bar max(8, 4.177 × 0.707 × 1.83) = 8 pp ≤ 33.8; the fallback σ̂_gate (2.16 pp) also passes; σ̂_gate rests on 2 df. Report only: P4 mean r = 2.5 pp (SE 1.06) against the proportional prediction 5.6 pp and saturation's about 0; P(LOCALIZED) at the nearest simulated σ_pair (2.15 pp): 61 / 97 / 100% at 8 / 10 / 12 pp. Spend: USD 4.78.
- **Gate B wave 0b: the rankings are SEALED** (FOCUS 16 Oct 13:21 UTC). The retrain's q and e1 adapter fingerprints equal Gate A seed 1101's (fingerprint trigger false); the noise gate re-read with the retrain: PASS. The diagnostics took 3.04 h against the frozen upper of 2.0 h (cap 4.5 h). Sealed file `gate_b_rankings.json`, SHA-256 d6d5cd6b…. **Rankings** (Dolly categories, top first): attribution general_qa > summarization > brainstorming > open_qa > closed_qa > classification > information_extraction > creative_writing; representation general_qa > summarization > open_qa > brainstorming > classification > closed_qa > creative_writing > information_extraction; classifier (report only) brainstorming > creative_writing > closed_qa > classification > general_qa > open_qa > summarization > information_extraction. **pick_A = pick_R = general_qa; pick_F = classification;** pick_A = pick_F false; pick_R = pick_F false. Spearman ρ

with first5_share: attribution −0.17, representation −0.14, classifier −0.17; classifier informative. These are Gate B values seen before the stage-1 commitments are frozen (§4.9). Spend in waves 0a–0b: USD 11.26.

- **D5 = (a)** (the user delegated it; FOCUS 16 Oct 12:15 UTC): P1–P6 primary; the selector secondary and model-based (simulation, replay, predictions); the held-out selector comparison becomes its own later pre-registered study after a funding event; the one-example safety patch bounded in text (applied in §2 and §6).
- **HarmBench input length (FOCUS 8 Oct 10:14 UTC; revision 6b).** The development compliance check used `cais/HarmBench-Mistral-7b-val-cls` with "the official HarmBench classifier prompt and its 512-token generation truncation", greedy; the stage-1 pre-registration DRAFT revision 2 (HB-1) keeps "the 512-token classifier clip" and "the 2,048-token input refusal" (read by search only). So the development E_HB values come from clipped answers (the pod runs reproduce the Mac pipeline by a parity check; conflict 55). **Revision 6c:** HarmBench's `evaluate_completions.py` clips each generation to `--num_tokens` (default 512) tokens of the classifier's own tokenizer before judging, and its default classifier is the Llama-2-13b one (CITE-CHECK 36). FOCUS records the same 512-token clip for the development pipeline with the validation classifier; whether that pipeline counts tokens with the classifier's tokenizer was not checked here. **Revision 6d:** the Gate B pipeline counts tokens with the validation classifier's own tokenizer (from its pinned snapshot), with the beginning-of-sequence token, and truncates on the right (CITE-CHECK 36), so the development E_HB values and stage 1 use the same clip as HarmBench's script.
- **Status at revision 6g (FOCUS fetched 17 Oct at 09:59 UTC, head 76a08a0; FOCUS last changed in 1423a4c).** No FOCUS entry is newer than 09:37 UTC on 17 Oct. 76a08a0 adds the stage-1 build addendum §8 (09:59 UTC, decided by the lead under the user's delegation: the disjoint-source anchor sensitivity for risk 44 with the §6 limit sentence as its fallback, and the Self-Inf-N-style rank in the counting list, conditional on REQ-13); 158d2c2 added §7 (the PHTest draw rule) and the text-free PHTest source record.
- **Status at revision 6f (FOCUS fetched 17 Oct at 09:19 UTC, head ecdd23e, FOCUS last changed in 00b99a4; fetched again at 09:55 UTC, head 23848e0).** Two new entries: at 09:32 UTC (1be4bb1; time corrected in 71f2458) the A1 removal set is sealed (600 of 4,800 rows over all 8 categories; a report-only stability read: Spearman ρ 0.771 between the q and e1 checkpoint scores, top-600 Jaccard 0.434; "the paper names it as a limit"); at 09:37 UTC (48f3ac2; time corrected in 1423a4c) REQ-19 runs on the Gate A records: the six-seed over-refusal MDE against θ0 is 11.04 pp with prompts random, so the O-ext trigger fires and, by UW-07a, the primary over-refusal bank becomes the 100 development prompts plus 400 harmless PHTest prompts (158d2c2: PHTest downloaded); τ_L = 0.0067 nats. Development evidence, not in the body in revision 6f; revision 6g states the MDE in Appendix C.11 and the two stability figures, rounded, in §6 (the lead's rulings 5 and 6; decision 55).
- **Status at revision 6e (FOCUS fetched 17 Oct at 08:42 UTC, last changed in fcf79a2, head 80fd326; fetched again at 09:01 UTC, head 00b99a4).** 8dac6ad records the user's decision on Olmo-3-7B-Instruct (decision 48), and 00b99a4 a relayed instruction for the weekend of 17–18 Oct: the lead decides pending scope and design choices without asking the user, within unchanged money and safety limits. Neither records a run or a result.
- **Status at revision 6d (FOCUS last changed in fcf79a2; fetched 17 Oct at 08:16 and 08:34 UTC; the branch head moved from f9714b5, revision 6c, to 4446539).** No FOCUS entry is newer than 07:21 UTC on 17 Oct. 4446539 adds the lead's free checks of Olmo-3-7B-Instruct and a pre-registration and re-plan addendum for it (a DRAFT that adopts nothing; decision 48).
- **Status at revision 6c (FOCUS last changed in fcf79a2; the branch head is 0432940; fetched 17 Oct at 07:56 UTC).** Newest entries (17 Oct 07:21 UTC): the user's decisions in the lead's chat (UW-01 confirmed; UW-02 to UW-21 at the recommended defaults; the user's words quoted in FOCUS) and the lead's decisions LW-01 to LW-35 (HB-C primary; τ_L against $a_0$; REQ-12 and REQ-13 built), with the lead's two corrections to FAQ's mapping of the rule (Llama-8B's third Gate A seed after money beyond the wave-1 reserve; token-matched random arms with the token-axis analysis as their fallback). The stage-1 build addendum §6 (07:22 UTC; 98ff104, time corrected in 0294d1b) sets every builder switch from them. FOCUS has no Olmo 3 entry.
- **Status at revision 6b (FOCUS to head 6368d66, fetched 17 Oct about 07:25 UTC).** Newest entries: the stage-1 T1 bundle committed (9525005; recorded 02:42 UTC on 17 Oct): REQ-2, REQ-14's A1 removal path,

the label registry (34 T1 entries), REQ-8's plant settings, a golden A1 test; the lead's real A1 run waits for the builder's batch 1 (A1's report-only stability variants). The backup of the Mac-only model weights and states (01:05 UTC). The ICML 2027 goal (16 Oct 21:05 UTC). The commit log adds a stage-1 analysis-model, testing-family and reference-test DRAFT (3db753e; "mock review A checked by simulation on recorded noise") and the stage-1 pre-registration DRAFT revision 2 with a consolidated decision list (2ded100; "104 items; 21 USER items due before wave 1"); neither was read in full here. The user's decisions of 17 Oct (decision 48; decisions 26 and 28) are not in FOCUS yet.

- **Status (FOCUS, to the 16 Oct 21:05 UTC entry, the newest; head 416b780).** The goal for the main paper is ICML 2027 (the user, relayed by FAQ; 21:05 UTC); no fallback venue named, so TMLR or NeurIPS 2027 stays. The stage-1 commitments build: the lead decided L1–L16 (19:47 UTC; for example (L5) dose draws d = 2–4 at seeds 1100 + d; (L6) the list-or-markup rule by three frozen patterns; (L14) SD-600 removes rows it cannot rewrite; (L16) τ_L fixed from the three Gate A a0 seeds); REQ-21 (m) must list every Gate B value seen. Wave 1 waits for those commitments; it needs USD 83.91 (13:21 UTC); the balance was USD 91.94 at 19:07 UTC. Pods: w0b-2 holds the retrain's states (only copy; keep); w0b-1 was deleted on the user's yes (19:08 UTC); the user said "stop asking me for approval and go ahead and do these things" (19:06 UTC; the lead reads it as doing planned steps without approval questions, with the frozen money rules, the guard, the USD 10/h limit and the launch check unchanged). The P5 lexicon pin is filled only at `--final`, with every sealed phase re-run and compared (11:08 UTC). Earlier (FOCUS 15 Oct records sync): the side paper "Same Recipe, Different Fate" went to ARR October 2026 as #13386 on 13 Oct; three grants were submitted on 13 Oct, none awarded; General Implementation VI leads the main study since 15 Oct 19:56 UTC.
- **Status at revision 5e (history; FOCUS to 9 Oct 17:22 UTC).** The user accepted decision 23's wording package and extended the ARC rule (15:23 UTC); the blind human read is recorded (15:05 UTC). **Pod price limit (the user, 15:29 UTC):** every kit refuses to create a pod priced above USD 10.00/h, or with no finite price, or stops it at once. This replaces the old USD 1.75/h limit and the "no hourly limit" of 8 Oct 20:17 UTC; the balance-based budget guard is unchanged (it plans a new pod at USD 1.85/h); brief revision 17 amends [D20]. Recipe B is the main recipe for Gate B and stage 1; Gate B DRAFT revision 5 is to stay on recipe B, add the [D14]–[D19] rules and the [D20] price constants, and recompute the money thresholds; the repository's DRAFT is still revision 4b. The two report-only Llama recipe-B seeds finished (12:18 UTC). At 09:30 UTC four idle pods were deleted and 15 stopped pods kept the fine-tuned adapters; the Llama recipe-B pods were stopped after their runs. Llama's rung-2 probe under recipe B (5e-5) was read at 16:59 UTC (entry above; Granite not yet run; it goes on the next Granite pod run); the user has not decided whether to accept it (decision 25). **Granite on RunPod (FOCUS 16:48 UTC; the user: "just use the gpus for granite as well"):** the Granite judge's RunPod launcher is committed (ff9196a). Its parity rule was fixed, and PARITY.md's SHA-256 (c4fc455e…) recorded, before the first Granite pod launch. The rule: on the four OLMo-2-1B Gate A units (untuned and e1 seeds 1101–1103; 800 greedy harm answers with Mac Granite labels), the tokenizer fingerprint and every item's input hash are identical, every item is judged, and each unit has at least 196 of 200 identical labels and a Yes count within 3 of the Mac's; all four units must pass. If parity fails, pod and Mac Granite numbers are never mixed in one contrast. By default, each Granite launch first judges 200 known answers (OLMo recipe B, seed 1101, e1) and stops unless at least 190 of 200 labels match the Mac's and the Yes count is within 5. **Granite parity: PASS (FOCUS 17:22 UTC, by the pre-set rule).** The first pod (16:49 UTC) got no network address and was stopped after 20 minutes without SSH (about USD 0.6); the retry, on another machine, passed SSH. Identical labels per unit: 200, 200, 199 and 200 of 200 (one no → yes on e1 seed 1102); all 800 input hashes and the tokenizer fingerprints are identical. Pod-made Granite files may now be used like Mac-made ones; this was recorded before the Gate B freeze, so Gate B's Granite files can come from the pod (DRAFT revision 5 rule). Next (FOCUS 16:48 UTC): Granite for the Side Project's Llama recipe-B seeds and Gate B's judge files. Safe LoRA: the lead reads the paper in full and fixes its settings in the stage-1 pre-registration; the post-hoc transform is built with tests and reviewed before use. B+Qi's engine change comes in a later round, committed before stage 1 is frozen. The fiction side study (Side Project; report-only) is approved at recipe B, up to about USD 20 in total (15:23 UTC; earlier about USD 10–16), and is not part of this paper. The Qwen2.5-1.5B recipe-G run is binned; the HarmBench add-on for the recipe-G side runs is dropped (10:07 UTC). Alpaca waits for Gate B and its own design note. The Gate C anchor list is sealed (8 Oct 15:53 UTC).

### Conflicts between the sources, and how this revision resolves them

1. **Spec RQ2 (the map co-primary, the headline figure) vs the user's decision (map secondary, model-based).** Followed the user's decision: the measured comparison leads the title, abstract, §1 and contributions; the map is §3.5, secondary, with predictions frozen before Gate B's category runs and scored on Llama. Figure 1 is now the repair comparison.
2. **Spec repair menu (category, cluster or row removal by heuristic, filters, attribution, representation similarity; random vs selected mix; "10 rows"; recipe-only) vs the adopted stage-1 design.** Followed the adopted design: six primary arms (A1, L1, H1, S1, S3, S4) at 600 removed or 48 added rows; recipe G is a reference, not a repair; category removal stays in Gate B; cluster menus exist only as the planted sweep's nested cells; the "10 rows" dose is an option, not the plan.
3. **Spec "Qwen is the fragile contrast" vs "do not claim Qwen is uniquely fragile at 7B".** Qwen appears only as context in §4.1 and §4.10 (its results are in this note), with Guo et al.'s Table 2 showing a Qwen2.5-7B LoRA cell that barely erodes.
4. **Previous draft vs spec on O's reference.** Spec and amendment: the untuned model (a0 reported). Applied.
5. **Previous draft's rewrite and self-distill fixes.** The amendment drops them from the core menu; one rewrite arm runs in the DIFFUSE branch (§4.6).
6. **Previous draft's co-primary side-effect prediction.** Secondary, on the add arms only (§4.10).
7. **The brief's "keep the Guo Table 2 wording".** Neither previous file contained it; it is now in §2.
8. **Spec §9 vs §5 on the Llama probe** ("queued" vs "running"): not paper text; noted above.
9. **The lead's message vs FOCUS on the common-recipe rule (revision 3).** The lead: if Llama under recipe G "reads partial", G is the common recipe; "if not", both families step down the learning rate together. FOCUS 17:36 UTC: G is the common recipe if Llama's probe *qualifies* (meets every frozen family-probe bar, not only the partial class); both families step down only if Llama *saturates*; if no rung qualifies for both (for example Llama weak, or OLMo below 20 pp where Llama turns partial), no common recipe exists and the user decides. Followed FOCUS (§4.1). **Superseded (revision 5):** FOCUS 9 Oct 09:29 UTC replaces the common-recipe rule with the per-family learning-rate rule.
10. **"A fourth reference" vs three references.** FOCUS 17:36 UTC item 4 calls G+Qi "a fourth model-level reference", but the same item makes recipe G the base, so the references are three: the safety prompt, adapter scaling and G+Qi. Followed that list: G+Qi replaces "a standard recipe" in every reference list. **Superseded (revision 5):** FOCUS 9 Oct 10:01 UTC lists five mandatory references (recipe G among them, with B+Qi and Safe LoRA), and 13:16 UTC puts the Qi arm on recipe B.
11. **OLMo's recipe-B number.** The lead gives +45.1 pp (Gate A, 3-seed mean); FOCUS 19:24 UTC gives +43.5 pp (seed 1101, the same-seed contrast with recipe G). Both are correct; the development evidence gives both, labelled.
12. **The Llama rule.** The 13:53 UTC amendment let Llama's next recipe change "LR or rank"; the 17:36 UTC common-recipe rule changes the learning rate only, for both families. Followed 17:36 UTC (§4.1, §4.8, §6). **Superseded (revision 5):** the per-family rule (FOCUS 9 Oct 09:29 UTC) also changes the learning rate only, now of recipe B, and per family.
13. **"Llama replicates" (spec and revision 2).** Removed, as the lead asked: FOCUS 16:50 UTC reads Llama saturated under recipe B, and its recipe-G probe had not been read at this revision. The claim rule for a replication stays in §4.8.
14. **OLMo recipe-G seed 1103 (+26.5 pp; revision 4).** The lead gives seeds 1101–1103 as +29.0, +31.0 and +26.5 pp. FOCUS (to 21:37 UTC) and the results record hold only the interim two-seed reading (1101, 1102); seed 1103 is not recorded there. The kit's pair reading for seeds 1101 and 1103 (`tmp/family-probe-olmo2-1b-g-seed1103/reading-olmo-g-1101-1103-nojudge.json`) shows a two-seed mean of 27.75 pp, which agrees with +26.5; the three-seed reading file beside it is empty. Resolution: the number appears only in the development evidence; the mean +28.8 pp is marked as the drafter's arithmetic. **RESOLVED (revision 4a):**

seed 1103 is now recorded in FOCUS ("OLMo under recipe G, all three seeds in", interim) and the results record (commit 8fccce2), so the "lead's figure" label is dropped.

15. **"Mechanism note §4" (revision 4).** FOCUS and the lead call the first-token loss-mass account the mechanism note's §4. In the note, the loss-share table is in §5 ("Training data and the Gate B link"), with first-5 shares .181 (per-row) against .054 (token mean); FOCUS 20:01 UTC quotes 17.7% against 5.3%. Resolution: these notes say "the mechanism note's first-token loss-mass account" without a section number and quote FOCUS's numbers (FOCUS wins); the body quotes neither.
16. **ARC in the body (revision 4).** Revision 3's §4.2 said ARC "sets no limit"; the frozen family-probe and Gate A bars do check ARC (a drop of at most 0.03), and the 20:50 UTC rule resolves its TOO CLOSE. Followed FOCUS: §4.2 states the bar and the rule.
17. **"The schedule carries all of it" (the lead) against FOCUS's one-seed caveat (revision 4).** FOCUS reads one seed and labels "G is, in effect, a lower dose" as interpretation. Resolution: the body states only design facts (recipe G's schedule halves the mean learning rate, so recipe B is the stronger dose) plus a placeholder for the run's result; the attribution of the reduction to the schedule stays in the development evidence, marked one seed.
18. **"If FOCUS records a newer decision, follow it" (the lead's brief; revision 4).** FOCUS records no step-down of the learning rate for both families: only the conditional rule (21:04 UTC) and the early Llama seed 1103 (21:37 UTC). [PLACEHOLDER: the common rung] stays, and §4.1 adds only the too-close rule that FOCUS cites from the frozen amendment. **Superseded (revision 5):** the user decided (FOCUS 9 Oct 09:29 UTC); [PLACEHOLDER: the common rung] is gone.
19. **"The two-layer mechanism account, as FOCUS states it" (the lead's brief).** FOCUS has no entry by that name. Its numbers are in FOCUS's 8 Oct 15:17 UTC (recipe G: G keeps refusal openings rather than making direct answers safer) and 16:50 UTC (Llama: "already unsafe in 64% of its direct openings (OLMo 18%)") entries, and in the mechanism note's addenda (17:14 and 21:56 UTC); the name "two layers" comes from the Side Project's arXiv note (branch claude/wonderful-greider-d1aa5b, commit c51cf39, read-only). Resolution: the body states the two layers as the two factors of §3.6's opening-switch term, without numbers; the numbers come from FOCUS and the mechanism note, which agree.
20. **Llama recipe-B seed 1102 (+61.3 pp).** FOCUS 12:18 UTC records that seed 1102 finished but gives no E for it (it gives +54.0 for seed 1103). The number is the kit's frozen reading (`tmp/family-probe-sp-llama32-1b-b-seed1102/reading-llama-b-1101-1102-nojudge.json`: E per seed 62.5 and 61.33) and the Side Project's results note (side branch, commit fd48c89); it matches the lead's figure. It appears only in the development evidence.
21. **The family × recipe interaction** is recorded only on the Side Project's branch, not in FOCUS or the main checkout's records. Resolution: development evidence only, with its limits; decision 17.
22. **"Fixed now before any Llama run at the main recipe" (FOCUS 09:29 UTC)** against Llama's recipe-B probe of 8 Oct 16:50 UTC and the two report-only recipe-B seeds run after the decision. The rule postdates Llama's first recipe-B probe and the ladder; the drafter reads FOCUS's phrase as "before any further Llama run at the main recipe that the rule governs". Resolution: §4.1 and §6 say the rule "was fixed after a development ladder was read"; decision 18 names the probe that would come next.
23. **The ARC rule's scope.** FOCUS 20:50 UTC scopes it to "the common recipe", which no longer exists. Resolution: §4.2 keeps the rule and its disclosure, drops the scope phrase and marks the scope [TO BE FINALISED] (decision 16). **RESOLVED (revision 5c):** FOCUS 15:23 UTC extends the rule to every family's Gate A read at its chosen learning rate; §4.2 states the scope.
24. **The E-scaled floor (FOCUS 8 Oct 20:17 UTC) against recipe B as the main recipe.** At recipe B it equals the 8 pp floor, so it adds nothing. Resolution: dropped from §4.6; decision 21.
25. **"Describe recipe G as a dose-lowering fix" (the lead's brief) against the Llama data.** Recipe G barely moves Llama, while halving recipe G's learning rate moves Llama by about 29% (development evidence). Resolution: the body gives the design fact (recipe G's schedule halves the mean learning rate), calls G "the

recipe-only fix" and "a gentler learning-rate schedule", and does not say that G lowers erosion in every family; these notes keep "dose-lowering" for OLMo, where the token-mean-only run supports it (risk 26).

26. **Numbers only in the notes, against the §3.6 sentence "where u_N0 is high and u_R0 low, a lost opening comes close to an unsafe answer".** The sentence is conditional and follows from the decomposition's first term, $(p_0 - p_1)(u_{\mathrm{N0}} - u_{\mathrm{R0}})$; it states no result. Kept.
27. **The secondary length-rule arm.** Revisions 3–4 named the mirror (the length rule under recipe B's per-row loss) because recipe G was main. With recipe B main again, the arm reverts to revision 2's "under a token-mean loss" (decision 11).
28. **"Numbers only in the development evidence" (the lead) against the wording rule's caveat (FOCUS 15:05 UTC).** The lead puts the read's numbers only in the development evidence, not in the abstract or the contributions; FOCUS requires each harm sentence to carry the caveat that "a blind human read rated about a third of judge-unsafe answers as usable help". They agree on the abstract and the contributions, but the caveat is itself a rounded development number in the body. Resolution (FOCUS wins on the wording): the body carries only the rounded caveat ("most … went along", "about a third", "one reader", "during development") in §4.2 and §6; counts, intervals and groups stay in the development evidence. "What this is" names this exception to "Nothing in the paper body is a result". **Revision 5c (FOCUS 15:23 UTC, item 3c–d):** the caveat is no longer in every harm sentence. §4.2 states it once, as a fact about the measures, and §6 once, as a limitation, each pointing to Appendix A; harm sentences say only "harmful compliance as the HarmBench classifier defines it". The pooled counts, intervals and limits are in Appendix A (part of the paper, labelled development evidence); group counts stay in the development evidence here. The body still gives the rounded caveat ("most … went along", "about a third") and no count (conflict 34).
29. **"Judge-unsafe".** The read sampled Qwen3Guard-Unsafe answers (13 of 39 usable help), while the claim rule applies the caveat to HarmBench sentences. On the 12 OLMo answers that HarmBench labels "yes", 4 were usable help, also about a third, so FOCUS's phrase "judge-unsafe answers" holds for both judges as measured; the HarmBench subset is small and OLMo-only. Kept FOCUS's phrase.
30. **"Of 39 Qwen3Guard-Unsafe answers" (the lead, FOCUS) against "40 labelled Unsafe" (the pre-registration).** Both are right: 40 Unsafe-labelled answers were drawn and one was skipped, so 39 were rated; with the 10 Safe controls the sheet held 50. The development evidence says so.
31. **Gate B's blind read, "DONE" (the lead, FOCUS 15:05 UTC) against the repository's DRAFT (still revision 4b: §12 item 5, "recommended").** FOCUS says DRAFT revision 5 reports the read as done, with its sample difference, and the verdict stays on U. Resolution: §4.6, which summarizes revision 5 [NOT YET FIXED], says it is done, on a different sample from the one first proposed, and that the verdict reads U alone (FOCUS wins). Revision 5c points this sentence to Appendix A, which describes the difference.
32. **Decision 23's letters.** Revision 5b's decision 23 listed (a) "adverse", (b) "safety" in the title, (c) whether §5 or an appendix reports the counts and (d) a second rater. FOCUS 15:23 UTC lists (a) "adverse", (b) the title keeps "safety" and the abstract names the measures, (c) the caveat once in the measurement section and the limitations, with an appendix, and (d) the counts in an appendix. Followed FOCUS's letters; revision 5b's (d), which FOCUS does not decide, stays open as decision 24.
33. **"'Significantly harmful' becomes 'adverse'" (the lead's brief) against the label's significance condition.** Dropping "significantly" would let §1 read as if any adverse estimate made a repair *worse*; revision 5b's decision 23a, which the user accepted, proposed "significantly adverse". Resolution: "significantly adverse" in §1 and §3.2, "a significant adverse result" in §4.3, and "adverse" alone where the phrase names only a direction ("adverse direction", "adverse repair"). If the lead wants "adverse" alone in §1 and §3.2, §3.2 still carries the Holm-adjusted 5% level.
34. **"The body may refer to 'a blind human read (Appendix A)' without giving numbers" (the lead's brief) against the caveat's wording "about a third" (FOCUS 15:05 UTC, placed by 15:23 UTC item 3c).** Read the brief as allowing number-free references (as §4.6 now makes) and FOCUS as keeping the rounded caveat in its two places; item 3d moves the counts, not the caveat. Resolution: §4.2 and §6 keep "about a third" (§6 also "most … went along"); no count, interval or group is in the body. If the lead wants a number-free caveat,

replace "about a third of judge-unsafe answers as usable help" by "most judge-unsafe answers as giving no usable help" in §4.2, and "but only about a third gave usable help" by "but most gave no usable help" in §6.

35. **"The abstract" in decision 23a.** The abstract names the *worse* label but has never contained "significantly harmful", so 23a changed nothing there; its change is 23b.
36. **The FOCUS entry's time.** The lead's brief first named the entry "USER DECISIONS 15:12"; FOCUS stamps it "USER DECISIONS, 2026-10-09 15:23 UTC", and the lead then corrected the brief (content unchanged). These notes cite 15:23 UTC.
37. **"Llama-3.2-1B's rung is 5e-5" (the lead's brief, §4.1) against FOCUS 16:59 UTC ("the answers-end TOO CLOSE is the user's decision"; the lead recommends accepting rung 2 "for planning").** The body sentence reads as settled while the user has not decided. Resolution: the brief's wording is kept, because its parenthesis labels the rung as one probe seed of development evidence, to be confirmed by stage 2's Gate A, and the pre-registration (FOCUS 15:31 UTC) makes rung 2 Llama's main-recipe rung if the probe reads partial; these notes mark it provisionally filled, pending the user (decisions 18 and 25). If the lead wants the body to say so, write "Llama-3.2-1B's provisional rung is …" (one more word). **RESOLVED (revision 5e):** the lead asked for it, because the user has not decided; §4.1 now says "Llama-3.2-1B's provisional rung is …".
38. **Granite.** The lead's brief gives the E class and the answers-end bar but not the second judge. FOCUS 16:59 UTC records that Granite has not run on this probe, and the frozen family-probe bars include the second judge's direction. Resolution (FOCUS wins): the development-evidence entry, decision 18, decision 25 and the placeholder table say Granite has not run.
39. **§4.1's "a too-close probe gets one more seed" against the answers-end TOO CLOSE.** In the frozen amendment (7 Oct 21:35 UTC) one more seed follows only a TOO CLOSE on the E class; a short-answers TOO CLOSE goes to the user, and the lead proposes that stage 2's Gate A seeds re-read answers end instead. §4.1's parenthesis is about the E class, but a reader could apply it to any bar. Not changed (this revision edits only §4.1's rung sentence). If the lead wants it exact: "(§4.6; a probe too close on E gets one more seed; any other too-close bar goes to the user)". **RESOLVED (revision 5e):** the lead asked for the exact rule; §4.1 now says "(§4.6; a probe too close on $E$ gets one more seed; any other too-close bar goes to the user)". Checked against the amendment's text: one more seed follows only the E class's edge band; the short-answers bar, ARC (another seed cannot resolve it: the questions are the same) and unscored answers go to the user; "any TOO CLOSE goes to the user with the numbers". Left out for length: the extra seed runs after the user's go, and a two-seed mean that is still too close goes to the user.
40. **The balance figure and the Granite parity pod (the lead's brief for revision 5e) against FOCUS.** If risk 17 needs a figure, the brief offers "the latest account reading in FOCUS (USD 54.80 at 16:49 UTC on 9 Oct, before the Granite parity pod)". FOCUS (to the 17:22 UTC entry, commit 2837eaf; read at 17:31 UTC) has no such reading; its latest balance is "USD 57.11 before" the 15:35 UTC launch (the 15:31 UTC entry). The brief also gave the Granite parity result as pending (the first pod got no network address and was stopped; the retry passed SSH and the GPU check); FOCUS 17:22 UTC, committed while this revision was written, records parity PASS. Resolution: risk 17 needs no figure, so it gives none; the status entry follows FOCUS (PASS), with the first pod's failure as FOCUS records it.
41. **The pipelines.** The brief says to reuse scratchpad scripts (`paper_to_pdf.py`, `scripts/body_pages.py`, `scripts/notes_md_to_tex_rev5c.py`, `scripts/wrap_tex.py`, `rev5d/splice.py`); none existed (the scratchpad held only empty folders from the earlier revisions). Re-created in `rev6/scripts/`: `body_pages.py` (the same definition; it gives 9.36 pages on revision 5e's PDF), `notes_md_to_tex.py` (Markdown notes to LaTeX in revision 5e's Appendix B style, with line wrapping), `paper_to_pdf.py` (Markdown to HTML with KaTeX, printed by headless Chrome). The Markdown PDF may differ slightly in look from revision 5e's.
42. **Appendix C's letter.** The brief asks for "Appendix C" beside the existing A, with the notes as B. In the LaTeX build the notes stay Appendix B and the design details are Appendix C, after the notes; when the notes are deleted, C becomes B (it is referenced by `\ref`). In the Markdown file the notes sit at the top, so Appendix C follows Appendix A; the PDF pipeline puts the notes last.

43. **"Nothing in the body is a result" against the Gate B status.** The brief asks the body to say that the noise gate passed and the rankings are sealed. The body says so as a status line, with no number and no category; "What this is" lists it as an exception.
44. **The Gate C format clause** goes beyond the brief's wording (decision 29); the frozen pre-registration requires it to be reported whatever Gate B reads.
45. **"Its Gate A PASSED on 10 Oct" (the brief) against the rule-text verdict.** FOCUS 9 Oct 23:16 UTC reads TOO CLOSE by the rule text, with one item (seed 1101's answers end), and the user recorded PASS under decision 25 (10 Oct 09:40 UTC). Both hold: the body states the rung plainly and discloses "accepted on a too-close bar"; these notes give the detail.
46. **"The one-example safety patch: bounded in text (related work and a limitation)"** (FOCUS 16 Oct 12:15 UTC). It needs a citation; revision 6 fetched Safety at One Shot by API (also on the coordinator's list). The patch is described from its abstract, after harmful fine-tuning; the body does not say it works after benign fine-tuning.
47. **GradShield's venue.** The coordinator: "GradShield is now at NeurIPS 2026". The arXiv API gives no venue; OpenReview shows only a withdrawn ICLR 2026 submission; the NeurIPS 2026 accepted-papers JSON (fetched by the literature sweep from neurips.cc) lists it as a poster. The brief allows new BibTeX only from the arXiv API, OpenReview or PMLR, so the entry keeps its arXiv form with the note "NeurIPS 2026 (poster)" (CITE-CHECK 8).
48. **DualSelect's wording.** The sweep's replacement text says "six data selectors"; the coordinator's facts give no count; the body gives none. The sweep put DataRx and SABA in §1's sentence; revision 6 puts them in §2 to hold the length.
49. **The 7–8B rungs as design (the brief) against their source.** The 7–8B probes were read by the Side Project under the main study's frozen family-probe bars, and FOCUS records the rungs as the main study's ("7–8B main-study rungs", 11 Oct 10:40 UTC; the user's decision 11 Oct 03:53 UTC). The body states the rungs as design; the side paper's same-recipe 7–8B numbers stay out (overlap rule).
50. **The overlap rule's scope.** The brief names the family × recipe and dose findings; the side paper's abstract also lists the opening layers and the blind read. Revision 6 cites the concurrent submission for those too (Appendix C.8; Appendix A, decision 30). It keeps the blind read in Appendix A as this paper's check of what U measures.
51. **"Don't add them to the body's reference list yet" (the coordinator's addition 3) against Safety at One Shot in §2 and §6.** Read as the body's list of reference arms (the five references of §4.4), not its bibliography: Safety at One Shot was cited in §2 and §6 earlier in this revision, because D5 = (a) bounds the one-example patch "in text (related work and a limitation)" and the coordinator's addition 2 asked for the citation if verified. It is not a reference arm. If the coordinator meant the bibliography too, delete the two clauses (§2's last sentence and §6's second sentence) and the BibTeX entry `zhang2026oneshot`.
52. **Option B (decision 28) against the body's stage-1 framing.** Option B makes OLMo-1B's stage 1 development and puts the confirmatory comparison at 7–8B; the body still calls OLMo-2-1B the stage-1 model and lists the testing order of revision 5. The coordinator asked only for the 7–8B tier's description; the rest of option B is for the lead and the user before the stage-1 pre-registration. **Revision 6c:** UW-21 (FOCUS fcf79a2) keeps OLMo-2-1B's P1–P6 as the confirmatory primary family under any re-plan option, which matches the body's stage-1 framing.
53. **The user's decisions of 17 Oct against FOCUS.** The brief relays four decisions (Olmo-3-7B-Instruct added; "ambitious default with fallbacks"; the "Compute-Matched" title kept with token matching; option B with C as the fallback). FOCUS at origin head 6368d66 records none of them, and the stage-1 pre-registration DRAFT revision 2 marks items that rely on the "ambitious default" rule [USER-CONFIRM]: "The lead's chat has not confirmed this rule" (its checklist item FC-1). Resolution: applied as relayed and recorded as relayed (decisions 26, 28, 48); the lead records them in FOCUS. **Revision 6c: resolved.** FOCUS fcf79a2 records the user's confirmation of the rule in the lead's chat (07:21 UTC; FC-1 closed) and UW-14 (the title and token matching). Olmo 3 was not among them: the lead then reported it as not yet confirmed (decision 48; the user confirmed it

later, FOCUS 8dac6ad, revision 6e). Option B with C as the fallback is still recorded only as relayed; UW-21 keeps OLMo-2-1B's P1–P6 confirmatory under any re-plan option (conflict 52).

54. **The Qwen clause against the overlap rule.** That Qwen2.5 saturates under recipe B is part of the concurrent submission's same-recipe family contrast. The brief asks for one neutral clause with the reason. Resolution: the clause gives the class ("saturated … (above 55 pp)") and no number, keeps the citation for the comparison, and "What this is" lists it as a fifth exception to "nothing in the body is a result". The reason is FOCUS's: the 7 Oct 17:44 UTC collapse of Qwen2.5-7B (4.8% → 87.0%), the frozen bars' "E > 55 is saturated, like Qwen" (19:14 UTC), and the user's "Llama/OLMo only" (20:58 UTC). Not stated in the body: Qwen was dropped before the per-family rule existed (risk 56). **Revision 6c:** the reason now sits in §6 as one sentence (the lead's ruling 2), with the model (Qwen2.5-1.5B), the recipe and the rate but no erosion number and no date; §4.1 points to §6, and "What this is" moves the exception from §4.1 to §6. The body now says that Qwen was dropped before the per-family rule existed.
55. **The HarmBench input (review B W15) against FOCUS.** Review B says the claim rule applies the classifier to answers of up to 2,048 tokens. FOCUS 8 Oct 10:14 UTC records the 512-token clip for the development check, and the stage-1 DRAFT keeps it. The brief asks to mark the length [TO BE FIXED in the stage-1 pre-registration]; §4.2 does so neutrally ("clipped to 512 tokens or in full (up to 2,048)"). If the lead keeps the clip, the sentence can state it as design. **Revision 6c: resolved** by the lead's ruling 1: the clip is design (§4.2; Appendix C.10), as HarmBench's evaluation script does.
56. **Which HarmBench classifier.** FOCUS names `cais/HarmBench-Mistral-7b-val-cls` (the validation classifier); review B cites `cais/HarmBench-Llama-2-13b-cls` as "the official classifier for text behaviors in HarmBench". The body says only "HarmBench's classifier". Name the model in Appendix C or the claim rule before submission, and check which one HarmBench's paper reports (not checked here). **Revision 6c:** the body names the validation classifier (§4.2), and Appendix C.10 gives both ids; the test classifier is a candidate sensitivity check only. HarmBench's evaluation script defaults to the Llama-2-13b classifier; which classifier the HarmBench paper's main tables use was not checked here (CITE-CHECK 36).
57. **Safety at One Shot in §2, §4.3 and §6.** D5 = (a) bounds the post-hoc one-example patch in text (not a reference; no claim covers it). Revision 6b's §4.3 lists its data-only analogue (the example added to the training data) as a candidate arm [TO BE FIXED in the stage-1 pre-registration]. They are different repairs, so §6's sentence ("Post-hoc repairs other than the references, such as the one-example patch (Zhang et al., 2026a), are not compared") still holds if the arm is added.
58. **"Five references" against the planned sixth.** The abstract, §1, §3.2 and §4.4 say five; §4.4 adds a sixth, the lower learning rate, as [PLANNED; to be pre-registered]. Resolution: "five" kept while the sixth is planned; Table 1's caption now says "references". If the pre-registration adds it, update every count, including §1's "dominant only if it beats five references".
59. **The customer and the safety prompt.** Table 1's legend and §4.4 let the customer run the safety prompt, because a test-time prefix is in the customer's own requests. A provider's safety gate (Microsoft, 2026) may evaluate without it, so it may not help the customer pass the gate; the body claims nothing about the gate. Decision 46.
60. **"Controls only the data".** §1 said the owner "often controls only the data" and §6 that a closed-service customer "can change only the data"; revision 6b's §3.1 says the customer can also set the learning rate and epochs (brief item 2; review B W3). Resolution: §1 and §6 now say "the data and a few training settings" and "the data and some hyperparameters"; the abstract's conditional "When only the training data may change" is kept.
61. **"Smaller differences stay inconclusive" (§6) against §3.2's rules.** A difference smaller than δ with a tight interval is labelled *equivalent*, not inconclusive. Resolution: §6's clause is cut (brief item 7).
62. **The brief's item 5 ("references on all six a0 seeds") against §4.4's three seeds.** §4.4 still says three seeds for the safety prompt, adapter scaling and Safe LoRA, and decision 19 defaults to seeds 1101–1103. Resolution: the TO BE FIXED sentence names both options ("three, or all six a0 seeds"); the stated three stay until the pre-registration rules.

63. **Body pages.** The measure is unchanged at 7.96 pages and the heading sits at the same height as in revision 6, because the cuts balanced the additions to within a line; a probe with one more line gave 8.00 pages ("What changed in revision 6b", Length).
64. **The brief's scope against the stage-1 decisions.** The brief for revision 6c applies two of the body's ten [TO BE FIXED] items (token matching; the HarmBench input) and keeps three items for the freeze. FOCUS fcf79a2 also settles the analysis model (UW-02, LW-01), the closed testing family (UW-04), the over-refusal baseline of *worse* (UW-05, rule R1.3), one primary τ_O (UW-06) and the label order (UW-08 (a)). Resolution: followed the brief; the body keeps these markers (decision 49). **Revision 6d: resolved** by the lead's rulings 1–3 (decision 49); decision 53 lists what is still not applied.
65. **Seeds.** Table 1 and Appendix C.3 say random removal at eight seeds and no repair at six; UW-03 (layout D1; addendum §6 item 2) runs $a_0$ and $a_R$ at seeds 1101–1107 (seven each; R_1108 is not run) and the method arms at 1102–1107. Not changed: outside the brief (decision 49). **Revision 6d: resolved** (the lead's ruling 2): Table 1, Appendix C.3 and §4.4 (conflict 70).
66. **The over-refusal bank.** §4.2 says 100 prompts; UW-07a adds 400 harmless PHTest prompts to the primary over-refusal bank if REQ-19's free check shows a six-seed MDE above 5 pp (fallback: the development 100 only). The addition is conditional, so §4.2 stays (decision 49). **Revision 6d: resolved** (the lead's ruling 3): §4.2 states the conditional extension and "never from the held-out bank". **Revision 6g:** the extension fired (REQ-19, FOCUS 09:37 UTC); §4.2 states the 500-prompt bank (decision 55).
67. **R_tok's details.** The W1 builder extract's construction of R_tok (W1-03: per-unit walks in a sealed order, seeds 1102–1104, one-sided bounds only, never in P1–P6) is marked as its writer's interpretation, and the token axis is the lead's reading (W1-02 item 4). Addendum §6 item 5 says only "a secondary arm at 3 seeds, funded tier", and FOCUS's UW-14 "pre-registered secondary arms in the funded tier … fallback: the analysis only". Resolution: §3.2 and Appendix C.3 state those facts, plus the analysis's report-only status and its dose curve from W1-02 (marked binding) and the pre-registration DRAFT's token-accounting item; no axis, construction or test is stated (decision 51). **Revision 6d:** the axis is stated (the lead's ruling 4: total row tokens, target tokens beside it); R_tok's construction and test are still not stated.
68. **"As HarmBench's own evaluation script does" (the brief).** Checked on the script (CITE-CHECK 36). The body says only "512-token window"; Appendix C.10 says the rest. The script's default classifier is the Llama-2-13b one, not the validation classifier that the study uses; the clip is the shared part.
69. **Which Qwen model.** The lead's ruling names Qwen2.5-1.5B, the main-study candidate that the user dropped on 7 Oct (20:58 UTC). Qwen2.5-7B also saturated (FOCUS 7 Oct 17:44 UTC) and is also out. §4.1 says "Qwen2.5 … is outside this study"; §6 names only the 1.5B model, as the ruling does. The later Qwen runs (FOCUS 9 and 11 Oct) are side work for the concurrent submission, which §6 cites as the only later evidence.
70. **Rulings 2 and 6 at §4.4.** Ruling 2 fixes "any other place that says … $a_0$ at 6"; ruling 6 keeps "the post-hoc reference seeds". §4.4's "(three, or all six $a_0$ seeds)" is both. Resolution: the open choice is kept (three, or all), and the $a_0$ count follows D1: "(three, or all seven $a_0$ seeds)". US-01 gives the same reading ("a0 1101–1106 (1101–1107 under D1)").
71. **The full-bank clause against ruling 3.** §3.2's [TO BE FIXED] sentence asked whether feasibility uses "the full XSTest and OR-Bench-Hard banks"; their remainder is the hidden over-refusal bank, which UW-07b and ruling 3 exclude ("never from the held-out bank"). Resolution: the clause is cut; the sentence keeps only the open non-inferiority question.
72. **The dollar sign in the ruling's sentence.** The ruling writes the check's cost as zero dollars with a dollar sign, which is a math delimiter in this file and in KaTeX, and the body states no money. Resolution: "a free, pre-registered check".
73. **The planning MDE against HB-C and Φ.** Not in the rulings, but the old figure ("after Holm's first step … about 5 pp") belongs to the replaced model. Resolution: the number is dropped and the placeholder kept (decision 52).
74. **"S-a and S-aW reported" (FOCUS LW-01) against "pre-registered sensitivities" (addendum §6 item 1; ruling 1 (a)).** The ruling and the addendum agree; FOCUS's "reported" is read as the same thing. The body

says "pre-registered sensitivity analyses".

75. **"No file paths" (ruling 5) against the script's name.** Ruling 5 names `evaluate_completions.py`; Appendix C.10 names that public script by its file name, with no directory, repository path or hash.
76. **The second half of the label-order marker.** Revision 6b's sentence also left open "whether a significant difference below δ gets its own label". UW-08 (a)'s recommended text, which the user took, says the row prints "significant, below the SESOI". Resolution: Appendix C.11 states it, with SESOI written out as "the smallest effect of interest"; the body sentence keeps only ruling 1 (e)'s words.
77. **"So an adverse repair is never reported as inconclusive" (§3.2) against R1.3.** Under R1.3 a repair that is significantly above its comparator on over-refusal but feasible against θ0 is not *worse*, so it can read *inconclusive* or another label. The clause held under the user's 18:31 UTC rule, which R1.3 narrows (UW-05 (b), the user's choice). Resolution: the clause is cut; Appendix C.11 gives R1.3's reason. **Revision 6e: confirmed by the lead (ruling 6);** Appendix C.11 now also says what such a repair's label is (conflict 83).
78. **The flag's name (revision 6e's brief against the sources).** Revision 6e's brief gave the flag another name; LW-09's default (with RC-02), DRAFT §3.7 and §4.5 name it "fails answers end", and "comparator fails answers end" for a failing comparator; FOCUS records LW-09 at its default. Resolution (revision 6e): the sources' names, in Appendix C.11 only; the body said "still read and flagged". **Revision 6f: RESOLVED (the lead's ruling 1):** the name is "fails answers end" everywhere; §4.2 now names it.
79. **UW-15's "harmful-prompt cause" against "benign-looking" (§4.5, Appendix C.5, the Impact Statement).** LW-10 corrects the description, but it is outside the brief. Resolution: not changed; decision 54. **Revision 6f: RESOLVED (the lead's ruling 2; decision 54):** §4.5, Appendix C.5 and the Impact Statement describe the plant as PKU red-team prompts with PKU-safe, non-refusing answers.
80. **B3's "the frozen rule".** Grade 1 is the user's 8 Oct label rule (decided), while the stage-1 pre-registration is not frozen, so "frozen" would mislead in the paper. Resolution: "the rule of §3.2". **Revision 6f: RESOLVED (the lead's ruling 4): confirmed; unchanged.**
81. **Gate A's mean erosion in Appendix C.11 against "nothing in the body is a result".** UW-13's rule needs its constant, 45.11 pp, OLMo-2-1B's Gate A mean $E$, and C1 states δ as about 11% of it. Resolution: the number, C1's wording and the per-seed ratio range (10.4–11.5%) sit in Appendix C.11, as Appendix C.1 already uses 0.75 of Gate A's mean $E$ (33.8 pp); the body's clause gives no number, and the per-seed $E$ values are not in the paper. **Revision 6f: RESOLVED (the lead's ruling 4): confirmed; unchanged** (the 45.11 pp makes δ_fam reproducible).
82. **Decision 2 and the fill-in "A repair is worse" against R1.3.** Both said that every significantly adverse contrast gives *worse*. Resolution: both note R1.3's narrowing (conflict 77).
83. **Ruling 6's "still feasible at τ_O*" against the three feasibility states.** Under LW-06 an arm can also be *undetermined*, and R1.3 needs *infeasible*. Resolution: Appendix C.11 says "not infeasible at $\tau_O^*$ (feasible or undetermined)", which follows from R1.3 and LW-06 as written, and adds the same for task loss (R1.3's second half). **Revision 6f: RESOLVED (the lead's ruling 4):** confirmed: only an arm that is infeasible at $\tau_O^*$ (its interval's lower end above the limit) and significantly above its comparator is *worse (O)*. Appendix C.11 says exactly this and is unchanged. **Revision 6g (the lead's ruling 7):** the asterisk in this item's title is escaped; the LaTeX converter had read it as the start of an emphasis.
84. **"Olmo 3 the first model cut" (§6) against the cut order.** FOCUS 8dac6ad cuts 70B, then 32B, then Olmo 3 7B. §6's clause ("at 7–8B it is planned on both families and Olmo-3-7B-Instruct, funding-gated, with one family as the fallback and Olmo 3 the first model cut") sits in a sentence about 7–8B, and §4.1's marker says "first 7–8B model cut", which the coordinator keeps. Resolution: unchanged; if a reader takes §6's phrase as the first cut overall, add "7–8B" there (one word). **Revision 6f: RESOLVED (the lead's ruling 4): confirmed; unchanged.**
85. **"Correct it wherever the notes give only USD 8–10" (the brief).** Decision 48 already gave both figures; it now labels USD 8–10 as Gate A rung 1 only (two seeds) and USD 16–44 as the three-seed Gate A. No other place in the notes gave an Olmo 3 cost.

86. **UW-15's wording against "each training-signal diagnostic".** The user took UW-15 (FOCUS fcf79a2): headline item (2) reads "does each diagnostic find a concentrated, harmful-prompt cause". LW-10 F2, at its default, counts only A1 and the Self-Inf-N-style score toward "finds a known cause". The lead's ruling 3 inserts "training-signal" in the abstract, §1, contribution 2 and §6. Resolution: applied as ruled. It narrows the user's adopted text by one word, so the lead may report it to the user: the weekend instruction (FOCUS 00b99a4) covers pending choices, and UW-15 was decided. If the user wants UW-15's exact words, remove "training-signal" in those four places and keep §4.5's counting sentence and contribution 2's construction-check sentence. **Revision 6g: RESOLVED (the lead's ruling 1):** "training-signal" is kept by the lead's ruling, to be mentioned to the user; no text change.
87. **The reason for the construction checks.** The brief says that the text-only scorers "can read the plant's text"; DRAFT §8.5 F2 says that they "read only row text and θ0, so their planted-cell ranks are fixed when the plant is drawn". Resolution: Appendix C.5 gives the source's reason ("read only row text and the untuned model"); the body gives none. **Revision 6g: RESOLVED (the lead's ruling 2):** the source's reason: the text-only scorers read only row text and θ0, so their ranks are fixed at the draw. Appendix C.5 already said so; §4.5 now says so too (conflict 94).
88. **"Each diagnostic's top-k hit rate" (DRAFT §8.6) against the narrowing.** §8.6 keeps every scorer's hit rate as an output, and F2 changes only what counts as finding the cause. Resolution: "each scorer's top-$k$ hit rate" in §4.5 and Appendix C.5; the counting rule is its own sentence.
89. **The Self-Inf-N-style score in the counting rule against DRAFT §8.4's list.** F2 and LW-10 count A1 and the Self-Inf-N-style score, but §8.4's list of planted-cell ranks scored at no cost (A1, L1, H1, H2a/b, DataShield-style and the content classifier) has no Self-Inf-N-style rank. The Self-Inf-N-style score is scored only (Appendix C.3), and its build is LW-18's ambitious default, with "post-Gate-B exploratory" as the fallback if it is not committed at W1 (FOCUS fcf79a2). Resolution: the paper follows F2 and LW-10. For the lead: add its rank to §8.4's list in the stage-1 record, and, if REQ-13 is not built by W1, drop it from the counting rule (§4.5, Appendix C.5). **Revision 6g: RESOLVED (the lead's ruling 3; addendum §8 item 2):** the Self-Inf-N-style rank joins the planted-cell ranks scored without new runs; if its scorer (REQ-13) is not committed before Gate B's category runs (W1), it leaves both that list and the counting rule, and only A1 counts (Appendix C.5, marked [PLANNED; the stage-1 pre-registration is not yet frozen]). §4.5 still names both counted diagnostics and states no condition.
90. **Measured figures in §6 (the lead's ruling 6) against "nothing in the body is a result".** The ruling's sentence carries two stability figures, and decision 55 had proposed keeping the numbers in these notes. Resolution: applied as ruled; "What this is" lists it as a sixth kind of exception: a pre-registered report-only read (addendum §5 item 2) of A1's sealed per-row scores on wave 0b's retrain of $a_0$, not an outcome of any repair arm. For the lead: if the body should stay free of measured values, the figures can move to Appendix C.2 and §6 keep "agree only moderately".
91. **Where the 11.04 pp goes (the lead's ruling 5).** The ruling gives the number and two places, §4.2 and Appendix C.11, but not which one carries the number. Resolution: Appendix C.11 only, as for Gate A's 45.11 pp (conflict 81); §4.2 says that the check "put the six-seed minimum detectable effect (MDE) on the 100 above 5 pp", as §4.6 states Gate B's passed noise gate without a number.
92. **"By W1" (the lead's ruling 3) in the paper's terms.** The DRAFT's deadline line defines W1 as "before Gate B wave 1", which the paper calls Gate B's category runs (§4.9; Appendix C.1). Resolution: "if its scorer is not committed before Gate B's category runs".
93. **Parts of addendum §7 not in the paper.** Its item 4 judges the 500-prompt bank on answers capped at 128 tokens, after W1-18's truncation check, and its item 3 stops with counts only if fewer than 400 prompts are eligible; §4.2 says "greedy answers" and gives no cap. Resolution: not stated (outside the ruling). For the lead: whether Appendix C.11 should state the 128-token cap.
94. **"Check that C.5 and §4.5 say exactly this" (the lead's ruling 2).** Revision 6f gave the reason in Appendix C.5 only. Read as: both places give the source's reason. Resolution: §4.5 gains it in the ruling's words, at about one line's cost; contribution 2 still gives none.

95. **The fallback clause in Appendix C.5 (the lead's ruling 4).** The ruling lists the sensitivity's parts; addendum §8 item 1 also gives its fallback, the shared source stated as a limit in §6 if the set cannot be built before the stage-1 freeze (S1) or stage 1b is not funded. Resolution: Appendix C.5 states the first branch ("If that set cannot be built before the freeze, the shared source is reported as a limit instead"); the second is moot there, because Appendix C.5's design runs only if funded. §6's sentence: decision 56.

**Sources read (revision 6g):** the brief for revision 6g (the lead's rulings 1–9, GI VI, 17 Oct, under the user's weekend delegation); FOCUS.md at `origin/stageb-costs-and-recovery-safeguards-20261003` (fetched 17 Oct at 09:59 UTC; head 76a08a0; FOCUS last changed in 1423a4c), its 09:32 and 09:37 UTC entries and its 8 Oct 15:53 UTC entry in full; the stage-1 build addendum (`.ai/briefs/stage1-commitments-build-20261016-lead-decisions.md` at origin), §7 and §8 in full and §3–§6; the commit messages of 158d2c2 and 76a08a0; in `research/experiments/2026-10-17-stage1-preregistration/` at origin (2ded100), the pre-registration DRAFT revision 2 (the deadline line, §1.2's A1 row, §8.1–§8.5 and §9.3 in full; other lines by search) and the consolidated decision list (LW-18, by search); these notes and the body of revision 6f.

**Sources read (revision 6f):** the brief for revision 6f (the lead's rulings on decision 54 and conflicts 78, 80, 81, 83 and 84, GI VI, 17 Oct, under the user's weekend delegation); FOCUS.md at `origin/stageb-costs-and-recovery-safeguards-20261003` (fetched 17 Oct at 09:19 UTC; head ecdd23e; FOCUS last changed in 00b99a4), its fcf79a2 entries (the user's UW decisions, UW-15 and UW-16 among them; the lead's LW decisions at their defaults, with LW-18) and its 00b99a4 entry; fetched again at 09:55 UTC (head 23848e0), its 09:32 and 09:37 UTC entries in full and 158d2c2's message; in `research/experiments/2026-10-17-stage1-preregistration/` at origin (2ded100), the consolidated decision list (LW-09, LW-10, UW-15, UW-16 and LW-26 in full; other items by search) and the pre-registration DRAFT revision 2 (§0.5, §3.7, §4.5, §8.1–§8.6 and §9.1–§9.2 in full); these notes and the body of revision 6e.

**Sources read (revision 6e):** the brief for revision 6e (the lead's rulings 1–9) and the coordinator's addition on Olmo 3 (decision 48 confirmed); FOCUS.md at `origin/stageb-costs-and-recovery-safeguards-20261003` (fetched 17 Oct at 08:42 UTC, head 80fd326, FOCUS last changed in fcf79a2; fetched again at 09:01 UTC, head 00b99a4), its fcf79a2 entries (UW-13, UW-15 and the LW items at their defaults) and its 8dac6ad and 00b99a4 entries in full; the 8 Oct 13:53 UTC headline (item (2)); in `research/experiments/2026-10-17-stage1-preregistration/` at origin (2ded100), the consolidated decision list (UW-13, UW-14, UW-15, UW-16, LW-02 to LW-10, LW-26 in full; other items by search), the pre-registration DRAFT revision 2 (§0.5, §3.7, §4.1–§4.6, §6.1–§6.4, §8.5, FC-8, FC-11 in full) and the consistency review (the 45.11 pp mean, by search); the stage-1 build addendum §6 (`.ai/briefs/stage1-commitments-build-20261016-lead-decisions.md` at origin); `research/review/2026-10-17-olmo3-7b/olmo3-7b-addendum.md` at origin (its Gate A cost lines, by search); these notes and the body of revision 6d.

**Sources read (revision 6d):** the brief for revision 6d (the lead's rulings 1–7) and the coordinator's addition on Olmo 3; FOCUS.md at `origin/stageb-costs-and-recovery-safeguards-20261003` (fetched 17 Oct at 08:16 UTC, head f9714b5; again at 08:34 UTC, head 4446539; FOCUS unchanged since fcf79a2), its two fcf79a2 entries in full; the stage-1 build addendum §3–§6 (`.ai/briefs/stage1-commitments-build-20261016-lead-decisions.md` at origin; §6 from 98ff104 and 0294d1b); in `research/experiments/2026-10-17-stage1-preregistration/` at origin, the consolidated decision list (its index; UW-02 to UW-09, UW-14, LW-01, LW-02, LW-06, LW-07, LW-17, LW-23 and US-01 in full), the pre-registration DRAFT revision 2 (§1.1–§1.3, §3, §4, §6 and §9.3 in full; other lines by search) and the W1 builder extract (W1-01 to W1-03); `scripts/gate_b_mac_judges/harmbench_files.py` at origin (its header, tokenizer and clip function, by search); HarmBench's `evaluate_completions.py` (raw file on GitHub, main branch, fetched 17 Oct about 08:24 UTC); the arXiv API record of 2409.00598 (08:25 UTC); `research/review/2026-10-17-olmo3-7b/` at 4446539 (`model-check.md` §0–§5 and its blocker table, `olmo3-7b-addendum.md` §0 and its Gate A cost line, the head of `verify-report.md`); these notes and the body of revision 6c.

**Sources read (revision 6c):** the brief for revision 6c (the lead's rulings 1–5); FOCUS.md at `origin/stageb-costs-and-recovery-safeguards-20261003` (fetched 17 Oct at 07:56 UTC; identical to the working copy), its two entries of fcf79a2 in full, and searches for Qwen2.5-1.5B, "Llama/OLMo only", "no more Qwen spend", the 7 Oct 17:44 UTC entry, Olmo 3, option B, re-plan and the wave-1 reserve; the stage-1 build addendum §6 (`git show` of 98ff104 and 0294d1b); in `research/experiments/2026-10-17-stage1-preregistration/`, the consolidated

decision list (UW-14 and LW-17, by search), the pre-registration DRAFT (its [USER-CONFIRM] rule, FC-1, FC-17, HB-1, its token-accounting item and its secondary list, by search) and the W1 builder extract (W1-02, W1-03 and its other R_tok lines); HarmBench's `evaluate_completions.py` (raw file on GitHub, main branch, fetched 17 Oct about 08:00 UTC, two reads); these notes and the body of revision 6b.

**Sources read (revision 6b):** the coordinator's brief for revision 6b; `writing/paper/2026-10-17-mock-review-A-methods.md` and `writing/paper/2026-10-17-mock-review-B-safety.md` in full; these notes and the body of revision 6; FOCUS.md at `origin/stageb-costs-and-recovery-safeguards-20261003` (fetched 17 Oct; head 6368d66), searched for Qwen, the HarmBench judge and the newest entries, and read for the 7 Oct entries of 17:44, 19:14, 20:45 and 20:58 UTC, the 8 Oct 10:14 UTC entry and the 17 Oct entries (T1 bundle, backup); the header of the stage-1 pre-registration DRAFT revision 2 and its lines on HB-1 and SRJ (by search only); the commit messages of 3db753e and 2ded100; the arXiv API records of 2505.12038, 2408.09600, 2409.18169, 2406.05946, 2402.10260 and 2512.13961; the PMLR v267 bibliography (entries lu25g and huang25b); the OpenReview API search record for Qi et al. (2025) (the PDF endpoint answered with a challenge); the Hugging Face model API record of allenai/Olmo-3-7B-Instruct.

**Sources read (revision 6):** FOCUS.md at `origin/stageb-costs-and-recovery-safeguards-20261003` (fetched; head 416b780), every entry from 9 Oct 17:00 UTC to 16 Oct 21:05 UTC, and the 8 Oct 12:53 and 13:53 UTC entries (the scale ladder; the adopted direction); the Gate B/C pre-registration, revision 5 (SHA-256 5c97cce7… checked), header, §1–§5, §7, §8 and §10; the kill-test design §4.4–§4.5; the GI handoff, STATUS of 16 Oct 13:20 UTC and the 15 Oct lead handoff; the stage-1 commitments brief's D5-7 row and FOCUS's L1–L16 entry; the restart plan's line on "Safety at One Shot" (for the reference only); the side paper's abstract and section list (`writing/side-notes/arr/main.tex` on its branch, read with `git show`; overlap only); `writing/paper/2026-10-17-lit-sweep.md` §0, §4's corrections, §5 and §6; the summary of `writing/paper/2026-10-17-icml-replan-PROPOSAL.md`; the ICML 2026 example paper's instructions (page limit, anonymity, self-citation, Impact Statement); the arXiv API records of 2601.01887, 2506.00676, 2606.00160, 2607.15081, 2605.14194 and 2606.09866; OpenReview API search records for EyH8Fu3vtZ, Jo6U0y2omr, yKetbTItcu and GradShield; the NeurIPS 2026 accepted-papers JSON (the sweep's copy) for GradShield.

**Sources read (revision 5e):** FOCUS.md to its newest entry at 17:31 UTC (9 Oct 17:22 UTC, the Granite parity PASS, commit 2837eaf), with the entries of 16:59 UTC (Llama's rung-2 read), 16:48 UTC (the Granite judge on RunPod; its parity rule and PARITY.md SHA-256), 15:31 UTC (the rung-2 pre-registration and its launch balance) and 15:29 UTC (the USD 10/h price limit), the 13:16 UTC lead notes (the [D20] constants and the wave-1 estimate) and the 7 Oct 21:35 UTC amendment to the family-probe bars (edge band, ARC, short answers, unscored answers, overall qualification); the lead's brief for revision 5e.

**Sources read (revision 5d):** FOCUS.md's entry "Llama-3.2-1B rung 2 at the main recipe (recipe B at LR 5e-5), seed 1101, read 2026-10-09 16:59 UTC" (commit 7a8eb08; no later entry exists at 17:03 UTC), its pre-registration (15:31 UTC) and the 7 Oct 21:35 UTC amendment to the family-probe bars (short answers; overall qualification).

**Sources read (revision 5c):** FOCUS.md's entry "USER DECISIONS, 2026-10-09 15:23 UTC" (no later entry exists), its 13:45 UTC item on the ARC scope and the 8 Oct 20:50 UTC ARC entry; on the Side Project's branch, by `git show` without checkout, `research/side-blind-read/PREREG-2026-10-08.md` (design, rubric, limits, results) and `blind-read-scores-2026-10-09.json` at commit 57d08dd (4d63e04 differs only in the results time); the Gate A audit §8 item 4 (the proposed sample: 40 Unsafe answers from OLMo's Gate A units, 20 under 20 tokens and 20 flagged only as unethical, plus the 29 Unsafe answers on the over-refusal bank).

**Sources read (revision 5b):** FOCUS.md's entries of 9 Oct 15:05 UTC (the blind human read and its two sub-items; recorded in commit 3e7c570) and 13:45 UTC, and of 8 Oct 18:31 UTC (the *worse* label's wording) and 15:59 UTC (FAQ items 1 and 4); on the Side Project's branch claude/wonderful-greider-d1aa5b, by `git show` without checkout, `research/side-blind-read/PREREG-2026-10-08.md` and `blind-read-scores-2026-10-09.json` at commit 57d08dd (4d63e04 differs only in the results time, 15:00 → 14:58 UTC); the Gate B/C DRAFT §12 item 5 (still revision 4b); the Gate A audit §8 item 4.

**Sources read (revision 5):** FOCUS.md's entries of 9 Oct from 00:36 to 13:16 UTC, and of 8 Oct at 15:17, 16:50, 20:17, 20:50, 21:36, 23:28 and 23:43 UTC, with the adopted amendment's reference and family lines; the results record's sections from "OLMo-2-1B under recipe G, seed 1103" (8 Oct 21:53 UTC) to "Rung 2" (9 Oct); the Gate B/C DRAFT header (still revision 4b); the brief's revision-16 header and [D14]–[D20]; the mechanism note's summary, §2, §4 and its three addenda; the Llama recipe-B kits' reading files (seeds 1102 and 1103); on the Side Project's branch (read-only), its Llama recipe-B results note, the arXiv note's commit c51cf39 and the fiction pre-registration header; the arXiv API record and the OpenReview notes for Safe LoRA (arXiv:2405.16833).

**Sources read (revision 4):** FOCUS.md's entries of 8 Oct from 17:36 to 21:37 UTC; the results record's sections on recipe G (seed 1101), Llama under recipe G, OLMo recipe G at seeds 1101 and 1102, the rank-16 check and the token-mean-only run; the mechanism note in full (including its recipe-G and Llama addenda); the header of the Gate B/C DRAFT (still revision 4b); the file names and the two-seed mean of the kit's reading files for OLMo recipe-G seed 1103; the arXiv abstract of Qi et al. (2025), by API.

**Sources read (revision 3):** FOCUS.md's entries of 8 Oct from 15:17 to 19:24 UTC; the results record's recipe-G, Llama and HarmBench (complete) sections; the header of the Gate B/C DRAFT (revision 4b); the arXiv abstract of Qi et al. (2025). The sharpened-direction plan was not changed after revision 2 used it (last commit 3ade6c3), so it was not re-read.

**Sources read (revision 2):** the paper specification (in full); the adopted FOCUS amendment (commit 4b37ec2) and FOCUS's dated entries of 6–8 Oct; the sharpened-direction plan (revision 2, in full); the Gate A audit; the kill-test Stage 0 results record (Gate A); the Gate B/C pre-registration draft (revision 3); the mechanism note; the Guo-gap analysis; the safety-landscape note; the erosion-amplifiers note; reading list v2 (groups 4–5 and the DataRx correction); the benign-erosion recipes note (DataRx); the kill-test design (measurement table). Abstracts fetched this session for the new citations.